
Manifold-Weighted Neural Networks

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Abstract

We establish universal approximation for neural networks whose weight matrices take values in matrix manifolds. The nonlinear matrix manifolds considered in this work arise as orbits of classical Lie group actions embedded in the ambient space $\mathbb{R}^{d \times k}$. Stiefel manifolds provide a prominent example with empirical precedent in prior neural-network models. We present an expanded catalog of such manifolds, drawing from well-studied constructions in physics and differential geometry. These manifolds are particularly appealing for intrinsic Riemannian optimization: the transitivity of the underlying group action yields explicit tangent-space descriptions and retractions. Moreover, the group-action structure enables parameter-efficient factorizations, reducing the number of trainable variables. This leads to a fundamental expressivity question: when weight matrices take values in nonlinear matrix manifolds, which architectures retain universal approximation power? We prove that manifold-weighted networks with residual connections and bounded scalar multipliers are $L^p(K)$ -universal. To establish universality across our manifold catalog, we introduce two independent sets of sufficient conditions: Register-Isolated Primitive Realizability (RIPR) and Cross-Axis Fold-and-Cut (CAFC). These conditions correspond to two distinct approximation constructions. By verifying RIPR or CAFC case by case for the matrix manifolds in our catalog, we establish universality for the corresponding residual manifold-weighted networks with bounded scalar multipliers.

1 Introduction

In a neural network, each layer applies a weight matrix $W_\ell \in \mathbb{R}^{d \times k}$ to its input, and the parameter space chosen for W_ℓ is a modelling decision. The conventional choice is the full ambient $\mathbb{R}^{d \times k}$, so the optimizer searches all dk dimensions. Another option is to draw W_ℓ from a smooth submanifold $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$; the manifolds studied here, including Stiefel and Grassmann manifolds, appear in Table 1. This is not merely an externally imposed restriction. A layer stores and applies a weight matrix but never adds two weights, so the vector-space structure of $\mathbb{R}^{d \times k}$ is not intrinsic to the layer. A smooth matrix manifold can therefore serve as the layer’s native parameter space, especially under intrinsic Riemannian optimization, where classical-group transitivity gives explicit tangent-space descriptions and retractions [1, 2].

Viewed against the ambient default, the same choice is also a constraint, hence a form of geometric regularization controlled by $\dim(\mathcal{M})$ and, for compact families bounded in operator norm, by uniform bounds on $\|W\|_{\text{op}}$ that can improve training stability and reduce reliance on weight decay. Although nonlinear manifolds need not have global injective coordinates, all manifolds we consider have global parametrizations induced by the group action. These parametrizations may be redundant because of gauge symmetries, but they often use far fewer trainable variables than ambient d^2 (or dk) matrices [3, 4]. These facts motivate the *manifold-weighted* view adopted here: the layer weight is a native point of a prescribed embedded matrix manifold. To our knowledge, this general viewpoint is new beyond the compact Stiefel family, including its orthogonal and unitary special cases. We use the constrained view when comparing with ambient dense networks or stating regularization and expressivity tradeoffs. For optimization, Edelman, Arias, and Smith [1] and Absil, Mahony, and

Sepulchre [2] systematized the matrix-manifold toolkit, and orthogonal initialization in deep linear networks [5] laid practical groundwork. Neural-network uses of this compact family mainly target stability, regularization, or parametrization: Stiefel constraints control gradient norms in unitary and orthogonal RNNs [6–8], parametrize orthogonal and unitary weights [9, 10], and implement orthogonality regularization and orthogonal convolutional layers [11, 12]. More recent work includes Muon-style matrix-sign optimizers and Stiefel-LoRA adaptation [14–16], alongside G -equivariant networks, where linear layers are structured by group-equivariance constraints [17, 30].

What has remained open beyond this compact-family precedent is the core theoretical question of whether manifold-weighted networks still universally approximate, which the present paper addresses.

The nonexpansive obstruction (§3.2). A simple Lipschitz argument rules out composition-only FFNs with $\|W\|_{\text{op}} \leq 1$ and 1-Lipschitz activation (Proposition 1). The obstruction applies uniformly to Stiefel, Grassmann, and every manifold in the operator norm unit ball.

Universality framework (§4.1–§4.3). Following Lin and Jegelka’s residual-block construction [18], we introduce a *Register-Isolated Primitive Realizability* (RIPR) criterion (Definition 4), which reduces the verification of $L^p(K)$ -universality for any manifold-weighted residual architecture to a single geometric interface: realizing four scalar-register primitives via the Register-Isolated Hinge (RIH) atom. Residual connections together with two scalar multipliers (an outer scalar multiplier s and an inner scalar multiplier t) break the nonexpansive barrier and restore $L^p(K)$ -universality on every compact $K \subset \mathbb{R}^d$.

Adapting Kidger and Lyons’s fold-and-cut technique [19] to a rotated orthonormal frame yields a complementary *Cross-Axis Fold-and-Cut* (CAFC) construction (§4.2) in which read and write directions of each block live on different axes; CAFC trades the inner scalar multiplier t for the rotated frame and improves the depth rate by one input dimension. The per-manifold multiplier budgets and proof routes are geometric; Table 2 records them, with proofs in §4.3.

Empirical complement (§5). A controlled synthetic study tests the Lipschitz separation and the recovery provided by residual scalar multipliers, and exposes an initialization-dependent optimization gap. Preliminary finite-scale CIFAR-10 [41] and nanoGPT [44] studies appear in Appendix C.

2 Related Work

Universality and constrained classes. Classical universality theorems [20–22] treat unconstrained feedforward networks. Lin and Jegelka [18] prove $L^1(\mathbb{R}^d)$ -universality for one-neuron residual blocks; our RIPR framework extends their proof to constrained manifolds via a minimal geometric interface. Murari, Furuya, and Schönlieb [23] prove universal approximation for 1-Lipschitz ResNets within the spectral-norm *ball* (a convex set) with scalar outputs; their follow-up [24] treats transformers via function-space gradient flows. Their setting is complementary: convex techniques cover the spectral-norm ball, while our universality results include nonconvex manifolds (Grassmann, $\mathcal{J}_n$, Gr_{Sp} , LGr). Conversely, the spectral-norm ball contains all rank-1 frames $W = uv^\top$, so our Stiefel RIPR proof recovers an overlapping spectral-norm-ball case by a different construction.

Tabuada and Gharesifard [25] and Ruiz-Balet and Zuazua [26] use control-theoretic methods that do not address manifold constraints; the latter’s half-space invariance lemma is the continuous-depth analogue of our read-axis separation in the CAFC proof (Theorem 66).

Elamvazhuthi et al. [27] obtain manifold-invariant Neural-ODE universality via bracket-generating controllability of vector fields; our setting is finite-block algebraic realizability for discrete-layer weight matrices in matrix manifolds, not continuous-time controllability of state flows. Min and Azizan [28] preserve universality under hard input-dependent inequality *output* constraints via a differentiable enforcement layer; our constraints sit in every weight matrix and arise from Lie-group orbits, not post-hoc output projection.

Obstructions and scalar multipliers. Anil, Lucas, and Grosse [13] achieve 1-Lipschitz universality via GroupSort, working *within* the 1-Lipschitz class; our Proposition 1 identifies why ReLU with weights of operator norm at most one is confined to that class under composition. Vorontsov

et al. [8] report a negative result for orthogonal RNN weights consistent with our obstruction. Our outer scalar multiplier s resembles the scalar weight of ReZero [29] but serves a different purpose: ReZero accelerates early training, while s (with t) recovers universality.

Manifold optimization in deep learning. Our results add an expressivity guarantee to recent work on manifold-constrained optimization: modular duality [14] motivates Newton–Schulz style dualization maps for linear and convolutional layers; the Polar Express [15] develops matrix-sign methods integrated into Muon; and StelLA [16] parametrizes LoRA adapters on Stiefel. Our framework also applies to G -equivariant architectures [17, 30] whose weight matrices lie in intertwining algebras: when the resulting manifold satisfies RIPR, the manifold-agnostic meta-theorem yields universality with zero depth overhead.

3 Preliminaries

This section fixes the matrix manifolds and residual blocks used by the two universality routes. We first specify the matrix manifolds and the residual blocks in which they appear; the later RIPR and CAFC conditions then ask whether a given manifold can realize a small set of register operations within those blocks. This keeps the general approximation criteria distinct from the manifold-specific witness checks carried out in the appendix.

3.1 Definitions and Architecture

Notation. $\mathbb{R}^{m \times n}$ denotes real matrices, $\|x\| = \|x\|_2$ the Euclidean norm, $\|A\|_{\text{op}}$ the spectral norm, $\|A\|_F$ the Frobenius norm. By convention d is the residual stream dimension and rectangular constrained weights have shape $\mathbb{R}^{d \times k}$. σ denotes the activation, applied coordinatewise to vectors. $\text{Lip}(f)$ is the Lipschitz constant. $K \subset \mathbb{R}^d$ is compact; for $1 \leq p < \infty$, $L^p(\mathbb{R}^d)$ and $L^p(K)$ denote the usual Lebesgue spaces, with Lebesgue measure restricted to K in the latter. e_i is the i -th standard basis vector.

Matrix manifolds. The manifolds considered here are realized as orbits $\mathcal{M} = G \cdot W_0$ of classical Lie groups acting on the ambient $\mathbb{R}^{m \times n}$. Equivalently, for the stabilizer $H = \{g \in G : g \cdot W_0 = W_0\}$, the orbit is an embedded homogeneous space $G/H \hookrightarrow \mathbb{R}^{m \times n}$ [2]. The orbit parametrization $G \rightarrow \mathbb{R}^{m \times n}$, $g \mapsto g \cdot W_0$, has fibers gH and gives a global parameter-efficient parametrization: the structured representative g often requires fewer stored parameters than the ambient embedding $\mathcal{M} \subset \mathbb{R}^{m \times n}$. For the concrete manifolds considered here, this recovers standard factors; for example, for $\text{Gr}(k, d) \subset \mathbb{R}^{d \times d}$, the map $Q \in \text{V}(k, d) \mapsto QQ^\top$ uses dk stored parameters versus d^2 in the ambient representation, with $R \in \text{O}(k)$ gauge equivalence $Q \leftrightarrow QR$; analogously, $U \in \text{St}(k, d) \mapsto UU^\top$ parametrizes $\text{PSD}(k, d) \subset \mathbb{R}^{d \times d}$ with the same ambient savings and gauge $U \leftrightarrow UR$. Table 1 summarizes the manifolds we treat. Full geometric definitions, parametrizations, and standard matrix-manifold references [2–4] appear in Appendix A.

Table 1: Matrix manifold definitions used in this paper. A generic weight matrix in each manifold is denoted W . The column $\dim(\mathcal{M})$ is the intrinsic parameter count, smaller than the dense ambient d^2 (square cases) or dk (rectangular Stiefel row). I denotes the identity matrix (subscripted as I_k, I_n when the size is not otherwise clear); J in the Gr_{Sp} row is the standard symplectic form $\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$; in the Sym_Λ row, Λ has m distinct eigenvalues λ_i with geometric multiplicities k_i ; the universality row assumes $m \geq 2$ and at least one nonzero eigenvalue.

$\mathcal{M}$	Shape	Range	$\dim(\mathcal{M})$	$\sup \ W\ _{\text{op}}$	Defining constraint
$\text{V}(k, d)$	$\mathbb{R}^{d \times k}$	$1 \leq k \leq d$	$dk - \frac{k(k+1)}{2}$	1	$W^\top W = I_k$
$\text{St}(k, d)$	$\mathbb{R}^{d \times k}$	$1 \leq k \leq d$	dk	∞	$\text{rank}(W) = k$
$\text{Gr}(k, d)$	$\mathbb{R}^{d \times d}$	$0 < k < d$	$k(d-k)$	1	$W^\top = W^2 = W, \text{tr } W = k$
$\text{PSD}(k, d)$	$\mathbb{R}^{d \times d}$	$0 < k < d$	$\frac{k(2d-k+1)}{2}$	∞	$W^\top = W, W \succeq 0, \text{rank}(W) = k$
$\text{Sym}_\Lambda(d)$	$\mathbb{R}^{d \times d}$	$m \geq 2, \Lambda \neq 0$	$\sum_{i < j} k_i k_j$	$\max_i \lambda_i $	$W^\top = W, \text{spec}(W) = \Lambda$
$\mathcal{J}_n$	$\mathbb{R}^{2n \times 2n}$	$n \geq 2$	$n(n-1)$	1	$W^\top W = I, W^2 = -I$
$\text{LGr}(\mathbb{R}^{2n})$	$\mathbb{R}^{2n \times 2n}$	$n \geq 1$	$\frac{n(n+1)}{2}$	1	$W = \begin{pmatrix} P & -Q \\ Q & P \end{pmatrix}, P, Q \text{ sym.}, P^2 + Q^2 = I_n, PQ = QP$
$\text{E}(d, k)$	$\mathbb{R}^{d \times d}$	$0 < k < d$	$2k(d-k)$	∞	$W^2 = W, \text{tr } W = k$
$\text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n})$	$\mathbb{R}^{2n \times 2n}$	$1 \leq k \leq n-1$	$4k(n-k)$	∞	$W^\top J W = J, W^2 = I, \text{tr } W = 4k - 2n$

Architecture. A *composition-only FFN* of depth L has the form $f(x) = W_L \sigma(W_{L-1} \sigma(\dots W_1 x + b_1 \dots) + b_{L-1}) + b_L$ with $W_i \in \mathcal{M}$. A *constrained scaled residual block* is

$$F_\ell(x) = x + s_\ell \cdot A_\ell \cdot \text{ReLU}(t_\ell \cdot B_\ell x + b_\ell), \quad A_\ell, B_\ell \in \mathcal{M}, \quad s_\ell, t_\ell \in \mathbb{R}, \quad b_\ell \in \mathbb{R}^d. \quad (1)$$

A *constrained ResNet* is $R = L_{\text{out}} \circ F_N \circ \dots \circ F_1$ with $L_{\text{out}}: \mathbb{R}^d \rightarrow \mathbb{R}$ unconstrained. When $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ ($k < d$), the read direction uses $B_\ell^\top x \in \mathbb{R}^k$ and $b_\ell \in \mathbb{R}^k$.

Bounded and unit-bounded scalar multipliers. We use *bounded scalar multipliers* in a fixed-tolerance constructive sense: for each fixed manifold $\mathcal{M}$, compact domain K , target f , and tolerance $\varepsilon > 0$, the construction emits finitely many explicit scalar values $\{(s_\ell, t_\ell)\}$, hence a finite envelope $B = \max_\ell \{|s_\ell|, |t_\ell|\}$. More generally, for target classes with uniform L^∞ and BV bounds, Remark 19 gives an envelope uniform over the class at fixed tolerance.

The strictly stronger *unit-bounded* regime imposes the uniform input-independent budget $|s_\ell|, |t_\ell| \leq 1$, independently of the target and tolerance. This is an architectural analogue of activation gates and affine rescalings: GLU-style gated convolutional language models [31], GLU variants in Transformer feed-forward layers [32], affine coupling in flow models [33], and highway gates [34]. Sigmoid gates are input-dependent and bounded in $[0, 1]$, whereas GLU variants and affine-coupling layers allow more general rescalings; our unit-bounded regime instead uses input-independent trainable scalars in $[-1, 1]$. The unit-bounded strengthening is established for the CAFC manifolds covered in Appendix E.

3.2 The Nonexpansive Obstruction

Proposition 1 (Nonexpansive obstruction). *Let each weight matrix, including the terminal linear map producing the scalar output, satisfy $\|W_i\|_{\text{op}} \leq 1$, and let $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ be 1-Lipschitz. Every depth- L composition-only scalar-output FFN $f: \mathbb{R}^d \rightarrow \mathbb{R}$ (no unconstrained terminal readout) has $\text{Lip}(f) \leq 1$. Consequently, on any compact $K \subset \mathbb{R}^d$ with at least two points, the uniform closure of such networks in $C(K)$ is contained in $\text{Lip}_1(K)$, a proper closed subset of $C(K)$; when K has positive Lebesgue measure, the same containment holds for their L^p -closure for every $p \in [1, \infty)$, and $\text{Lip}_1(K)$ is a proper closed subset of $L^p(K)$.*

The proof composes two elementary lemmas: 1-Lipschitz σ extended coordinatewise is 1-Lipschitz on $\mathbb{R}^d$, and composition multiplies Lipschitz constants. Full proof in Appendix B.1. Proposition 1 requires the operator-norm bound $\|W_i\|_{\text{op}} \leq 1$, so it does *not* apply to noncompact manifolds (PSD, $E(d, k)$, Gr_{SP}); those manifolds are simply skipped when invoking the obstruction.

Corollary 2. *Stiefel $V(k, d)$, Grassmann $\text{Gr}(k, d)$, $\mathcal{J}_n$, LGr , and every constrained weight family contained in the operator-norm unit ball satisfy the obstruction of Proposition 1. Composition-only scalar-output FFNs over any of these manifolds with a 1-Lipschitz activation are globally 1-Lipschitz and cannot be dense in $C(K)$ for any compact K with at least two points, nor in $L^p(K)$ for any $p \in [1, \infty)$ when K has positive Lebesgue measure.*

Corollary 3. *Allow each layer of a composition-only FFN to carry scalar multipliers: $x \mapsto s_\ell \sigma(t_\ell W_\ell x + b_\ell)$ with $|s_\ell|, |t_\ell| \leq 1$, $\|W_\ell\|_{\text{op}} \leq 1$, and σ 1-Lipschitz, and no unconstrained terminal linear readout. Each layer is 1-Lipschitz ($|s_\ell| \text{Lip}(\sigma) |t_\ell| \|W_\ell\|_{\text{op}} \leq 1$), so the composition is in $\text{Lip}_1(K)$ and the conclusion of Proposition 1 extends.*

4 Main Results

The main results give the positive counterpart to the obstruction: skip connections plus scalar multipliers restore $L^p(K)$ -universality for constrained weights. We prove this through two reusable geometric reductions: RIH realizes the scalar-register primitives needed for the piecewise-constant approximation program, while CAFC realizes rotated fold-and-cut atoms that reduce the scalar-multiplier budget when the manifold has the required fold geometry.

4.1 The Register-Isolated Hinge (RIH) Framework

For a constrained residual block $F(x) = x + s \cdot A \cdot \text{ReLU}(t \cdot Bx + b)$, fix coordinate i (throughout, “register” is shorthand for a single coordinate of the residual stream used as a working scalar). F

isolates register i on K if $F(x) = x + c_i(x) e_i$ with $F(x)_j = x_j$ for $j \neq i$. We abstract Lin and Jegelka’s unconstrained residual-block construction [18] into a reduction of L^1 -universality to four scalar-register primitives:

Definition 4 (RIPR: Register-Isolated Primitive Realizability). Let

$$\mathcal{B}_{\mathcal{M}} := \{x \mapsto x + s \cdot A \cdot \text{ReLU}(t \cdot Bx + b) : A, B \in \mathcal{M}, s, t \in \mathbb{R}, b \in \mathbb{R}^d\}$$

denote the constrained residual block family. The family $\mathcal{B}_{\mathcal{M}}$ satisfies RIPR with primitive cost $c_{\text{prim}} \geq 1$ if the following two statements hold for every compact state set $K_{\text{st}} \subset \mathbb{R}^d$.

- (a) **Same-coordinate primitive (signed-amplitude P3).** For every coordinate $i \in \{1, \dots, d\}$ and every $\gamma, a, b \in \mathbb{R}$, there is a composition $G_{i,\gamma,a,b}$ of at most c_{prim} blocks from $\mathcal{B}_{\mathcal{M}}$ such that, for every $x \in K_{\text{st}}$,

$$G_{i,\gamma,a,b}(x)_i = x_i + \gamma \text{ReLU}(ax_i + b), \quad G_{i,\gamma,a,b}(x)_m = x_m \quad (m \neq i).$$

(P1 nonnegative shift and P2 max-with-constant are special cases: $\gamma = 1, a = 0, b = c \geq 0$ for P1; $\gamma = 1, a = -1, b = c$ for P2.)

- (b) **Register coupling (P4).** For every ordered pair $i \neq j$, there is a composition $G_{i \leftarrow j}$ of at most c_{prim} blocks from $\mathcal{B}_{\mathcal{M}}$ such that, for every $x \in K_{\text{st}}$ with $x_j \geq 0$,

$$G_{i \leftarrow j}(x)_i = x_i + x_j, \quad G_{i \leftarrow j}(x)_m = x_m \quad (m \notin \{i, j\}).$$

No condition is imposed on $G_{i \leftarrow j}(x)_j$.

(For rectangular manifolds $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ with $k < d$, e.g. $\mathcal{M} = V(k, d)$, the same conditions apply after substituting $Bx \rightarrow B^\top x$ in the residual block; see Remark 16.)

Theorem 5 (RIPR $\Rightarrow L^p$ -universality). *If the constrained block family satisfies RIPR with primitive cost c_{prim} , the constrained ResNet class $L_{\text{out}} \circ F_N \circ \dots \circ F_1$ is dense in $L^p(K)$ for every $p \in [1, \infty)$ and every compact $K \subset \mathbb{R}^d$. For approximation tolerance $\varepsilon > 0$, the depth may be chosen so that $D_{\text{constr}}(\varepsilon) \leq N(\varepsilon) \cdot c_{\text{prim}}$, where $N(\varepsilon)$ is the depth required by the unconstrained P1–P4 program of Theorem 18.*

Theorem 5 is a compact-domain density statement (cf. Section 6). The extension from $L^1(K)$ to $L^p(K)$ uses the truncation argument of Remark 20. The proof replaces each primitive with the $\mathcal{M}$ -constrained gadget delivered by RIPR; the resulting per-manifold RIH constructions emit bounded scalar multipliers computed from the P1–P4 parameters (γ_r, a_r, b_r) and the manifold-witness coefficients. At fixed tolerance, the bound is explicit and uniform over target classes with bounded L^∞ and BV norms (Remark 19); Appendix E separately proves the tolerance-uniform unit-bounded strengthening $|s_\ell|, |t_\ell| \leq 1$ for the literal and macro CAFC manifolds it covers. The per-manifold RIPR details are in Appendix B.2.

Approximate RIPR. For some manifolds, the available gadgets approximate, rather than exactly realize, the projected RIH primitives on the compact state set. Approximate RIPR (Definition 21) fixes the finite P1–P4 schedule and tracks only the live coordinates that will be read later or enter the terminal readout. If the constrained gadget matches the ideal primitive on those coordinates to tolerances η_r , the live error satisfies a recurrence $e_r \leq \eta_r + \ell_r e_{r-1}$; choosing the η_r sufficiently small preserves the final $L^p(K)$ tolerance. Corollary 22 therefore gives the same density conclusion for these approximate-RIPR cases, with per-manifold verifications in Appendix B.5.

The RIH atom. The Register-Isolated Hinge realizes each primitive in a single block. Fix register i and pick witnesses $A, B \in \mathcal{M}$ so that $B^\top$ has its q -th row reading only x_i ($\text{row}_q(B^\top) = \nu \cdot e_i^\top, \nu \neq 0$), with write $Ae_q = e_i$. Choose biases that zero out inactive rows by making their preactivations strictly negative on the compact state set (Lemma 32). The block becomes

$$F(x) = x + s \cdot \text{ReLU}(\nu t x_i + b_q) \cdot e_i, \quad (2)$$

which realizes signed-amplitude P3(i, γ, a, b) at $\nu t = a, b_q = b, s = \gamma$ (with the read coefficient sign flipped via t where ν is fixed by the manifold witness); P4 is realized analogously using a separate read/write witness. Each manifold supplies its own witness pair (A, B) ; §4.3 records the table of witnesses and §B contains the per-manifold proofs.

Worked example: Grassmann. For $\mathcal{M} = \text{Gr}(k, d)$ with $1 \leq k \leq d-1$, the RIH atom is realized by the rank- k projector $A = B = e_i e_i^\top + \sum_{m=2}^k q_m q_m^\top \in \text{Gr}(k, d)$ with $\{q_2, \dots, q_k\}$ orthonormal in $e_i^\perp$. The active read component $(Bx)_i = x_i$ gives surviving coefficient $\nu = 1$ (so $\nu t = t$). Dead-zone biasing $b_j = -C$ for $j \neq i$ with $C > |t| M \sqrt{d} + |b|$ where $M = \sup_K \|x\|_\infty$ kills inactive ReLUs. The write $Ae_i = e_i$ deposits the active output on register i . Setting $s = \gamma$, $t = a$, $b_i = b$ gives $F(x)_i = x_i + \gamma \text{ReLU}(ax_i + b)$, realizing signed P3 exactly in one Grassmann block (Appendix B.5.2, Lemma 45). P4 uses an analogous $A \neq B$ pair with A rotated through the $(e_i + e_j)$ direction (Appendix B.5.2, Lemma 46).

From RIH atoms to L^p approximants. A finite sequence of RIH atoms (each realizing one P1–P4 primitive by Theorem 5) builds a piecewise-constant approximant of any compactly supported step function on $\mathbb{R}^d$ via the trapezoid grid-indicator construction recast in our normal form (Appendix B.2, Theorem 18). Each one-dimensional trapezoid level is produced by signed P3 updates: $\max\{R, \alpha R + \beta\}$ for the build-up, $\min\{R, \alpha R + \beta\}$ for the fold, and $\min\{R, c\}$ for the cap. Each update is rewritten as $R \mapsto R + \gamma \text{ReLU}(aR + b)$ with explicit (γ, a, b) , then a descending value-adjustment pass with $\gamma_m \leq -1$ maps each level to its target value. In dimension d , the construction combines per-coordinate indicators in separate registers, couples them into the output register via P4 with mixed-radix offsets so distinct cells take distinct levels, and applies a cut-and-shift to zero out the exterior. Density of compactly supported rectangular step functions in $L^1(\mathbb{R}^d)$ [35, Theorem 2.41] closes the argument.

Depth rates. For BV targets, $D_{\text{constr}}(\varepsilon) = O(\varepsilon^{-d})$ for every $c_{\text{prim}} = 1$ RIH manifold. For Lipschitz targets, CAFC (§4.2) saves one dimension to $O(\varepsilon^{-(d-1)})$ on the manifolds where it applies (via the Maiorov ridge approximation rate in $d-1$ variables [36]), with $O(d^2)$ macro-CAFC overhead on $\mathcal{J}_n, \text{Gr}_{\text{Sp}}$. Rate derivations and the macro-CAFC overhead are in Remark 29 (Appendix B.3).

4.2 The Cross-Axis Fold-and-Cut (CAFC) Framework

The RIH route uses both scalar multipliers on several manifolds. The complementary fold-and-cut construction trades the inner scalar multiplier t for a rotated frame: each block reads and writes along *different* axes of a totally-nonnzero orthonormal basis. We call this the *cross-axis fold-and-cut* construction (CAFC).

Definition 6 (CAFC-fold condition). Let

$$\mathcal{B}_{\mathcal{M}} := \{x \mapsto x + s \cdot A \cdot \text{ReLU}(Bx + b) : A, B \in \mathcal{M}, s \in \mathbb{R}, b \in \mathbb{R}^d\}$$

denote the constrained residual block family with $t = 1$ and $\mathcal{M} \subseteq \mathbb{R}^{d \times d}$. The family $\mathcal{B}_{\mathcal{M}}$ satisfies the **CAFC-fold condition** if there exist a totally-nonnzero orthonormal basis $\{w_1, \dots, w_d\}$ of $\mathbb{R}^d$ (every entry of every w_p nonzero) with w_d having all-positive standard-basis coordinates, and a dense set $\mathcal{D} \subseteq \{v \in w_d^\perp : \|v\| = 1\}$, such that the following two statements hold for every compact $K \subset \mathbb{R}^d$, every $c \in \mathbb{R} \setminus \{0\}$, and every $\beta \in \mathbb{R}$. The case $c = 0$ is represented by omitting the block.

- (a) For each $i \in \{1, \dots, d-1\}$, there exist $A, B \in \mathcal{M}$, $b \in \mathbb{R}^d$, and $s \in \mathbb{R}$ such that, for every $x \in K$,

$$x + s \cdot A \cdot \text{ReLU}(Bx + b) = x + c \cdot w_i \cdot \text{ReLU}(w_d^\top x + \beta).$$

- (b) For each $v \in \mathcal{D}$, there exist $A, B \in \mathcal{M}$, $b \in \mathbb{R}^d$, and $s \in \mathbb{R}$ such that, for every $x \in K$,

$$x + s \cdot A \cdot \text{ReLU}(Bx + b) = x + c \cdot w_d \cdot \text{ReLU}(v^\top x + \beta).$$

The CAFC atom. The CAFC atom is a single piecewise-affine fold in the rotated frame:

$$F(x) = x + c \cdot w_p \cdot \text{ReLU}(w_q^\top x + \beta), \quad p \neq q. \quad (3)$$

Iterating folds on a finite cover of the support (the “cut” phase) builds a piecewise-constant approximant of any BV target.

Worked example: Stiefel. For $\mathcal{M} = V(k, d)$ (rectangular, $1 \leq k \leq d - 1$), the CAFC atom is realized by Stiefel frames whose first columns supply the desired write/read directions. Given write w_i , read $v \in w_i^\perp$, set $\hat{v} = v/\|v\|$, extend $\{w_i\}$ to a Stiefel frame $A = [w_i, q_2, \dots, q_k] \in V(k, d)$ and $\{\hat{v}\}$ to $B = [\hat{v}, r_2, \dots, r_k] \in V(k, d)$. Then $Ae_1 = w_i$ and $\text{row}_1(B^\top) = \hat{v}^\top$, so the active read is $\hat{v}^\top x = \|v\|^{-1} v^\top x$. Dead-zone biasing $b_m = -C$ ($m \geq 2$) for C exceeding $\sup_K \|x\|$ kills the remaining preactivations. Setting s and b_1 so that $c = s\|v\|^{-1}$ and $\beta = \|v\| b_1$ gives $F(x) = x + c w_i \text{ReLU}(v^\top x + \beta)$, exactly the CAFC atom. The map $(s, b_1) \mapsto (c, \beta)$ is a bijection $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, so all CAFC fold parameters are realizable (Appendix B.5.1, Proposition 38). The dual fold family (write w_d , read $v \in \mathcal{D}$) is symmetric.

From CAFC atoms to L^p approximants. Given a target $f \in L^p(K)$, the CAFC construction approximates f in three phases (Appendix B.3, Theorem 28): (i) *Strip-and-cut.* Use folds with read direction w_d (and write directions w_i for $i < d$) to translate each w_d -strip of K into a disjoint transverse box, so the $(d - 1)$ -dimensional slices live in non-overlapping coordinate regions. (ii) *Transverse ridge approximation.* On each shifted box, approximate the per-strip target by a finite ridge sum $\sum_m c_m \text{ReLU}(a_m^\top x + \eta_m)$ along directions in the dense set $\mathcal{D} \subseteq w_d^\perp$. (iii) *Output assembly.* Use folds with write direction w_d to deposit the ridge sum into w_d ; the terminal linear readout $L_{\text{out}}(y) = w_d^\top y$ extracts the scalar value. Boundary-strip error vanishes as the strip width $\rightarrow 0$, and the density theorem of Leshno et al. [22, Theorem 1] controls the per-strip error, giving $L^1(K)$ -density; the extension to $L^p(K)$ -density for $p \in [1, \infty)$ uses the truncation argument of Remark 20.

Theorem 7 (CAFC $\Rightarrow L^p$ -universality). *Let $d \geq 2$. If $\mathcal{M}$ satisfies the CAFC-fold condition (Definition 6) and the outer scalar multiplier $s \in \mathbb{R}$ is available, then the $\mathcal{M}$ -constrained ResNet class with an unconstrained terminal linear readout $L_{\text{out}}: \mathbb{R}^d \rightarrow \mathbb{R}$ is dense in $L^1(K)$ for every compact $K \subset \mathbb{R}^d$, and in $L^p(K)$ for every $p \in [1, \infty)$ via the truncation argument of Remark 20. Depth $D_{\text{constr}}(\varepsilon) = O(\varepsilon^{-(d-1)})$ for Lipschitz targets [36].*

The proof adapts the Lin–Jegelka piecewise-constant argument to the rotated $\{w_q\}$ frame; see Appendix B.3 (Theorem 28). Vector-output density is not implied by CAFC alone: the fold uses the residual state as working memory, so auxiliary coordinates are not restored after each fold, which other CAFC programs running in parallel would require.

Macro-CAFC. For $\mathcal{J}_n$ and Gr_{Sp} the literal CAFC-fold condition fails (partner-zero and zero-diagonal obstructions prevent an active write). A *macro-CAFC* variant (Theorem 31) replaces each ideal fold by $O(d^2)$ block-diagonal fold components, delivering the same scalar-output approximant at an $O(d^2)$ block overhead per ideal fold.

On Grassmann manifolds, the inner scalar multiplier t required by the single-block RIH primitive comes from that particular mechanism rather than from the architecture itself: a complementary CAFC construction (Appendix B.6) proves $\text{Gr}(k, d)$ -constrained ResNets with $t_\ell \equiv 1$, $d \geq 2$, $1 \leq k \leq d - 1$, outer scalar multiplier $s_\ell \in \mathbb{R}$, and an unconstrained terminal scalar readout remain $L^1(K)$ -universal.

4.3 Per-Manifold Coverage

Table 2 records the multiplier prescriptions established by RIH and CAFC across the surveyed matrix submanifolds.

Reading Table 2. The *Route* column names the geometric witness used by the RIH verification (for example, the Grassmann “rank- k projector” route chooses a projector whose image contains the active register direction, giving the read/write used in the worked RIH atom above). The *Status* labels refer to the mechanisms explained above: exact and approximate RIPR in §4.1, and literal and macro-CAFC in §4.2. The *Best Multipliers* column is the smallest scalar multiplier budget our current theory proves sufficient after comparing the RIPR/RIH and CAFC constructions. Per-manifold witnesses, status labels, and absorption arguments appear in Appendix B.5.

Theorem 8 (Universal approximation of manifold-weighted ResNets). *For every manifold listed in Table 2 satisfying the range and spectrum hypotheses in Table 1, and every compact $K \subset \mathbb{R}^d$, the corresponding constrained ResNet class with bounded scalar multipliers and an unconstrained*

Table 2: Multiplier prescriptions and route exactness across the surveyed matrix families. The Grassmann entry shows a mechanism separation: RIH requires s, t (Proposition 65), whereas CAFC needs only s (Theorem 66). $\text{PSD}(k, d)$ inherits the Grassmann constructions through $W = QQ^\top$ [3, 4, 37, 38]. The $\text{Sym}_\Lambda(d)$ entry uses the non-scalar, nonzero-spectrum hypothesis from Table 1. All entries are scalar-output compact-domain L^p -density statements with a terminal readout; column definitions follow.

Manifold $\mathcal{M}$	RIPR			CAFC		Best
	Route	Scalar multipliers	Status	Scalar multipliers	Status	Multipliers
$V(k, d)$	orthonormal frame	s, t	exact	s	literal	s
$\text{St}(k, d)$	full-rank frame	$\emptyset$	exact	$\emptyset$	literal	$\emptyset$
$\text{Gr}(k, d)$	rank- k projector	s, t	exact	s	literal	s
$\text{PSD}(k, d)$	Grassmann inclusion	s, t	exact	s	literal	s
$\text{Sym}_\Lambda(d)$	eigenvector embedding	s, t	exact	?	open	s, t
$\mathcal{J}_n$	partner-pair coupling	s, t	exact	s	macro	s
LGr	symmetric-unitary witnesses	s, t	exact	s	literal	s
$E(d, k)$	oblique idempotent	s	approx	s	literal	s
Gr_{Sp}	symplectic-involution diagonal	s	approx	s	macro	s

terminal linear readout $L_{\text{out}} : \mathbb{R}^d \rightarrow \mathbb{R}$ is dense in $L^p(K)$ for every $p \in [1, \infty)$. Only the general fixed-spectrum symmetric family $\text{Sym}_\Lambda(d)$ requires both an outer multiplier s and an inner multiplier t .

Unit-bounded multipliers on the unit-ball manifolds. For the manifolds with operator norm at most one (Gr , $V(k, d)$ with $k < d$, $\mathcal{J}_n$, LGr), the residual class achieves $L^p(K)$ -universality even when the scalar multipliers are further restricted to $|s_\ell|, |t_\ell| \leq 1$. The route-specific unit-bounded theory is in Appendix E: Theorem 75 handles the literal-CAFC manifolds, and Theorem 76 handles the macro-CAFC cases such as $\mathcal{J}_n$. This is a genuine separation from the composition-only case: under the identical unit-bounded multiplier and $\|W\|_{\text{op}} \leq 1$ budget, composition-only FFNs remain confined to $\text{Lip}_1(K)$ (Corollary 3). The general fixed-spectrum symmetric family $\text{Sym}_\Lambda(d)$ remains open under the unit-bounded multiplier constraint.

5 Experiments

We report a controlled synthetic Lipschitz-separation diagnostic that probes Proposition 1 and Theorem 5 directly. Appendix C reports preliminary studies of an MLP classifier on CIFAR-10 and nanoGPT on OpenWebText; the universality theorem itself is asymptotic.

5.1 Controlled Synthetic Lipschitz Separation

To test the obstruction and residual escape mechanism directly, we run an 8-block residual stack at width 128 with rank fraction $r/d = 0.5$, in which every constrained layer lies on $\text{Gr}(k, d)$ and is parametrized as $W = QQ^\top$ with $Q \in V(k, d)$ (3 seeds). On Lip-2 linear, Lip-8 sawtooth, and Lip-0.94 sinusoid targets, Grassmann composition-only plateaus at $\text{MSE} \approx 0.135$ on Lip-2 ($1.6 \times$ the Lip-1 floor; empirical product upper bound 1.0008, consistent with the cap of Proposition 1); adding a residual connection and outer scalar multiplier s recovers $\text{MSE} \approx 3 \times 10^{-4}$ (Theorem 5), and unconstrained controls succeed throughout, isolating the constraint as the cause of failure.

At Lip-8 the Grassmann residual+ s model with default $s_0 = 0.1$ also plateaus at the Lip-1 floor ($\text{MSE} \approx 0.081$; outer scalars stay at $s \approx 0.13$, well below the $\prod (1 + s)^8 = 8$ requirement) although a fresh-init sweep at $s_0 = 1.0$ (3 seeds) recovers $\text{MSE} \approx 0.007$ and post hoc multiplier rescaling from $s_0 = 0.1$ fails (9/9 runs), exposing an initialization-dependent optimization gap distinct from the expressivity barrier.

5.2 Direct CAFC Construction Validation

This deterministic experiment is a constructive diagnostic of the CAFC atom, not full $L^1(K)$ approximation evidence. It instantiates the CAFC atom explicitly for $\text{Gr}(1, 2)$ at width $d = 2$ (no training; rank-1 orthogonal projectors verified) and reports the empirical depth versus L^1 -error rate on a univariate read-direction slice for both a smooth target and a 7-jump BV staircase on $[0, 1]$. The BV target exhibits the expected N^{-1} decay consistent with Theorem 7’s $O(N^{-(d-1)})$ rate at $d = 2$; the smooth target exhibits the faster C^2 -rate N^{-2} of CPWL interpolation, consistent with the proof providing only an upper bound.

5.3 Preliminary Larger-Scale Studies

Beyond the synthetic diagnostic, two larger-scale empirical studies appear in Appendix C: an 8-layer width-512 C-series MLP sweep on CIFAR-10 (§C.2, n shown per row) and nanoGPT-scale constrained- Q/K and constrained-FFN comparison studies at $d_{\text{model}}=384$, 6 layers, 20,000 steps (§C.5 and following). As a heuristic summary, $\text{Gr}(k, d)$ implemented through $Q \in V(k, d)$ shows potential in the CIFAR-10 MLP, while $\text{PSD}(k, d)$ implemented as $UU^\top$ with $U \in \text{St}(k, d)$ shows potential in the low-rank nanoGPT setting. Detailed configurations are in Appendix C.

6 Discussion

Universality shows promise. We have established $L^p(K)$ -universality for residual manifold-weighted networks with bounded scalar multipliers across Table 2 via RIPR or CAFC verification. Placing W_ℓ on a manifold therefore still retains expressive potential. Whether read as a regularizing constraint or as a principled native parameter space for layers that apply but never add weights, the choice between these views and between manifolds rests on optimization, generalization, and Riemannian-geometry considerations.

Our study does not, however, completely settle the theoretical and practical significance of manifold-weighted neural networks; the limitations below are both scope boundaries and openings for future work (Appendix D).

Limitations and future work.

1. *Vector-output universality.* Theorems 5 and 7 give scalar-output density via a terminal unconstrained readout, and vector targets follow by stacking one scalar approximant per output coordinate. A direct route using only constrained blocks at augmented width is sketched in Appendix D.1; a tighter construction without width augmentation is left to future work.
2. *Activation regime.* Theorems use ReLU; the activation-substitution argument in Appendix D sketches the GELU extension underlying our empirical runs, and a finite-depth GELU analogue is left to future work.
3. *L^p vs $C(K)$.* We prove $L^p(K, \mu)$ -density; a $C(K)$ sup-norm lift in the style of Liu et al. [39] remains open.
4. *Compact domain.* Density holds on each compact $K \subset \mathbb{R}^d$, with the per-target architecture depending on $\sup_K \|x\|$ via Lemma 32; a single architecture uniform across subdomains, or density on noncompact targets, is left to future work.
5. *Expressivity vs. optimization vs. generalization.* Universality is existential; the synthetic study (§5.1) exposes the optimization side and Appendix D.4 sketches a covering-number heuristic on the generalization side. Quantitative theory for either effect is left to future work; see Appendix D.
6. *Experiment scale.* Our empirical findings are potentially encouraging but preliminary (synthetic Lipschitz-separation, an 8-layer width-512 CIFAR-10 C-series sweep with condition-dependent seed count $n \leq 7$, and nanoGPT at $d_{\text{model}} = 384$, 6 layers, with seed counts $n = 2$ for convergence controls and $n = 4$ for low-rank panels; Appendix C). Assessing practical value will require experiments at larger data, model, and architecture scales.

Reproducibility. Architectural specs, hyperparameters, seeds, and per-experiment provenance are in Appendix C; code will be released with the public code repository upon acceptance.

References

- [1] Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM J. Matrix Anal. Appl.*, 20(2):303–353, 1998.
- [2] P.-A. Absil, R. Mahony, and R. Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.
- [3] Samuel Burer and Renato D. C. Monteiro. A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization. *Mathematical Programming*, 95(2):329–357, 2003.
- [4] Michel Journée, Francis Bach, P.-A. Absil, and Rodolphe Sepulchre. Low-rank optimization on the cone of positive semidefinite matrices. *SIAM Journal on Optimization*, 20(5):2327–2351, 2010.
- [5] Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *ICLR*, 2014.
- [6] Martin Arjovsky, Amar Shah, and Yoshua Bengio. Unitary evolution recurrent neural networks. In *ICML*, 2016.
- [7] Scott Wisdom, Thomas Powers, John R. Hershey, Jonathan Le Roux, and Les Atlas. Full-capacity unitary recurrent neural networks. In *NeurIPS*, 2016.
- [8] Eugene Vorontsov, Chiheb Trabelsi, Samuel Kadoury, and Christopher Pal. On orthogonality and learning recurrent networks with long term dependencies. *Proceedings of the International Conference on Machine Learning*, 2017. arXiv:1702.00071.
- [9] Lei Huang, Xianglong Liu, Bo Lang, Adams Wei Yu, Yongliang Wang, and Bo Li. Orthogonal weight normalization: Solution to optimization over multiple dependent Stiefel manifolds in deep neural networks. In *AAAI*, 2018.
- [10] Mario Lezcano-Casado and David Martínez-Rubio. Cheap orthogonal constraints in neural networks: A simple parametrization of the orthogonal and unitary group. In *ICML*, 2019.
- [11] Nitin Bansal, Xiaohan Chen, and Zhangyang Wang. Can we gain more from orthogonality regularizations in training deep cnns? In *NeurIPS*, 2018.
- [12] Asher Trockman and J. Zico Kolter. Orthogonalizing convolutional layers with the Cayley transform. In *ICLR*, 2021.
- [13] Cem Anil, James Lucas, and Roger Grosse. Sorting out Lipschitz function approximation. *Proceedings of the International Conference on Machine Learning*, 2019. arXiv:1811.05381.
- [14] Jeremy Bernstein and Laker Newhouse. Modular Duality in Deep Learning. *arXiv preprint arXiv:2410.21265*, 2024.
- [15] Noah Amsel, David Persson, Christopher Musco, and Robert M. Gower. The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm. *arXiv preprint arXiv:2505.16932*, 2025.
- [16] Zhizhong Li, Sina Sajadmanesh, Jingtao Li, and Lingjuan Lyu. Stella: Subspace Learning in Low-rank Adaptation using Stiefel Manifold. *arXiv preprint arXiv:2510.01938*, 2025.
- [17] Taco Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the International Conference on Machine Learning*, pages 2990–2999, 2016.
- [18] Hongzhou Lin and Stefanie Jegelka. ResNet with one-neuron hidden layers is a universal approximator. *Advances in Neural Information Processing Systems*, 31, 2018. arXiv:1806.10909.
- [19] Patrick Kidger and Terry Lyons. Universal approximation with deep narrow networks. In *Proceedings of the Conference on Learning Theory (COLT)*, 2020. arXiv:1905.08539.
- [20] George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4):303–314, 1989.
- [21] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.
- [22] Moshe Leshno, Vladimir Ya. Lin, Allan Pinkus, and Shimon Schocken. Multilayer feedforward networks with a nonpolynomial activation function can approximate any function. *Neural Networks*, 6(6):861–867, 1993.

- [23] Davide Murari, Takashi Furuya, and Carola-Bibiane Schönlieb. Approximation Theory for 1-Lipschitz ResNets. *arXiv preprint arXiv:2505.12003*, 2025.
- [24] Takashi Furuya, Davide Murari, and Carola-Bibiane Schönlieb. Approximation Theory for Lipschitz Continuous Transformers. *arXiv preprint arXiv:2602.15503*, 2026.
- [25] Paulo Tabuada and Bahman Ghahsarpour. Universal approximation power of deep residual neural networks via nonlinear control theory. In *International Conference on Learning Representations*, 2021. arXiv:2007.06007.
- [26] Domènec Ruiz-Balet and Enrique Zuazua. Neural ODE control for classification, approximation, and transport. *SIAM Review*, 65(3):735–773, 2023. arXiv:2104.05278.
- [27] Karthik Elamvazhuthi, Xuechen Zhang, Samet Oymak, and Fabio Pasqualetti. Learning on manifolds: Universal approximation properties using geometric controllability conditions for neural odes. In *Learning for Dynamics and Control Conference (LADC)*, PMLR, 2023.
- [28] Youngjae Min and Navid Azizan. HardNet: Hard-constrained neural networks with universal approximation guarantees. *arXiv preprint*, 2024. arXiv:2410.10807.
- [29] Thomas Bachlechner, Bodhisattwa Prasad Majumder, Huanru Henry Mao, Garrison W. Cottrell, and Julian McAuley. ReZero is all you need: Fast convergence at large depth. *Proceedings of the Conference on Uncertainty in Artificial Intelligence*, 2021. arXiv:2003.04887.
- [30] Risi Kondor and Shubhendu Trivedi. On the generalization of equivariance and convolution in neural networks to the action of compact groups. *Proceedings of the International Conference on Machine Learning*, 2018. arXiv:1802.03690.
- [31] Yann N. Dauphin, Angela Fan, Michael Auli, and David Grangier. Language modeling with gated convolutional networks. In *ICML*, 2017.
- [32] Noam Shazeer. GLU variants improve transformer. In *arXiv preprint arXiv:2002.05202*, 2020.
- [33] Laurent Dinh, Jascha Sohl-Dickstein, and Samy Bengio. Density estimation using Real NVP. In *ICLR*, 2017.
- [34] Rupesh Kumar Srivastava, Klaus Greff, and Jürgen Schmidhuber. Training very deep networks. In *NeurIPS*, 2015.
- [35] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, 2nd edition, 1999.
- [36] Vitaly Maiorov. On best approximation by ridge functions. *Journal of Approximation Theory*, 99(1): 68–94, 1999.
- [37] Bart Vandereycken, P.-A. Absil, and Stefan Vandewalle. Embedded geometry of the set of symmetric positive semidefinite matrices of fixed rank. *Proceedings of the IEEE Workshop on Statistical Signal Processing*, pages 389–392, 2009.
- [38] Estelle Massart and P.-A. Absil. Quotient geometry with simple geodesics for the manifold of fixed-rank positive-semidefinite matrices. *SIAM Journal on Matrix Analysis and Applications*, 41(1):171–198, 2020.
- [39] Hude Liu, Jerry Yao-Chieh Hu, Zhao Song, and Han Liu. Attention Mechanism, Max-Affine Partition, and Universal Approximation. *arXiv preprint arXiv:2504.19901*, 2025.
- [40] Tokio Kajitsuka and Issei Sato. Are transformers with one layer self-attention using low-rank weight matrices universal approximators? *arXiv preprint arXiv:2307.14023*, 2024.
- [41] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>.
- [42] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. OpenAI technical report, 2019. https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.
- [43] Aaron Gokaslan and Vanya Cohen. OpenWebText Corpus. 2019. <https://skylion007.github.io/OpenWebTextCorpus/>.
- [44] Andrej Karpathy. nanoGPT. GitHub repository, 2022. <https://github.com/karpathy/nanoGPT>.

A Architecture and Manifold Definitions

This appendix collects the defining equations and operator-norm facts for the matrix manifolds used in the main body, plus the constrained residual block specification ((4), (5), (6)). Per-manifold proof gadgets that realize the RIPR primitives (read/write frames, dead-zone biasing conditions, multiplier prescriptions) are constructed in the per-manifold verification of Appendix B.5.

A.1 Matrix manifolds

Each entry below gives a concrete parametrization or factorization and the corresponding embedded constraint form.

Orthonormal Stiefel manifold.

$$\begin{aligned} V(k, d) &= \{Q \begin{pmatrix} I_k \\ 0 \end{pmatrix} : Q \in O(d)\} \\ &= \{W \in \mathbb{R}^{d \times k} : W^\top W = I_k\}, \quad 1 \leq k \leq d. \end{aligned}$$

For $k = d$: $V(d, d) = O(d)$ (orthogonal group). Every $W \in V(k, d)$ satisfies $\|W\|_{\text{op}} = 1$.

Noncompact Stiefel manifold.

$$\begin{aligned} \text{St}(k, d) &= \{S \begin{pmatrix} I_k \\ 0 \end{pmatrix} : S \in \text{GL}(d)\} \\ &= \{W \in \mathbb{R}^{d \times k} : \text{rank}(W) = k\}, \quad 1 \leq k \leq d. \end{aligned}$$

This full column rank manifold is a noncompact open subset of $\mathbb{R}^{d \times k}$ and contains $V(k, d)$. Since multiplying a full column rank matrix by a nonzero scalar preserves full column rank, the nonzero scalar multipliers in the rectangular residual blocks can be absorbed into the adjacent read/write factors:

$$sA \text{ReLU}(tB^\top x + b) = \tilde{A} \text{ReLU}(\tilde{B}^\top x + b), \quad \tilde{A} = sA, \quad \tilde{B} = tB.$$

If a requested primitive has zero gain, its residual update is the identity map, so the corresponding block can be omitted. The scalar-free full column rank results below use finite nonzero reads; inactive coordinates on a fixed compact state set are suppressed by sufficiently negative biases, and RIPR constant shifts use a two-block active-zone cancellation. Thus no rank-dropping zero read matrix is needed.

Grassmann manifold.

$$\begin{aligned} \text{Gr}(k, d) &= \{QQ^\top : Q \in V(k, d)\} \\ &= \{\Pi \in \mathbb{R}^{d \times d} : \Pi^\top = \Pi, \Pi^2 = \Pi, \text{tr}(\Pi) = k\}, \quad 0 < k < d. \end{aligned}$$

The factorization has the gauge $Q \mapsto QR$ for $R \in O(k)$. Every $\Pi \in \text{Gr}(k, d)$ satisfies $\|\Pi\|_{\text{op}} = 1$.

Convention. Throughout we write the rank or subspace dimension first: $V(k, d) \subset \mathbb{R}^{d \times k}$ and $\text{Gr}(k, d) \subset \mathbb{R}^{d \times d}$, following Absil, Mahony, and Sepulchre [2].

Fixed-rank symmetric positive semidefinite manifold.

$$\begin{aligned} \text{PSD}(k, d) &= \{UU^\top : U \in \text{St}(k, d)\} \\ &= \{W \in \mathbb{R}^{d \times d} : W = W^\top, W \succeq 0, \text{rank}(W) = k\}, \quad 0 < k \leq d. \end{aligned}$$

This is the smooth rank-exactly- k stratum of the rank-bounded PSD variety, a smooth manifold of dimension $k(2d - k + 1)/2$. The Burer–Monteiro factorization $W = QQ^\top$, restricted to the open full column rank domain $Q \in \mathbb{R}_*^{d \times k} := \{Q : \text{rank}(Q) = k\}$, provides a smooth submersion onto $\text{PSD}(k, d)$ modulo the $O(k)$ gauge $Q \mapsto QU$; fixed-rank PSD optimization is a standard tool in semidefinite programming and matrix-completion literatures [3, 4, 37, 38]. The unconstrained- Q parametrization only guarantees $\text{rank}(W) \leq k$ mathematically (the closure includes lower-rank strata), but with random initialization and training on real data Q is almost surely full column rank, so $W = QQ^\top$ stays in the smooth top stratum throughout training. $\|W\|_{\text{op}}$ equals the largest eigenvalue and is unbounded above on $\text{PSD}(k, d)$, so the nonexpansive obstruction (Proposition 1) does not apply. The Grassmann manifold is the binary-spectrum substratum

$\text{Gr}(k, d) = \{W \in \text{PSD}(k, d) : W^2 = W\}$. The general geometric definition above admits $0 < k \leq d$; the universality-inheritance entry of Table 2 via $\text{Gr}(k, d) \subset \text{PSD}(k, d)$ is restricted to $0 < k < d$ throughout the main results, since $\text{Gr}(d, d) = \{I_d\}$ supplies no register-isolation primitives.

To our knowledge, $\text{PSD}(k, d)$ has not previously been studied as a forward-pass weight manifold in transformer/ResNet blocks; the existing ML practice with “ $QQ^\top$ factorization” (e.g., the kernel literature, LoRA variants with symmetric updates) typically does not place the rank constraint on the same matrix used as a forward weight.

Oblique projector manifold.

$$\begin{aligned} E(d, k) &= \{S \text{diag}(I_k, 0) S^{-1} : S \in \text{GL}(d)\} \\ &= \{P \in \mathbb{R}^{d \times d} : P^2 = P, \text{tr}(P) = k\}. \end{aligned}$$

We use E for *idempotent*; this is non-standard notation. The same idempotent can also be parametrized as $P = XY^\top$ with $X, Y \in \text{St}(k, d)$ and $Y^\top X = I_k$. Elements need not be symmetric and in general $\|P\|_{\text{op}} \geq 1$ (the manifold is non-compact with unbounded operator norm).

Fixed-spectrum symmetric orbit.

$$\begin{aligned} \text{Sym}_\Lambda(d) &= \{Q\Lambda Q^\top : Q \in \text{O}(d)\} \\ &= \{W \in \mathbb{R}^{d \times d} : W^\top = W, \text{spec}(W) = \Lambda\}, \end{aligned}$$

for fixed diagonal $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $\lambda_1 \geq \dots \geq \lambda_d$. Two-eigenvalue fixed-spectrum orbits are obtained by taking $\Lambda = aI_k \oplus bI_{d-k}$ with $a \neq b$; the standard Grassmann projector case is $\Lambda = I_k \oplus 0_{d-k}$. More generally, $\text{Sym}_\Lambda(d)$ allows Λ to have $m \geq 3$ distinct eigenvalues with multiplicities $k_1, \dots, k_m$ ($\sum_i k_i = d$). The block-constant choices of Λ give the standard embedded real flag-manifold models. The universality catalog uses the non-scalar, nonzero-spectrum case: $m \geq 2$ and at least one λ_i is nonzero. $\|M\|_{\text{op}} = \max_i |\lambda_i|$ for any $M \in \text{Sym}_\Lambda(d)$, so $\|M\|_{\text{op}} \leq 1$ exactly when $|\lambda_i| \leq 1$ for all i .

Complex-structure manifold $\mathcal{J}_n$. For even $d = 2n$:

$$\begin{aligned} \mathcal{J}_n &= \{QJ_0Q^\top : Q \in \text{SO}(2n)\} \\ &= \{J \in \mathbb{R}^{2n \times 2n} : J^\top J = I_{2n}, J^2 = -I_{2n}\}, \quad J_0 = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}. \end{aligned}$$

Intrinsically, J is simultaneously skew-symmetric and orthogonal ($J^\top = -J$, $J^\top J = I$). Every $J \in \mathcal{J}_n$ satisfies $\|J\|_{\text{op}} = 1$. The set is empty for odd d ($\det(J)^2 = \det(-I_d) = (-1)^d$ forces d even). Geometrically, $\mathcal{J}_n$ is the manifold of orthogonal complex structures on $\mathbb{R}^{2n}$, equivalently the homogeneous space $\text{SO}(2n)/\text{U}(n)$, the type-**DIII** compact Riemannian symmetric space in Cartan’s classification.

Symplectic Grassmannian (involution embedding). Let $J_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$ and $\text{Sp}(2n, \mathbb{R}) = \{X \in \mathbb{R}^{2n \times 2n} : X^\top J_{2n} X = J_{2n}\}$. For even $d = 2n$ and $1 \leq k \leq n - 1$, let $D_k = \text{diag}(I_k, -I_{n-k}, I_k, -I_{n-k})$. Then

$$\begin{aligned} \text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n}) &= \{gD_k g^{-1} : g \in \text{Sp}(2n, \mathbb{R})\} \\ &= \{X \in \text{Sp}(2n, \mathbb{R}) : X^2 = I_{2n}, \text{tr}(X) = 4k - 2n\}. \end{aligned}$$

Elements are *symplectic involutions* of prescribed trace; the $+1$ -eigenspace $\ker(X - I)$ is a $2k$ -dimensional symplectic subspace, giving a bijection with the abstract symplectic Grassmannian. Unlike the compact families $(V, \text{Gr}, \mathcal{J}_n)$, $\|X\|_{\text{op}}$ is *unbounded* on $\text{Gr}_{\text{Sp}} : \text{Sp}(2n, \mathbb{R})$ is non-compact, and conjugation by ill-conditioned symplectic matrices produces elements of arbitrarily large operator norm. The oblique projector manifold $E(d, k)$ is also non-compact with unbounded $\|P\|_{\text{op}}$.

Lagrangian Grassmannian (realified). For even $d = 2n$, the real Lagrangian Grassmannian $\text{LGr}(\mathbb{R}^{2n}) \cong \text{U}(n)/\text{O}(n)$ parametrizes n -dimensional Lagrangian subspaces of $\mathbb{R}^{2n}$. Via the fac-

torization $A + iB = ZZ^\top$ with $Z \in \text{U}(n)$ and the realification $A + iB \mapsto \begin{bmatrix} A & -B \\ B & A \end{bmatrix}$:

$$\begin{aligned} \text{LGr}(\mathbb{R}^{2n}) &\cong \left\{ \begin{pmatrix} A & -B \\ B & A \end{pmatrix} : A + iB = ZZ^\top, Z \in \text{U}(n) \right\} \\ &= \left\{ \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \in \mathbb{R}^{2n \times 2n} : A, B \text{ symmetric, } A^2 + B^2 = I_n, AB = BA \right\}. \end{aligned}$$

Every $M \in \text{LGr}(\mathbb{R}^{2n})$ is orthogonal, so $\|M\|_{\text{op}} = 1$. Notably, $I_{2n} \in \text{LGr}(\mathbb{R}^{2n})$ (take $A = I_n$, $B = 0$).

Relation to equivariant networks. In a G -equivariant neural network, the weight matrices are chosen from the intertwining algebra

$$\{A \in \mathbb{R}^{n \times n} : A\rho(g) = \rho(g)A \text{ for every } g \in G\},$$

a linear subspace of $\mathbb{R}^{n \times n}$. The manifolds in this paper arise differently: they are *orbits* of classical groups ($\text{V}(k, d) \cong \text{O}(d)/\text{O}(d-k)$, $\text{Gr}(k, d) \cong \text{O}(d)/(\text{O}(k) \times \text{O}(d-k))$), not commutants; their stabilizers are non-normal subgroups, which is what makes the orbits genuinely nonlinear (a normal stabilizer would collapse the orbit into a linear subspace). Our results apply whenever the constraint set is a smooth manifold, whether arising from group orbits, algebraic conditions, or other constructions.

A.2 Architecture (full specification)

Composition-only FFN (depth L).

$$f(x) = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 x + b_1) \cdots) + b_{L-1}) + b_L, \quad (4)$$

with $W_i \in \mathcal{M}$ (constrained weight set), $b_i \in \mathbb{R}^d$, and $\sigma = \text{ReLU}$ applied coordinatewise. More generally, $W_i \in \mathbb{R}^{d_i \times d_{i-1}}$ may be rectangular with compatible dimensions; the fixed-width case $d_i = d$ is the default notation in the main body.

Constrained scaled residual block (one block).

$$F_\ell(x) = x + s_\ell \cdot A_\ell \cdot \text{ReLU}(t_\ell \cdot B_\ell x + b_\ell), \quad (5)$$

where $A_\ell, B_\ell \in \mathcal{M}$, $s_\ell, t_\ell \in \mathbb{R}$ are *scalar multipliers*, and $b_\ell \in \mathbb{R}^d$ is a bias. When $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ with $k < d$ (e.g., $\mathcal{M} = \text{V}(k, d)$), the read direction uses $B_\ell^\top x \in \mathbb{R}^k$ and the write direction uses $A_\ell \in \mathbb{R}^{d \times k}$, with $b_\ell \in \mathbb{R}^k$.

Full constrained ResNet.

$$R(x) = L_{\text{out}} \circ F_N \circ \cdots \circ F_1(x), \quad (6)$$

where $L_{\text{out}} : \mathbb{R}^d \rightarrow \mathbb{R}$ is an *unconstrained* linear readout. Each block F_ℓ already contains an additive identity (Eq. 5), matching the $\text{Id} + \mathcal{T}_\ell$ form of Lin and Jegelka [18]. The unconstrained readout is standard in manifold-constrained architectures; it combines the per-coordinate registers built by the constrained blocks and is essential for our proofs. The augmented-width construction sketched in §D.1 would eliminate the readout at the cost of two scratch coordinates ($D = d + 2$), reading the output directly from the first d coordinates of the forward pass using only constrained blocks.

nanoGPT-scale model

The nanoGPT-scale study (§C.5) uses a nanoGPT-scale GPT-2 [42, 44] architecture with $d_{\text{model}} = 384$, $n_{\text{layer}} = 6$, $n_{\text{head}} = 2$, sequence length 512, OpenWebText [43].

Q/K constraint families.

- **Q/K Stiefel $QQ^\top$ (ID: G1q).** Per-head, the standard query/key bilinear form $W_Q^{h\top} W_K^h \in \mathbb{R}^{D \times D}$ in the attention score logits is replaced by a rank- k Stiefel parametrization $Q_Q^h (Q_K^h)^\top$ with $Q_Q^h, Q_K^h \in \text{V}(D, k)$ ($k = 128$ in our experiments; $D = 384$). When $Q_Q^h = Q_K^h = Q^h$, the bilinear form reduces to the symmetric rank- k orthogonal projector $Q^h (Q^h)^\top \in \text{Gr}(k, D)$. Each Q^h is polar-projected onto $\text{V}(D, k)$ after each AdamW step.

- **Shared-head Q/K Stiefel projector (ID: G1qs).** Single $P_Q \in \text{Gr}(k, D)$ shared across heads, output split into H chunks of D/H via interleaved index pattern (head h takes positions $[h, h+H, h+2H, \dots]$).
- **Q/K free $QQ^\top$ control (ID: G1u).** Same shape as the Q/K Stiefel condition, no polar projection (free Q). Isolates the rank- k bilinear-form effect from the Stiefel constraint.

FFN families.

- **Square Grassmann atom stack (ID: G4q).** N residual atoms, each $y = x + s_n A_n \text{ReLU}(t_n B_n x + b_n)$ with $A_n = Q_n^A (Q_n^A)^\top$, $B_n = Q_n^B (Q_n^B)^\top$.
- **Free $UU^\top$ atom-stack control (ID: G4u).** Same shape as the square Grassmann atom stack, with Q_n^A, Q_n^B trained without polar projection (free $UU^\top$). Isolates the symmetric-projector effect from the Stiefel constraint.
- **Combined Q/K + atom-stack variants (IDs: G5q, G5u).** Stack-composed variants pairing the Q/K constraint with the FFN constraint: the Stiefel version pairs Stiefel Q/K with the Stiefel atom stack, and the free version pairs free Q/K with the free atom stack. Used as joint-site comparisons.
- **Block-Grassmann GELU MLP (ID: G6A).** Residual MLP $y = x + \sum_{i=0}^3 s_i B_i \text{GELU}(t_i P_i x + b_i)$ with $P_i = Q_i Q_i^\top$, $Q_i \in \text{V}(D, k)$, B_i dense.
- **Free $UU^\top$ block-FFN control (ID: G6Au).** Same shape as the block-Grassmann GELU MLP, with Q_i trained without polar projection. Isolates the symmetric-projector constraint from the rank- k bottleneck and from the residual-MLP architecture.
- **Rank-matched FFN $UV^\top$ control (ID: G6Alr).** Same residual architecture as the block-Grassmann GELU MLP, but each P_i replaced by an unconstrained low-rank factor $U_i V_i^\top$ with rank k . Parameter-matched to the block-Grassmann condition; isolates the symmetric/projector restriction from generic rank- k factorization. The “^{lr}-half” variant uses rank $k/2$ to match parameter count under doubled-factor counting.
- **Rank-matched Q/K $UV^\top$ control (ID: G6lir).** Q/K analogue of the FFN $UV^\top$ control: replaces the $QQ^\top$ bilinear form with $UV^\top$ at matched rank.

Provenance IDs. Result-table IDs match experiment logs and result directories. In particular, B_0 is the unconstrained dense baseline (standard GPT-2 attention and MLP, no manifold constraint, no rank bottleneck), and B_0^R is the same baseline architecture run at augmented residual width $D = d_{\text{model}} + 2$, used for the augmented-width atom-stack panel. ID S1q denotes the synthetic augmented-width test for the $\text{Gr}(k, d)$ atom (no nanoGPT runtime; see §C.6). Within the nanoGPT IDs, the suffix q denotes a projected Stiefel $QQ^\top$ factor, u denotes the same $QQ^\top$ factor with the Stiefel projection removed, lr denotes an unconstrained $UV^\top$ low-rank control, and s denotes a shared-head Q/K projector. The leading numbers are historical experiment bookkeeping rather than theorem numbers: G1 is Q/K-only, G4 is an FFN atom stack, G5 is the joint Q/K+FFN atom stack, and G6 is the block-FFN/GELU family used in the final 20,000-step panel; the A in G6A is an internal variant label with no separate mathematical meaning. G2 and G3 belonged to an earlier smoke-test matrix: G2 added a seventh constrained layer for a depth check at matched parameter count, and G3 removed the inner scalar multiplier. They are not shown in the final tables because the final nanoGPT studies use site-isolated and low-rank controls instead; in particular, the old G3 ablation is not a test of the paper’s RIH/ReLU single-block necessity statements.

Polar projection. Constrained Stiefel factors Q are polar-projected back to $\text{V}(D, k)$ after each AdamW optimizer step: $Q \leftarrow UV^\top$ where $Q = U\Sigma V^\top$ is the thin SVD. This is implemented in fp32 with TF32 disabled. The fp32-polar floor for the $D = 384, 386$ matrices used here (at $k \in \{96, 128, 192\}$) keeps the Stiefel violation $\|Q^\top Q - I\|_\infty$ in $6\text{--}10 \cdot 10^{-5}$ during the reported nanoGPT runs (full per-row values in Table 6).

LM head. The LM head reads the first d_{model} coordinates of the final residual stream, after a boundary LayerNorm. Standard nanoGPT weight tying is used (`lm.head.weight = tok_emb.weight`).

B Proofs

This appendix collects the proofs of the theorems stated in the main body. The organization mirrors Section 4: the RIPR primitive construction, RIPR theorem, approximate-RIPR corollary, single-block separation, fold-and-cut theorem and CAFC machinery, and per-manifold proofs in the order Stiefel, Grassmann, fixed-spectrum symmetric orbits, oblique projectors, $\mathcal{J}_n$, Gr_{Sp} , Lagrangian Grassmannian. Stiefel and Grassmann are presented in full detail as representative rows; the remaining rows are handled by shorter checks that use the witness table at the start of Appendix B.5. Additional notes on the activation regime, the augmented-width sketch, constrained- Q/K attention, block-Grassmann FFN containment, and a covering-number generalization heuristic are collected in Appendix D.

Proof-dependency map. The main-text theorems depend on the appendix proofs as follows:

- **Proposition 1 (Nonexpansive obstruction)** $\leftarrow$ §B.1 (Lemmas 9, 10, 11, then Proposition 1 proof + Corollary 2 proof).
- **Theorem 5 (RIPR $\Rightarrow L^1$ -density)** $\leftarrow$ §B.2: P1–P4 primitive definitions, self-contained Theorem 18 (P1–P4 programs are $L^1(\mathbb{R}^d)$ -dense), dead-zone biasing (Lemma 32), then Theorem 5 proof and the approximate-RIPR Corollary 22.
- **Theorem 7 (CAFC $\Rightarrow$ scalar-output L^1 -density)** $\leftarrow$ §B.3 (Definition 26, Lemma 27 (dense-direction ridge density), then Theorem 28 proof). The macro-CAFC variant (§B.4) handles $\mathcal{J}_n$ and Gr_{Sp} via the same outer fold-and-cut argument with $O(d^2)$ block-replacement.
- **Proposition 65 (single-block RIH needs t on Grassmann)** and **Theorem 66 (Grassmann fold-and-cut without inner multiplier)** $\leftarrow$ §B.6.
- **Theorem 8 (catalog-wide universal approximation)** $\leftarrow$ §B.5: row-by-row witnesses in §§B.5.1–B.5.7. The consolidated witness table opens §B.5.

The scoped notes in Appendix D do not enter the proof of any main-text theorem.

B.1 Nonexpansive Obstruction

Two elementary lemmas give the obstruction by direct composition.

Lemma 9 (Coordinatewise σ is 1-Lipschitz). *If $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is 1-Lipschitz, then its coordinatewise extension to $\mathbb{R}^d$, also denoted σ , is 1-Lipschitz.*

Proof. For each coordinate, $|\sigma(u_i) - \sigma(v_i)| \leq |u_i - v_i|$. Squaring and summing:

$$\|\sigma(u) - \sigma(v)\|^2 = \sum_i |\sigma(u_i) - \sigma(v_i)|^2 \leq \sum_i |u_i - v_i|^2 = \|u - v\|^2.$$

Taking square roots gives $\|\sigma(u) - \sigma(v)\| \leq \|u - v\|$. $\square$

Lemma 10 (Composition multiplies Lipschitz constants). *If g is L_g -Lipschitz and h is L_h -Lipschitz, then $h \circ g$ is $L_h L_g$ -Lipschitz.*

Proof. $\|h(g(x)) - h(g(y))\| \leq L_h \|g(x) - g(y)\| \leq L_h L_g \|x - y\|$. $\square$

Lemma 11 (Uniform limits preserve Lipschitz constants). *Let $f_n : K \rightarrow \mathbb{R}^p$ be 1-Lipschitz and suppose $f_n \rightarrow f$ uniformly. Then f is 1-Lipschitz.*

Proof. Fix $x, y \in K$ and $\varepsilon > 0$. Choose n large enough that $\|f - f_n\|_\infty < \varepsilon/2$. Then

$$\|f(x) - f(y)\| \leq \|f(x) - f_n(x)\| + \|f_n(x) - f_n(y)\| + \|f_n(y) - f(y)\| \leq \varepsilon + \|x - y\|.$$

Since ε was arbitrary, $\|f(x) - f(y)\| \leq \|x - y\|$. $\square$

Proof of Proposition 1. Each affine layer $x \mapsto W_i x + b_i$ is $\|W_i\|_{\text{op}}$ -Lipschitz (≤ 1 by hypothesis), and each coordinatewise σ is 1-Lipschitz by Lemma 9. By Lemma 10 the scalar-output composition $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has

$$\text{Lip}(f) \leq \prod_{i=1}^L \|W_i\|_{\text{op}} \leq 1.$$

Uniform closure. If $f_n \rightarrow f$ uniformly with $\text{Lip}(f_n) \leq 1$, then by Lemma 11 f is itself 1-Lipschitz. If K contains distinct points x_0, y_0 , then $g(x) = 2\|x - x_0\|$ is continuous on K but not 1-Lipschitz, since $|g(y_0) - g(x_0)| = 2\|y_0 - x_0\| > \|y_0 - x_0\|$. Thus $\text{Lip}_1(K)$ is a proper closed subset of $C(K)$ whenever K has at least two points.

L^p -closure. Assume K has finite positive Lebesgue measure μ , fix $p \in [1, \infty)$, and suppose $f_n \in \text{Lip}_1(K)$ converge to f in $L^p(K)$. L^p -convergence implies L^p -boundedness of $\{f_n\}$, so the means $f_n = \mu(K)^{-1} \int_K f_n d\mu$ are bounded by Hölder. For any anchor $x_0 \in K$, $|f_n(x_0) - f_n| \leq \sup_{x \in K} |f_n(x) - f_n(x_0)| \leq \text{diam}(K)$, hence $\{f_n(x_0)\}$ is bounded; combined with $\text{Lip}(f_n) \leq 1$ this gives a uniform bound $\sup_n \sup_{x \in K} |f_n(x)| < \infty$. By Arzelà–Ascoli the equicontinuous, uniformly bounded family $\{f_n\}$ admits a uniformly convergent subsequence $f_{n_k} \rightarrow \tilde{f} \in \text{Lip}_1(K)$; uniform convergence on the compact K implies L^p -convergence, so $f = \tilde{f}$ a.e., and $f \in \text{Lip}_1(K)$. The same scalar function $g(x) = 2\|x - x_0\|$ used above lies in $L^p(K)$ but not in $\text{Lip}_1(K)$, so the inclusion is proper.

Vector output (remark). For a vector-valued composition-only FFN, each output coordinate is itself a scalar-output composition-only FFN, so the same scalar 1-Lipschitz / closure argument applies coordinatewise. $\square$

Proof of Corollary 2

The Stiefel manifold $V(k, d)$, the Grassmann manifold $\text{Gr}(k, d)$, the slice of $\text{Sym}_\Lambda(d)$ with operator norm at most one (when $\max_i |\lambda_i| \leq 1$), $\mathcal{J}_n$, and $\text{LGr}(\mathbb{R}^{2n})$ all consist of matrices with $\|W\|_{\text{op}} \leq 1$ (Appendix A.1). Proposition 1 therefore applies, giving $\text{Lip}(f) \leq 1$ for any composition-only FFN whose weight matrices lie in any of these families and whose activation is 1-Lipschitz. The conclusion follows since $\text{Lip}_1(K, \mathbb{R}^p)$ is a proper closed subset of $C(K, \mathbb{R}^p)$ on any compact K with at least two points.

Proof of Corollary 3

For a layer

$$T_\ell(x) = s_\ell \sigma(t_\ell W_\ell x + b_\ell)$$

with $|s_\ell|, |t_\ell| \leq 1$, $\|W_\ell\|_{\text{op}} \leq 1$, and $\text{Lip}(\sigma) \leq 1$, Lemmas 9–10 give

$$\text{Lip}(T_\ell) \leq |s_\ell| \text{Lip}(\sigma) |t_\ell| \|W_\ell\|_{\text{op}} \leq 1.$$

Thus every finite composition of such layers is 1-Lipschitz on K . There is no unconstrained terminal linear readout, so no final map can increase this Lipschitz constant. The uniform-closure and L^p -closure arguments in the proof of Proposition 1 therefore apply verbatim: the class lies in the proper closed subset $\text{Lip}_1(K)$ and cannot be dense in $C(K)$ when K has at least two points, nor in $L^p(K)$ when K has positive Lebesgue measure.

B.2 The RIPR Framework

RIPR primitives and register isolation

Definition 12 (RIPR primitives (P1–P4)). On a scalar register $R \in \mathbb{R}$, the *RIPR primitives* (P1–P4) are:

- **(P1) Nonnegative shift:** $R \mapsto R + c$ for $c \geq 0$ (special case of P3 with $\gamma = 1$, $a = 0$, $b = c$).
- **(P2) Max with constant:** $R \mapsto \max\{R, c\} = R + \text{ReLU}(c - R)$ (special case of P3 with $\gamma = 1$, $a = -1$, $b = c$).

- **(P3) Signed-amplitude hinge (master):** $R \mapsto R + \gamma \text{ReLU}(aR + b)$ for any $\gamma, a, b \in \mathbb{R}$. The sign of γ selects between max-style ($\gamma > 0$) and min/cap-style ($\gamma < 0$) updates: e.g. $\min\{R, c\} = R - \text{ReLU}(R - c)$ corresponds to $\gamma = -1, a = 1, b = -c$.
- **(P4) Register coupling:** given registers $i \neq j$ with $x_j \geq 0$ on the current state set, perform $x_i \mapsto x_i + x_j$ while preserving all registers $m \notin \{i, j\}$; the source j is unrestricted.

P3 is the master same-coordinate primitive. Both signs of γ are realized in a single residual block by the outer scalar multiplier s : one γ -instance corresponds to one block $F(x) = x + sA \text{ReLU}(tBx + b)$ with $s = \gamma\mu$ for some manifold-witness write gain μ (sign of μ depends on the witness; sign of γ is always controllable by the sign of s).

Remark 13 (The negative- γ case is essential). The piecewise-affine grid-indicator construction in §B.2 uses both signs: $\gamma > 0$ realizes max-with-affine build-up steps; $\gamma < 0$ realizes folds, caps, and the descending value-adjustment pass (each level adjustment has $\gamma_k \leq -1$ strictly).

Remark 14 (Why P4 is needed for $d > 1$). Lin and Jegelka’s d -dimensional construction uses a dimensional induction: a 1D grid indicator $R_1(x_1)$ and a $(d-1)$ -dimensional indicator $R_{d-1}(x_{2:d})$ are built independently (using P1–P3), then *coupled* by summing registers: $R_1^+ = R_1 + [R_{d-1}]_+$. This sum is followed by a nonlinear cut-and-shift (P2 on the coupled register), so it cannot be deferred to the final readout L_{out} . There are $d-1$ such couplings in total (one per induction step). Since $x_j \geq 0$ is ensured by a preceding $\max\{R_{d-1}, 0\}$ step, and register j is never used again after the coupling, the “unrestricted aftermath” condition of P4 is satisfied.

Definition 15 (Register isolation). A constrained block $F(x) = x + s \cdot A \cdot \text{ReLU}(t \cdot Bx + b)$ ($A, B \in \mathcal{M}$, $s, t \in \mathbb{R}$, $b \in \mathbb{R}^d$) *isolates register coordinate i* on a compact $K \subset \mathbb{R}^d$ if there exists $c_i : K \rightarrow \mathbb{R}$ with $F(x) = x + c_i(x) e_i$ for all $x \in K$ (equivalently, $F(x)_j = x_j$ for all $j \neq i$).

Remark 16 (Rectangular manifolds). When $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ with $k < d$ (e.g., $\mathcal{M} = \mathcal{V}(k, d)$), the constrained block reads $F(x) = x + s \cdot A \cdot \text{ReLU}(t \cdot B^\top x + b)$ with $A, B \in \mathcal{M}$, $b \in \mathbb{R}^k$ (Eq. 5). The conditions in Definition 4 apply verbatim after the substitution $Bx \rightarrow B^\top x$, since the primitives depend only on the effective read/write directions, not on the matrix dimensions.

Self-contained L^1 density via P1–P4 programs

Definition 17 (P1–P4 programs). Fix an input dimension d . For $D \geq d$, write $\iota_D(x) = (x_1, \dots, x_d, 0, \dots, 0) \in \mathbb{R}^D$. A *P1–P4 program* of width D is a finite composition $P = P_N \circ \dots \circ P_1$ of register primitives, each acting on the current state $z \in \mathbb{R}^D$, drawn from P1–P4 of Definition 12. A scalar-output P1–P4 program is a pair (P, L_{out}) with $L_{\text{out}} : \mathbb{R}^D \rightarrow \mathbb{R}$ linear; it realizes the scalar function $x \mapsto L_{\text{out}}(P(\iota_D(x)))$. P1 and P2 are retained as named primitives for exposition; both are special cases of signed P3 (hinge). In particular, $\max\{R, c\} = R + \text{ReLU}(c - R)$ is P3 with $(\gamma, a, b) = (1, -1, c)$, and $\min\{R, \alpha R + \beta\} = R - \text{ReLU}((1 - \alpha)R - \beta)$ is P3 with $(\gamma, a, b) = (-1, 1 - \alpha, -\beta)$.

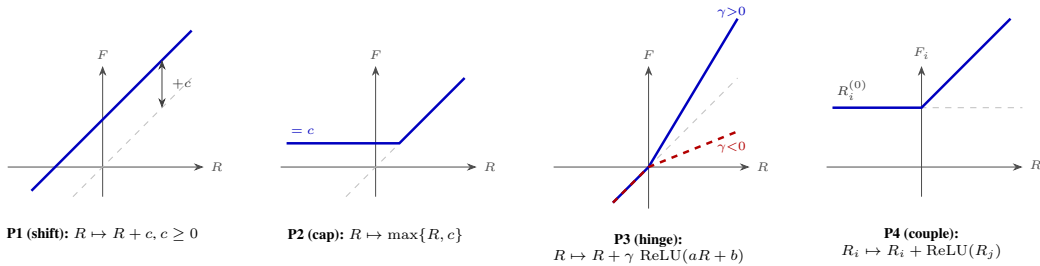


Figure 1: **P1–P4 primitives.** P1 (shift) and P2 (cap) are special cases of signed P3 (hinge); P4 (couple) reads register j and adds to register i . The P4 panel shows the gadget form $R_i \mapsto R_i + \text{ReLU}(R_j)$, which agrees with the register-coupling primitive $R_i \mapsto R_i + R_j$ on the source-nonneg side $R_j \geq 0$ enforced by the program invariant.

Theorem 18 (P1–P4 programs are $L^1(\mathbb{R}^d)$ -dense). *For every $f \in L^1(\mathbb{R}^d)$ and every $\varepsilon > 0$, there exist $D \geq d$, a finite P1–P4 program P of width D , and a linear readout $L_{\text{out}} : \mathbb{R}^D \rightarrow \mathbb{R}$ such that*

$$\|f - L_{\text{out}} \circ P \circ \iota_D\|_{L^1(\mathbb{R}^d)} \leq \varepsilon.$$

The construction below uses width $D = d$ and a readout supported on register 1.

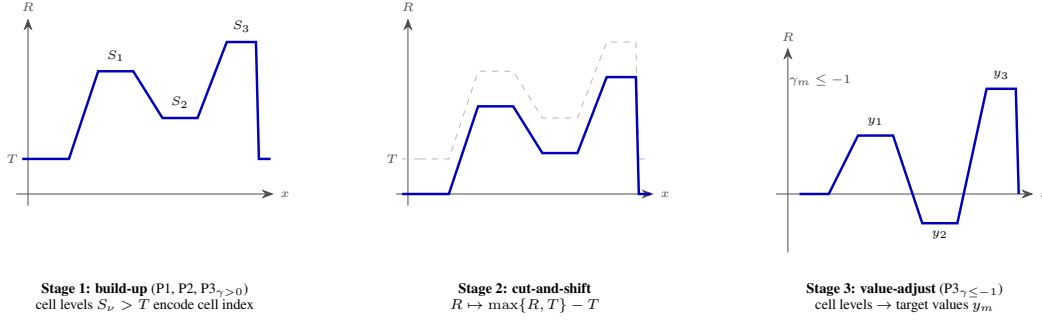


Figure 2: **P1–P4 (RIH) program** (1D slice). **Stage 1:** build-up via P1, P2, signed-P3 with $\gamma > 0$, producing distinct cell levels S_ν above the cut threshold T . **Stage 2:** cut-and-shift $R \mapsto \max\{R, T\} - T$ lowers cells by T and zeroes the exterior. **Stage 3:** descending value-adjustment via signed-P3 with $\gamma_m \leq -1$ maps cell levels to target values y_m . The full d -dim program builds 1D codes per coordinate (Stages 1–2), couples them via P4 with mixed-radix off-sets, then applies Stage 3 in register 1.

Proof. The proof recasts the trapezoid grid-indicator construction of Lin–Jegelka [18, §3, Eq. (2)] into our P1–P4 normal form. We use their construction shape only: every operation is rewritten explicitly as one of the primitives in Definition 12.

Step 1: density reduction. By the standard density of compactly supported rectangular step functions in $L^1(\mathbb{R}^d)$ [35, Theorem 2.41], choose a rectangular step function

$$h(x) = \sum_{\nu \in \mathcal{J}} h_\nu 1_{Q_\nu}(x), \quad Q_\nu = \prod_{p=1}^d [a_{p,\nu_p-1}, a_{p,\nu_p}),$$

with finite index set $\mathcal{J} = \prod_{p=1}^d \{1, \dots, M_p\}$, such that $\|f - h\|_{L^1(\mathbb{R}^d)} < \varepsilon/2$. Let $H := \max\{1, \max_\nu |h_\nu|\}$. It remains to build a P1–P4 program approximating h in L^1 to accuracy $\varepsilon/2$.

Step 2: one-dimensional trapezoid codes. Fix a partition $a_0 < a_1 < \dots < a_M$ and prescribed nonnegative levels $\ell_1, \dots, \ell_M$. For $0 < \delta < \frac{1}{2} \min_k (a_k - a_{k-1})$, the trapezoid string produces a continuous piecewise-affine function T_δ with $T_\delta(x) = \ell_k$ on $[a_{k-1} + \delta, a_k - \delta]$, $T_\delta(x) = 0$ outside $[a_0, a_M]$, and $0 \leq T_\delta(x) \leq C_\ell$ globally. Its elementary updates are the three forms

$$R \leftarrow \max\{R, \alpha R + \beta\}, \quad R \leftarrow \min\{R, \alpha R + \beta\}, \quad R \leftarrow \min\{R, c\},$$

which become signed P3 instances

$$\begin{aligned} R &\leftarrow R + \text{ReLU}((\alpha - 1)R + \beta) & (\gamma, a, b) &= (1, \alpha - 1, \beta), \\ R &\leftarrow R - \text{ReLU}((1 - \alpha)R - \beta) & (\gamma, a, b) &= (-1, 1 - \alpha, -\beta), \\ R &\leftarrow R - \text{ReLU}(R - c) & (\gamma, a, b) &= (-1, 1, -c). \end{aligned}$$

No cross-register primitive is used in dimension one.

After the trapezoid produces a register taking distinct levels $0 = \lambda_0 < \lambda_1 < \dots < \lambda_N$ on the cells, we adjust to target values $y_0 = 0, |y_m| \leq H$ as follows. Choose code levels with $\lambda_1 > 2H$ and thresholds $\tau_m \in (\lambda_{m-1}, \lambda_m)$, $\tau_m > H$. Process m in descending order $N, N-1, \dots, 1$, applying

$$R \leftarrow R + \gamma_m \text{ReLU}(R - \tau_m), \quad \gamma_m := \frac{y_m - \lambda_m}{\lambda_m - \tau_m}. \quad (7)$$

This is signed P3 with $(\gamma, a, b) = (\gamma_m, 1, -\tau_m)$. Since $y_m \leq H < \tau_m$, we have $\gamma_m < -1$. The operation acts only on the level λ_m (sending it to y_m); levels below τ_m are inactive, and previously processed levels (now in $[-H, H]$) are also below τ_m and untouched. Hence the 1D trapezoid+adjustment string is a finite same-register P1/P2/P3 program with the claimed interior agreement, zero exterior, and a finite uniform bound.

Step 3: a d -dimensional grid code by register coupling. Define $B_1 = 1$, $B_p = \prod_{q < p} M_q$, and the mixed-radix digit $\omega(\nu) = \sum_p (\nu_p - 1) B_p \in \{0, \dots, W\}$ with $W = \prod_p M_p - 1$. Set $H_0 = 4H + 1$, $L = (d - 1)W + H_0 + 1$, $T = dL - H_0$. For each coordinate p , apply the one-dimensional code program of Step 2 to register p with levels $\ell_{p,k} = L + (k - 1)B_p$. Away from δ -boundary strips, register p equals ℓ_{p,ν_p} on $[a_{p,\nu_p-1}, a_{p,\nu_p})$ and 0 outside; since the construction is same-register, the other registers are preserved.

Registers $2, \dots, d$ are nonnegative everywhere after this stage. Apply the coupling primitives $P4(1 \leftarrow 2), P4(1 \leftarrow 3), \dots, P4(1 \leftarrow d)$. The source condition $z_j \geq 0$ holds; after coupling, register j is never read again (the live-register invariant). On the δ -interior of Q_ν , register 1 now equals $S_\nu = dL + \omega(\nu)$. If at least one coordinate is outside its range while no coordinate is in a strip, register $1 \leq (d - 1)(L + W) \leq T$. Apply the cut-and-shift $R \mapsto \max\{R, T\} - T$, i.e. P2 followed by P3 with $(\gamma, a, b) = (-T, 0, 1)$. Register 1 is now 0 outside the grid, and equals $\lambda_\nu := H_0 + \omega(\nu) > 2H$ on interior(Q_ν). Levels are separated by at least 1 and uniquely identify the cells.

Sort the cell levels $\lambda_{\nu(1)} < \dots < \lambda_{\nu(N)}$ ($N = |\mathcal{J}|$) and apply the descending value adjustment (7) with target values $y_m = h_{\nu(m)}$, threshold $\tau_m \in (\lambda_{\nu(m-1)}, \lambda_{\nu(m)})$ for $\tau_m > H$ (with $\lambda_{\nu(0)} = 0$). Each step is signed P3 with $\gamma_m < -1$, and as in Step 2 levels are processed exactly once. Outside the boundary strips the final register 1 equals h_ν on Q_ν and 0 outside the rectangular support of h .

The construction maintains three invariants used in the error estimate: (i) *level separation* (every pre-adjustment cell has a distinct level $\lambda_\nu > 2H$, exterior is 0); (ii) *boundedness* (register 1 is bounded by a finite constant C_h depending only on the grid and H); (iii) *source consumption* (each P4 source register is nonnegative when read and never read after the coupling).

Step 4: boundary-strip error. Let $B_\delta := \bigcup_{p,k} \{x : |x_p - a_{p,k}| < \delta\} \cap \prod_q [a_{q,0}, a_{q,M_q}]$ denote the union of boundary strips inside the rectangular support box. There is a constant C_{grid} depending only on the grid such that $|B_\delta| \leq C_{\text{grid}}\delta$ (e.g. $|B_\delta| \leq 2dM\delta$ when all side lengths ≤ 1 and $M_p \leq M$). Outside B_δ the program output g_δ equals h exactly; on B_δ both $|h| \leq H$ and $|g_\delta| \leq C_h$, so $\|h - g_\delta\|_{L^1} \leq (H + C_h)C_{\text{grid}}\delta$. Choose $\delta > 0$ with this bounded by $\varepsilon/2$; combining with Step 1 gives $\|f - g_\delta\|_{L^1} < \varepsilon$. The function g_δ is realized by the finite P1–P4 program above with $L_{\text{out}}(z) = z_1$. $\square$

Proof of Theorem 5

Proof of Theorem 5. Step 1: existence of an unconstrained P1–P4 approximant. Fix any compact input box $K_0 \subset \mathbb{R}^d$ containing the support of the target f up to mass $\varepsilon/4$ (truncate f outside K_0 if necessary; standard truncation argument). By Theorem 18, for every $\varepsilon > 0$ there exist a width $D = d$, a finite P1–P4 program $P = P_N \circ \dots \circ P_1$ with $N = N(\varepsilon)$ primitives, and a linear readout $L_{\text{out}}: \mathbb{R}^d \rightarrow \mathbb{R}$, such that

$$\|f - L_{\text{out}} \circ P \circ \iota_d\|_{L^1(\mathbb{R}^d)} \leq \varepsilon.$$

The construction in the proof of Theorem 18 satisfies the *source-consumption invariant*: each P4 coupling $P4(i \leftarrow j)$ has $z_{r-1}(x)_j \geq 0$ on the ideal state set $S_{r-1} := P_{r-1} \circ \dots \circ P_1(K_0)$, and the source register j is not read by any subsequent primitive $P_{r+1}, \dots, P_N$ nor by L_{out} . We refer below to the *live-coordinate* set $\Lambda_r \subseteq \{1, \dots, d\}$, defined recursively backward from $\Lambda_N = \{q : L_{\text{out}} \text{ has nonzero coefficient on register } q\}$ by adding the read-coordinates of P_{r+1} . By construction, every read coordinate of P_r lies in Λ_{r-1} , and every P4 source j satisfies $j \in \Lambda_{r-1} \setminus \Lambda_r$.

Step 2: per-primitive constrained replacement on the actual state set. Set $T_0 = K_0$ (compact). Recursively, suppose $G_1, \dots, G_{r-1}$ have been chosen as compositions of at most c_{prim} constrained residual blocks each, and that $T_{r-1} = (G_{r-1} \circ \dots \circ G_1)(K_0)$ is compact. Apply Definition 4 (RIPR) on the compact state set T_{r-1} to the primitive P_r : it delivers a constrained gadget G_r of at most c_{prim} blocks such that $G_r(z) = P_r(z)$ for every $z \in T_{r-1}$ (in the same-coordinate case) or $G_r(z)_m = P_r(z)_m$ for $m \neq j$ with $G_r(z)_j$ unconstrained (in the P4 case, requiring the source condition $z_j \geq 0$ on T_{r-1}). Set $T_r = G_r(T_{r-1})$, compact as the image of a compact set under a continuous map. This builds a constrained ResNet of depth $\leq N \cdot c_{\text{prim}}$.

Step 3: live-coordinate induction. Write $z_r(x) = P_r(z_{r-1}(x))$ for the ideal trajectory and $y_r(x) = G_r(y_{r-1}(x))$ for the constrained trajectory, with $z_0(x) = y_0(x) = x$. We claim $\pi_{\Lambda_r} y_r(x) = \pi_{\Lambda_r} z_r(x)$ for every $x \in K_0$, by induction on r . The base case $r = 0$ is immediate. For the inductive step:

- *Same-coordinate P_r on coordinate i .* G_r agrees with P_r on T_{r-1} . For $q \in \Lambda_r$, q is also live at time $r-1$ (since P_r reads only coordinate $i \in \Lambda_{r-1}$ if $i \in \Lambda_r$). The induction hypothesis at the relevant read coordinate i gives $y_{r-1}(x)_i = z_{r-1}(x)_i$, hence $y_r(x)_q = z_r(x)_q$ for every $q \in \Lambda_r$.
- *Coupling $P4(i \leftarrow j)$.* By the source-consumption invariant, $j \notin \Lambda_r$ and $j \in \Lambda_{r-1}$. The induction hypothesis gives $y_{r-1}(x)_j = z_{r-1}(x)_j \geq 0$, so the P4 source condition holds at $y_{r-1}(x) \in T_{r-1}$, and the RIPR clause delivers $y_r(x)_i = y_{r-1}(x)_i + y_{r-1}(x)_j$ and $y_r(x)_m = y_{r-1}(x)_m$ for $m \notin \{i, j\}$. For $q \in \Lambda_r$, $q \neq j$, and we conclude $y_r(x)_q = z_r(x)_q$ by the same matching at $q \in \{i\} \cup (\Lambda_r \setminus \{j\})$.

The induction is complete.

Step 4: readout and final error. $L_{\text{out}} = L_{\text{out}} \circ \pi_{\Lambda_N}$ by definition of Λ_N , so the live-coordinate identity at time N gives $L_{\text{out}}(y_N(x)) = L_{\text{out}}(z_N(x))$ for every $x \in K_0$. Hence the constrained ResNet matches the unconstrained P1–P4 program pointwise on K_0 , so the $L^1(K_0)$ error of the constrained approximant equals the $L^1(K_0)$ error of the unconstrained P1–P4 program, which is bounded by ε by Step 1. Depth: $D_{\text{constr}}(\varepsilon) \leq N(\varepsilon)c_{\text{prim}}$. $L^p(K)$ density for $p \in [1, \infty)$ on every compact $K \subseteq K_0$ follows by Remark 20. The density statement is compact-domain: outside the compact state set T_{r-1} (and hence outside K_0), the dead-zone biasing inside each constrained gadget need not hold, so the constrained network is not controlled on $\mathbb{R}^d \setminus K_0$. $\square$

Remark 19 (Multiplier bound at fixed tolerance). Fix an internal L^1 tolerance $\delta > 0$ and any target class $\mathcal{F}$ of functions on a fixed compact $K_0 \subset \mathbb{R}^d$ with uniform bounds $\sup_{f \in \mathcal{F}} \|f\|_{L^\infty(K_0)} \leq H$ and $\sup_{f \in \mathcal{F}} \text{TV}(f) \leq V$. For an $L^p(K)$ target tolerance, δ is the internal tolerance chosen in Remark 20. At this fixed δ , the construction in Theorem 18 produces P1–P4 primitive parameters (γ_r, a_r, b_r) uniformly bounded over $f \in \mathcal{F}$ by an explicit constant

$$M(\delta, H, V, d) = C_d(W(\delta) + H), \quad W(\delta) = \prod_{q=1}^d M_q(\delta) - 1,$$

where $M_q(\delta)$ is the per-coordinate grid-cell count needed to drive $\|f - h\|_{L^1} \leq \delta/2$ uniformly over $\mathcal{F}$ (for BV-bounded $\mathcal{F}$ this is $M_q = O(\delta^{-1})$ and $W(\delta) = O(\delta^{-d})$; for $W^{k,1}$ -bounded $\mathcal{F}$ this is $M_q = O(\delta^{-1/\min(k,2)})$ and $W(\delta) = O(\delta^{-d/\min(k,2)})$). The per-manifold RIH and approximate-RIH verifications use finitely many algebraic witness templates. Hence, for each fixed manifold $\mathcal{M}$ satisfying its dimensional and spectral hypotheses, there is a finite manifold-dependent function $P_{\mathcal{M}}$ such that all scalar multipliers obey

$$|s_\ell|, |t_\ell| \leq P_{\mathcal{M}}(M(\delta, H, V, d))$$

uniformly over $f \in \mathcal{F}$. The constants inside $P_{\mathcal{M}}$ include only fixed witness quantities, such as reciprocals of nonzero eigenvalues in the fixed-spectrum family. This is the bound, uniform over f at fixed tolerance, used in Theorem 5. The stronger tolerance-uniform statement $|s_\ell|, |t_\ell| \leq 1$ is proved separately for the literal and macro CAFC manifolds covered in Appendix E.

Remark 20 (L^p extension via truncation). Fix $f \in L^p(K)$, $p \in [1, \infty)$, on a compact $K \subset \mathbb{R}^d$, and $\varepsilon > 0$. Truncate: set $f_M := f \cdot \mathbf{1}_{|f| \leq M} \in L^\infty(K) \cap L^p(K)$; by dominated convergence $\|f - f_M\|_{L^p(K)} \rightarrow 0$ as $M \rightarrow \infty$, so choose M with $\|f - f_M\|_{L^p(K)} < \varepsilon/2$. Suppose g is an $L^1(K)$ approximant of f_M that is uniformly bounded on K by some C_g (the constrained ResNets produced elsewhere in this appendix supply such g as compositions of continuous maps on a compact set). Set $B := M + C_g$. Since $|f_M - g| \leq B$ pointwise on K ,

$$\|f_M - g\|_{L^p(K)}^p \leq B^{p-1} \|f_M - g\|_{L^1(K)}.$$

Choosing g with $\|f_M - g\|_{L^1(K)} < (\varepsilon/2)^p B^{1-p}$ gives $\|f_M - g\|_{L^p(K)} < \varepsilon/2$, hence $\|f - g\|_{L^p(K)} < \varepsilon$. The same argument extends to $L^p(K, \mu)$ for any σ -finite μ absolutely continuous with respect to Lebesgue measure (replace $|K|$ by $\mu(K) < \infty$).

Approximate RIPR corollary

Definition 21 (Approximate RIPR on a fixed P1–P4 schedule). Fix a finite P1–P4 program $P_1, \dots, P_N$ supplied by Theorem 18, compact input set K_0 , the live-coordinate sets $\Lambda_r \subseteq \{1, \dots, d\}$ defined recursively backward from $\Lambda_N = \{q :$

L_{out} has nonzero coefficient on register q by adding the read coordinates of P_{r+1} , and tolerances $\eta_1, \dots, \eta_N > 0$. The constrained block family satisfies *approximate RIPR with primitive cost c_{prim} on this schedule* if, with $T_0 = K_0$, the following recursive replacement is possible. After $G_1, \dots, G_{r-1}$ have been chosen and $T_{r-1} = (G_{r-1} \circ \dots \circ G_1)(K_0)$ has been formed, there is a composition G_r of at most c_{prim} constrained blocks such that

$$\sup_{u \in T_{r-1}} \left\| \pi_{\Lambda_r}(G_r(u) - \hat{P}_r(u)) \right\|_{\infty} \leq \eta_r.$$

Here $\hat{P}_r = P_r$ for same-coordinate primitives. For a coupling primitive $i \leftarrow j$,

$$\hat{P}_r(u)_i = u_i + \text{ReLU}(u_j), \quad \hat{P}_r(u)_m = u_m \quad (m \neq i).$$

On ideal trajectory states, $\pi_{\Lambda_r} \hat{P}_r = \pi_{\Lambda_r} P_r$, because P4 sources are nonnegative and the source coordinate is not live after coupling (source-consumption invariant of Theorem 18).

Corollary 22 (Approximate RIPR $\Rightarrow L^p(K)$ -universality). *Suppose the constrained block family satisfies Definition 21 with primitive cost c_{prim} for every finite P1–P4 schedule, compact input set, live-coordinate family, and positive tolerance vector described there. Then the constrained ResNet class (Eq. 6) with unconstrained linear readout is dense in $L^p(K)$ for every $p \in [1, \infty)$ and every compact $K \subset \mathbb{R}^d$ (cf. Theorem 5; Remark 20).*

Proof of Corollary 22. Fix a compact truncation K_0 and a P1–P4 program $P_1, \dots, P_N$ produced by Theorem 18 whose final readout approximates the compactly supported target to accuracy $\varepsilon/2$ in L^1 . Let $z_r(x)$ and $y_r(x)$ be the ideal and constrained trajectories:

$$z_0(x) = y_0(x) = x, \quad z_r(x) = P_r(z_{r-1}(x)), \quad y_r(x) = G_r(y_{r-1}(x)).$$

For a coordinate set Λ , write $\|v\|_{\Lambda} := \|\pi_{\Lambda} v\|_{\infty}$, and define the live error

$$e_r := \sup_{x \in K_0} \|y_r(x) - z_r(x)\|_{\Lambda_r}.$$

Then $e_0 = 0$.

For each r , the projected primitive $\hat{P}_r$ is Lipschitz from the previous live seminorm to the next live seminorm:

$$\|\hat{P}_r(u) - \hat{P}_r(v)\|_{\Lambda_r} \leq \ell_r \|u - v\|_{\Lambda_{r-1}} \quad \text{for every } u, v \in \mathbb{R}^d.$$

One may take $\ell_r = 1 + |\gamma_r| |a_r|$ for a same-coordinate signed P3 primitive with parameters (γ_r, a_r, b_r) , and $\ell_r = 2$ for a coupling primitive. These bounds use only coordinates in Λ_{r-1} : a same-coordinate primitive reads its own live coordinate, and a coupling primitive reads the live target and the live source before discarding the source.

Since $y_{r-1}(x) \in T_{r-1}$, the approximate RIPR property gives

$$\|G_r(y_{r-1}(x)) - \hat{P}_r(y_{r-1}(x))\|_{\Lambda_r} \leq \eta_r.$$

Since $z_{r-1}(x)$ is an ideal trajectory state, the projected primitive $\hat{P}_r$ agrees with the ideal P_r on the live coordinates at step r : $\pi_{\Lambda_r} \hat{P}_r(z_{r-1}(x)) = \pi_{\Lambda_r} P_r(z_{r-1}(x)) = \pi_{\Lambda_r} z_r(x)$ (off-live coordinates of $\hat{P}_r$ may differ, but they are ignored by $\|\cdot\|_{\Lambda_r}$). Therefore

$$\begin{aligned} \|y_r(x) - z_r(x)\|_{\Lambda_r} &= \|G_r(y_{r-1}(x)) - P_r(z_{r-1}(x))\|_{\Lambda_r} \\ &\leq \|G_r(y_{r-1}(x)) - \hat{P}_r(y_{r-1}(x))\|_{\Lambda_r} \\ &\quad + \|\hat{P}_r(y_{r-1}(x)) - \hat{P}_r(z_{r-1}(x))\|_{\Lambda_r} \\ &\leq \eta_r + \ell_r \|y_{r-1}(x) - z_{r-1}(x)\|_{\Lambda_{r-1}}. \end{aligned}$$

Taking the supremum over $x \in K_0$ gives the recurrence

$$e_r \leq \eta_r + \ell_r e_{r-1}.$$

Hence

$$e_N \leq \sum_{r=1}^N \eta_r \prod_{q=r+1}^N \ell_q.$$

Let $B_{\text{out}} := \max\{1, \|L_{\text{out}}\|_{\ell^\infty(\Lambda_N) \rightarrow \mathbb{R}}\}$. If $\text{vol}(K_0) = 0$ the claim is trivial (the integral is over a measure-zero set); otherwise choose tolerances as follows. Choose the tolerances so that

$$\sum_{r=1}^N \eta_r \prod_{q=r+1}^N \ell_q \leq \frac{\varepsilon}{2B_{\text{out}} \text{vol}(K_0)}.$$

Then the final readout error introduced by approximate gadget replacement is at most

$$\int_{K_0} |L_{\text{out}}(y_N(x)) - L_{\text{out}}(z_N(x))| dx \leq B_{\text{out}} \text{vol}(K_0) e_N \leq \varepsilon/2.$$

Adding the $\varepsilon/2$ approximation error from Theorem 18 gives total L^1 error at most ε .

The proof never compares $G_r(y_{r-1})$ with $G_r(z_{r-1})$. Full input-space Lipschitzness of the constrained gadget is therefore not used to propagate errors across coordinates that are no longer live. $\square$

Remark 23 (P4 source contamination). The Grassmann P4 gadget (Lemma 46) contaminates register j : $F(x)_j = 2x_j$. Register j is consumed and never reused after the coupling, so the contamination is local; the live-coordinate update at most doubles in the exact P4 step, giving a multiplicative inflation of at most 2^{d-1} in the dead-zone bias constant over $d-1$ couplings. For Stiefel and $\mathcal{J}_n$, the P4 gadget is exact (no contamination, factor 1).

B.3 Cross-axis (CAFC) fold-and-cut machinery

The RIH route requires both scalar multipliers s and t for Grassmann manifolds (§B.6). The CAFC route circumvents this by changing the *read–write pattern*: instead of reading and writing the *same* coordinate (as in P3), it reads one rotated coordinate and writes another, exploiting the orthogonality of a rotated basis. The construction adapts the fold-and-cut technique of Kidger and Lyons [19] to the manifold-constrained setting.

Definition 24 (Fold). A *fold* with write direction $w \in \mathbb{R}^d$, read direction $v \in \mathbb{R}^d$, amplitude $c \in \mathbb{R}$, and threshold $\beta \in \mathbb{R}$ is the map

$$F(x) = x + c \cdot w \cdot \text{ReLU}(v^\top x + \beta).$$

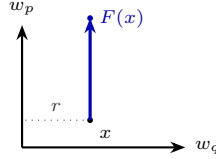


Figure 3: **CAFC atom**: a single fold $F(x) = x + c w_p \text{ReLU}(w_q^\top x + \beta)$ with $w_p \perp w_q$ in a totally-nonzero ONB. The read coordinate $r := w_q^\top x + \beta$ is the signed distance from x to the hinge line $\{r = 0\}$ along w_q . The displacement $F(x) - x$ is along w_p with magnitude $c \cdot r$ on the active side ($r > 0$).

Lemma 25 (Block reduction). Let $\mathcal{M} \subseteq \mathbb{R}^{d \times d}$ and consider a block $F(x) = x + s \cdot A \cdot \text{ReLU}(Bx + b)$ with $A, B \in \mathcal{M}$. Suppose there exists an index $j \in \{1, \dots, d\}$ such that:

- (i) $Ae_j = \alpha_A w_i + z$ with $\alpha_A \neq 0$ and $z \in w_i^\perp$;
- (ii) $\text{row}_j(B) = \nu \cdot v^\top$ for some nonzero scalar $\nu \in \mathbb{R}$ and nonzero $v \in w_i^\perp$ (not necessarily unit);
- (iii) for every $m \neq j$, the bias b_m can be chosen so that $\text{ReLU}((Bx + b)_m) = 0$ for all $x \in K$.

Then on K , setting $\eta = \text{sgn}(\nu)$:

$$F(x) = x + c \cdot w_i \cdot \text{ReLU}(\eta(v^\top x + \beta)) + s \cdot z \cdot |\nu| \text{ReLU}(\eta(v^\top x + \beta)),$$

where $c = s \cdot \alpha_A \cdot |\nu|$ and $\beta = b_j/\nu$. If additionally $z = 0$ (i.e., $Ae_j \propto w_i$), the block realizes an exact fold (Definition 24) with write direction w_i and read direction ηv , and the map $(s, b_j) \mapsto (c, \beta)$ is a bijection $\mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Proof. Apply dead-zone biasing ((iii)): for $m \neq j$, set $b_m < -(\|B\|_{\text{op}} \sup_K \|x\| + 1)$ to ensure $(Bx + b)_m < 0$ on K , hence $\text{ReLU}((Bx + b)_m) = 0$. Only $\text{ReLU}((Bx + b)_j)$ survives. By (ii), $(Bx + b)_j = \nu(v^\top x) + b_j$. Then $A \cdot \text{ReLU}(Bx + b) = Ae_j \cdot \text{ReLU}(\nu v^\top x + b_j)$. Factoring: $\text{ReLU}(\nu v^\top x + b_j) = |\nu| \text{ReLU}(\eta(v^\top x + b_j/\nu))$ since $\text{ReLU}(\alpha t) = |\alpha| \text{ReLU}(\text{sgn}(\alpha)t)$ for $\alpha \neq 0$. Substituting (i): $F(x) - x = s(\alpha_A w_i + z) |\nu| \text{ReLU}(\eta(v^\top x + \beta))$. When $z = 0$, the second term vanishes. $\square$

Definition 26 (CAFC-fold condition (square and rectangular cases)). For square $\mathcal{M} \subseteq \mathbb{R}^{d \times d}$, this is Definition 6. For rectangular $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ with $k < d$, impose the same condition, including the universal quantifiers over the desired nonzero fold amplitude c and threshold β , after replacing the block family by

$$\mathcal{B}_{\mathcal{M}}^{\text{rect}} := \{x \mapsto x + s \cdot A \cdot \text{ReLU}(B^\top x + b) : A, B \in \mathcal{M}, s \in \mathbb{R}, b \in \mathbb{R}^k\}$$

and replacing Bx by $B^\top x$ and $b \in \mathbb{R}^d$ by $b \in \mathbb{R}^k$ in the two fold-realization clauses.

Lemma 25 gives the verification template for both square and rectangular cases: it suffices to find an index j with $Ae_j \propto w$ (write column proportional to the write direction), the surviving read row of the linear part proportional to $v^\top$ with positive coefficient $\nu > 0$, and dead-zone biasing forcing $\text{ReLU}(\cdot)$ to vanish on the non-surviving rows for all $x \in K$.

Lemma 27 (Dense-direction ridge density). *Let $\Omega \subset \mathbb{R}^m$ be compact and let $\mathcal{D} \subseteq S^{m-1}$ be a dense subset of the unit sphere. Finite sums of the form $\psi(z) = \sum_{m=1}^M c_m \text{ReLU}(a_m^\top z + \eta_m)$ with $a_m/\|a_m\| \in \mathcal{D}$, $\eta_m \in \mathbb{R}$, $c_m \in \mathbb{R}$ are sup-dense in $C(\Omega)$.*

Proof. By Theorem 1 of Leshno et al. [22], finite *unrestricted* ridge sums $\tilde{\psi}(z) = \sum_{m=1}^{\tilde{M}} \tilde{c}_m \text{ReLU}(\tilde{a}_m^\top z + \tilde{\eta}_m)$ ($\tilde{a}_m \in \mathbb{R}^m \setminus \{0\}$, $\tilde{\eta}_m \in \mathbb{R}$, $\tilde{c}_m \in \mathbb{R}$) are sup-dense in $C(\Omega)$. Fix $\tilde{\psi}$ with $\|\tilde{\psi} - h\|_{\infty, \Omega} \leq \varepsilon/2$ for the target $h \in C(\Omega)$. By positive homogeneity of ReLU we may normalize each direction to $\hat{a}_m := \tilde{a}_m/\|\tilde{a}_m\| \in S^{m-1}$ at the cost of absorbing $\|\tilde{a}_m\|$ into the coefficient and bias: $\tilde{c}_m \text{ReLU}(\tilde{a}_m^\top z + \tilde{\eta}_m) = \tilde{c}_m \|\tilde{a}_m\| \text{ReLU}(\hat{a}_m^\top z + \tilde{\eta}_m/\|\tilde{a}_m\|)$. Let $B := \max_{1 \leq m \leq \tilde{M}} |\tilde{\eta}_m|/\|\tilde{a}_m\|$. The map $(a, \eta) \mapsto c \text{ReLU}(a^\top z + \eta)$ is uniformly continuous in (a, η) on the compact set $S^{m-1} \times [-B, B] \times \Omega$ (the modulus follows from $|a^\top z + \eta - a'^\top z - \eta'| \leq \|z\| \|a - a'\| + |\eta - \eta'|$ and the 1-Lipschitz property of ReLU). By density of $\mathcal{D}$ in S^{m-1} , choose $a_m \in \mathcal{D}$ with $\|a_m - \hat{a}_m\|$ small enough that $\sum_{m=1}^{\tilde{M}} |\tilde{c}_m| \|\tilde{a}_m\| \sup_{z \in \Omega} |\text{ReLU}(a_m^\top z + \tilde{\eta}_m/\|\tilde{a}_m\|) - \text{ReLU}(\hat{a}_m^\top z + \tilde{\eta}_m/\|\tilde{a}_m\|)| \leq \varepsilon/2$. Setting $c_m = \tilde{c}_m \|\tilde{a}_m\|$, $\eta_m = \tilde{\eta}_m/\|\tilde{a}_m\|$, the resulting $\psi = \sum c_m \text{ReLU}(a_m^\top z + \eta_m)$ has $\|\psi - h\|_{\infty, \Omega} \leq \varepsilon$. $\square$

Theorem 28 (CAFC $\Rightarrow L^p(K)$ -universality). *Let $d \geq 2$. If $\mathcal{M}$ satisfies the CAFC-fold condition (Definition 26), each constrained block has the outer scalar multiplier $s \in \mathbb{R}$ available (so arbitrary fold amplitudes are realizable), and the ResNet has an unconstrained linear readout, then the constrained ResNet class is dense in $L^1(K)$ for every compact $K \subset \mathbb{R}^d$.*

Proof. Since continuous functions are dense in $L^1(K)$, it suffices to approximate any continuous $f : K \rightarrow \mathbb{R}$. Let $\{w_1, \dots, w_d\}$ and $\mathcal{D}$ be supplied by Definition 26, and write $\xi_i = w_i^\top x$.

Let $K_\xi := \{(w_1^\top x, \dots, w_d^\top x) : x \in K\}$ and define $g_0(\xi) = f(\sum_i \xi_i w_i)$ on K_ξ . For each coordinate let $a_i := \min_{K_\xi} \xi_i$ and $b_i := \max_{K_\xi} \xi_i$, and set $Q := \prod_{i=1}^d [a_i, b_i]$. By Tietze's theorem, extend g_0 to a continuous function $g \in C(Q)$. Let ω_g be the modulus of continuity of g on Q .

Assume $b_d > a_d$ (the case $b_d = a_d$ is trivial). Partition $[a_d, b_d]$ into strips I_p of width $\delta = (b_d - a_d)/n$, with midpoints $\bar{\xi}_d^{(p)}$. Choose $S > 1 + \max_{i < d} (b_i - a_i)$ and define disjoint rectangles $R_p := \prod_{i=1}^{d-1} [a_i + pS, b_i + pS]$. On R_p define $h_p(z_1, \dots, z_{d-1}) := g(z_1 - pS, \dots, z_{d-1} - pS, \bar{\xi}_d^{(p)}) - \bar{\xi}_d^{(p)}$. The argument lies in Q , so each h_p is well-defined and continuous. Since the R_p are pairwise disjoint compact sets, $h := h_p$ on $\bigcup_p R_p$ is continuous; extend by Tietze to $\tilde{h} \in C(\Omega)$ on a compact box $\Omega \supset \bigcup_p R_p$.

The set $\mathcal{D} \subset w_d^\perp$ is identified with a subset of the unit sphere S^{d-2} ; by hypothesis it is dense. By Lemma 27 (dense-direction ridge density), finite ridge sums with directions restricted to $\mathcal{D}$ are

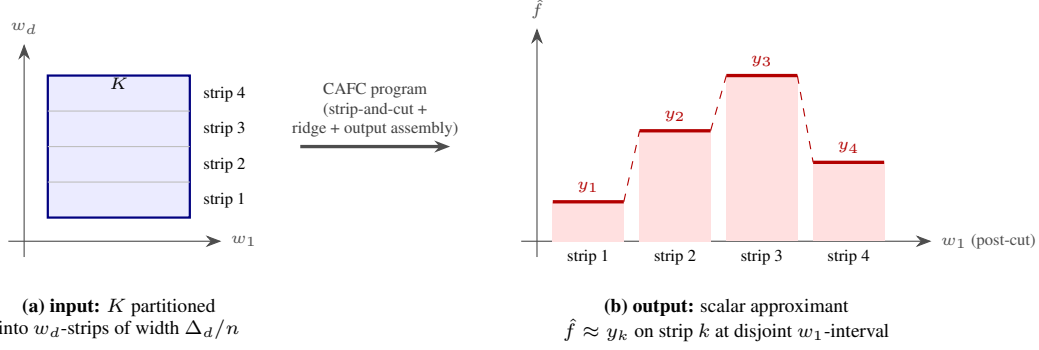


Figure 4: **CAFC program.** (a) **Input:** compact K partitioned into w_d -strips of width Δ_d/n . (b) **Output:** the scalar approximant $\hat{f}$ realized by the constrained ResNet, plotted against the post-strip-and-cut w_1 coordinate. The proof realizes three internal stages: strip-and-cut (folds with read w_d , write w_i , $i < d$), ridge approximation (along $\mathcal{D} \subseteq w_d^\perp$), and output assembly (folds with write w_d); the arrow summarizes them.

sup-dense in $C(\Omega)$. Choose

$$\psi(z) = \sum_{m=1}^M c_m \text{ReLU}(a_m^\top z + \eta_m), \quad c_m \in \mathbb{R}, \quad a_m / \|a_m\| \in \mathcal{D},$$

such that $\|\psi - \tilde{h}\|_{\infty, \Omega} \leq \varepsilon_2$.

Choose CPWL staircases ϕ_i with transition width $\tau \ll \delta$ such that $\phi_i(\xi_d) = pS$ on the core of strip I_p and interpolates linearly inside transition slabs. Each ϕ_i is a finite sum of terms $c \text{ReLU}(\xi_d + \beta)$. The ideal fold construction is: for $i < d$, write in direction w_i and read w_d to add $\phi_i(\xi_d)$ to ξ_i ; then write in direction w_d and read $\sum_{i < d} a_{m,i} w_i$ to add the ridge terms of ψ to ξ_d . Orthogonality of the ONB gives exact summation within each phase.

Enumerate the finitely many ideal folds as $\mathcal{F}_0, \dots, \mathcal{F}_{N-1}$, where $\mathcal{F}_m(x) = x + c_m w_{i_m} \text{ReLU}(v_m^\top x + \beta_m)$. Define cumulative state sets $K_0 := K$ and $K_{m+1} := K_m \cup \mathcal{F}_m(K_m)$. With $R_m := \sup_{x \in K_m} \|x\|$, $R_{m+1} \leq (1 + |c_m| \|v_m\|) R_m + |c_m| |\beta_m|$, so $R_N < \infty$. Let $K_* := K_N$. By the CAFC-fold condition applied on K_* and Lemma 25 with $z = 0$, $\nu > 0$, each ideal fold is realized exactly on K_* , so the constrained ResNet realizes the whole ideal map on K .

Take readout $L_{\text{out}}(y) = w_d^\top y$. On a strip core,

$$|L_{\text{out}}(R(x)) - g(\xi)| \leq \varepsilon_2 + \delta/2 + \omega_g(\delta/2).$$

Transition slabs have d -dimensional measure $\leq 2n\tau \prod_{i < d} (b_i - a_i)$, with bounded error. Thus

$$\|L_{\text{out}} \circ R - f\|_{L^1(K)} \leq \text{vol}(K) (\varepsilon_2 + \delta/2 + \omega_g(\delta/2)) + C_{\text{tr}} n\tau.$$

Choose n large, then ε_2 small, then $\tau/\delta \rightarrow 0$. $\square$

Remark 29 (Depth). The staircase phases use $O(n)$ blocks each, contributing $O((d-1)n)$. Phase d uses M ridge blocks. Total depth is $O((d-1)n + M)$. If h is Lipschitz, $M = O(\varepsilon^{-(d-1)})$ by classical ridge approximation rates [36] and $n = O(1/\varepsilon)$ suffices, giving $O(d/\varepsilon + \varepsilon^{-(d-1)})$. This saves one dimension in the exponent compared to single-register RIH depth $O(\varepsilon^{-d})$. For $\mathcal{J}_n$ and Gr_{Sp} , the literal single-index CAFC gadget is unavailable (zero diagonal for $\mathcal{J}_n$; partner-zero obstruction for Gr_{Sp}); the macro-CAFC variant (Theorem 31) provides the CAFC-style construction with $O(d^2 N_{\text{fold}})$ blocks, equivalently $O(d^2)$ blocks per ideal fold.

B.4 Macro-CAFC replacement

Definition 30 (Macro-CAFC fold). Let $\mathcal{M} \subseteq \mathbb{R}^{d \times d}$ and consider residual blocks $R_\ell(x) = x + s_\ell A_\ell \text{ReLU}(B_\ell x + b_\ell)$ with $A_\ell, B_\ell \in \mathcal{M}$ and outer scalar multipliers $s_\ell \in \mathbb{R}$. We say that $\mathcal{M}$ admits *exact macro-CAFC folds* if, for every rank-one ReLU fold $\mathcal{F}(x) = x + c w \text{ReLU}(u^\top x + \beta)$, every

nonempty compact state set $K \subset \mathbb{R}^d$, and all $w, u \in \mathbb{R}^d$, $c, \beta \in \mathbb{R}$, there is a finite block program $P = R_m \circ \dots \circ R_1$ such that $P(x) = \mathcal{F}(x)$ for all $x \in K$. This is an inflated-block analog of Definition 26; it does not identify any manifold as satisfying Definition 26.

Theorem 31 (Macro-CAFC replacement for the CAFC proof). *Let $d \geq 2$. Suppose $\mathcal{M}$ admits exact macro-CAFC folds on every nonempty compact state set, with only outer scalar multipliers as in Definition 30, and suppose the scalar CAFC proof above uses a finite ideal fold sequence of length N_{fold} . Then replacing each ideal fold by its macro realization gives the same scalar-output $L^1(K)$ -universality conclusion as Theorem 28, with the same terminal unconstrained linear readout. If each ideal fold is realized by $O(d^2)$ constrained blocks, the resulting realization uses $O(d^2 N_{\text{fold}})$ constrained blocks in total, equivalently $O(d^2)$ blocks per ideal fold. This is a block-count statement only, not a depth-efficiency, conditioning, or trainability claim.*

Proof. Run the proof of Theorem 28 up to the finite ideal fold list $\mathcal{F}_0, \dots, \mathcal{F}_{N_{\text{fold}}-1}$. Set $K_0 = K$. At step m , the current state set K_m is compact because it is the image of a compact set under a finite composition of continuous residual blocks. By Definition 30, choose a finite constrained block program P_m that equals $\mathcal{F}_m$ on K_m , and set $K_{m+1} = \mathcal{F}_m(K_m) = P_m(K_m)$. Induction over the finite list shows that the composed macro program agrees exactly with the ideal CAFC fold program on the original compact set. The terminal readout and the transition-slab L^1 estimate are therefore unchanged from Theorem 28. Summing the per-fold block counts gives the stated $O(d^2 N_{\text{fold}})$ total. $\square$

B.5 Per-Manifold Verification

The per-manifold verifications below establish, for each constrained family in Table 2, exact RIPR (RIH route) with finite primitive cost (usually $c_{\text{prim}} = 1$; scalar-free noncompact Stiefel uses $c_{\text{prim}} \leq 2$) or approximate RIPR with $c_{\text{prim}} = 1$, with the scalar multiplier prescriptions recorded in Table 2. The CAFC-route manifolds where literal CAFC fails ($\mathcal{J}_n, \text{Gr}_{\text{Sp}}$) additionally admit macro-CAFC folds at $O(d^2)$ blocks per ideal fold (Theorem 31); this is a CAFC-route per-fold block cost separate from RIPR’s c_{prim} . All proofs are self-contained in this appendix.

Witness table. Table 3 consolidates, for each manifold, the witness pairs $(A, B) \in \mathcal{M} \times \mathcal{M}$, the read direction (the row of $B^\top$ that survives the dead-zone step on the active register), the write direction (the column of A), the scalar multiplier use in the RIH construction, and the route exactness label of Table 2. The table points to the per-manifold proofs in §§B.5.1–B.5.7; each manifold’s full construction is found in the cited subsubsection.

Table 3: Per-manifold witness summary for the RIPR verifications. “P3 (read $\rightarrow$ write)” records the row of B (or $B^\top$ for rectangular Stiefel) surviving the dead-zone step and the corresponding column of A depositing the hinge into the register. “P4” uses an analogous witness for the register-coupling primitive (§B.2). “Scalars” is the RIH multiplier budget actually used in the per-manifold construction. Full constructions in the cited proof locations.

Manifold	Mech.	P3 (read $\rightarrow$ write)	P4 (read $\rightarrow$ write)	Scalars	Route	Proof
$V(k, d)$, $k < d$	frame embed.	$\text{row}_q(B^\top) = \nu e_i^\top$; $Ae_q = e_i$	coupled rectangular witness	s, t	RIH/CAFC exact	§B.5.1
$\text{St}(k, d)$	full-rank frame	full-rank witness; const. shift two blocks	full-rank witness	$\emptyset$	RIH exact ($c_{\text{prim}} \leq 2$); CAFC literal	§B.5.1
$\text{Gr}(k, d)$	projector row	$\text{row}_q(B) = e_i^\top$; $Ae_q = e_i$	$\text{row}_q(B) = e_j^\top$; $Ae_q = \frac{1}{2}(e_i + e_j)$, $s=2$	s, t	CAFC/RIH exact	§B.5.2
$\text{PSD}(k, d)$, $k < d$	Gr incl. ($W=QQ^\top$)	inherited Gr witness	inherited Gr witness	s, t	RIH exact; CAFC literal via Gr	§B.5.2
Sym_Λ	fixed-spec. or-bit	eigvec at chosen level; sign-routed	coupled-eigvec at chosen level	s, t	RIH exact (fixed-spec.)	§B.5.3
$\mathcal{J}_n$	partner-pair	$\text{row}_i(B) = e_i e_i^\top$; $Ae_i = \delta_i e_i$	aux $\text{row}_r(B) = e_j^\top$; $Ae_r = e_i$	s, t	macro-CAFC exact	§B.5.5
LGr	symm.-unitary	symm.-unitary col. completion	symm.-unitary col. completion	s, t	literal CAFC exact	§B.5.7
$E(d, k)$	oblique idemp.	$\text{row}_q(B) = e_i^\top$; $Ae_q = e_i$	oblique-coupled witness	s	RIH approx (signed via $s = \gamma$)	§B.5.4
Gr_{Sp}	symp. involution	approx. $\text{row}_i(B) = (\alpha-1)e_i^\top + \epsilon e_j^\top$; $Ae_i = e_i$	signed-swap $e_j \rightarrow e_i$	s	RIH approx; macro-CAFC exact	§B.5.6

Lemma 32 (Dead-zone biasing). *Let $K \subset \mathbb{R}^d$ be compact and let $B \in \mathbb{R}^{d' \times d}$. For any coordinate m , the choice*

$$b_m < - \sup_{x \in K} (Bx)_m$$

ensures $(Bx + b)_m < 0$ for every $x \in K$, hence $\text{ReLU}((Bx + b)_m) = 0$ on K . In particular, any $b_m < -\|B\|_{\text{op}} \cdot \sup_{x \in K} \|x\|_2$ suffices, and any pointwise upper bound on $(Bx)_m$ over K may be used in place of the operator-norm estimate.

This lemma is invoked throughout the per-manifold proofs to zero out inactive preactivation coordinates; we cite it inline as “dead-zone biasing (Lemma 32)” rather than rewriting the inequality each time.

B.5.1 Orthonormal Stiefel $V(k, d)$ and noncompact Stiefel $\text{St}(k, d)$ (V : $c_{\text{prim}} = 1$ with s, t ; St : scalar-free $c_{\text{prim}} \leq 2$; s -only literal CAFC)

Every signed P3 primitive (with $a \in \mathbb{R}$, including the constant-shift $a = 0$ case via $t = 0$) and P4 register coupling are realized exactly in a single $V(k, d)$ -constrained block, using both scalar multipliers $s, t \in \mathbb{R}$ as needed. The key observation is that $V(k, d)$ admits columns in any direction, a property we formalize as *directional richness*. Grassmann projectors do *not* satisfy directional richness (Proposition 41), necessitating the weaker projector-richness condition of §B.5.2. For the noncompact manifold $\text{St}(k, d)$, nonzero Stiefel RIH scalars are absorbed into full column rank factors, and the constant-shift primitive is handled by a two-block active-zone cancellation. Its CAFC route similarly absorbs nonzero fold gains, paralleling the way $\text{PSD}(k, d)$ inherits from $\text{Gr}(k, d)$ in §B.5.2, but by absorption rather than inclusion.

Definition 33 (Directional richness). A matrix set $\mathcal{M} \subseteq \mathbb{R}^{d \times k}$ (with $k \leq d$) satisfies *directional richness* if for every unit vector $v \in S^{d-1}$ and every column index $j \in \{1, \dots, k\}$, there exists $W \in \mathcal{M}$ with $We_j = v$.

Proposition 34 (Stiefel and $O(d)$ are directionally rich). $V(k, d)$ and $O(d)$ satisfy *directional richness*.

Proof. For $O(d)$: given unit $v \in S^{d-1}$, extend to an orthonormal basis $\{v, q_2, \dots, q_d\}$ of $\mathbb{R}^d$ and set W to be the orthogonal matrix with v as its j -th column; then $We_j = v$. For $V(k, d)$ with $k \leq d$: given unit $v \in S^{d-1}$, extend to an orthonormal set $\{v, q_2, \dots, q_k\}$; define $W = [q_2 \cdots q_j \ v \ q_{j+1} \cdots q_k] \in V(k, d)$ (with v in column j) so that $We_j = v$. $\square$

Corollary 35 (Orthonormal Stiefel exact RIPR, $c_{\text{prim}} = 1$). *The $V(k, d)$ -constrained block family with both scalar multipliers $s, t \in \mathbb{R}$ satisfies exact RIPR with $c_{\text{prim}} = 1$, for any $1 \leq k \leq d$.*

Proof. Fix register i and a compact state set K_{st} . We realize each RIPR primitive (signed P3 with $a \in \mathbb{R}$, and P4) in one Stiefel block.

Signed P3 with $a \neq 0$. If $\gamma = 0$, the primitive is the identity and can be omitted; assume $\gamma \neq 0$. Let $u_0 = \text{sign}(a)e_i$ and $v_0 = \text{sign}(\gamma)e_i$. Set $B = [u_0, q_2, \dots, q_k] \in V(k, d)$ with $\{q_2, \dots, q_k\} \subset e_i^\perp$ orthonormal, and $A = [v_0, v_2, \dots, v_k] \in V(k, d)$ with $\{v_2, \dots, v_k\} \subset e_i^\perp$ orthonormal. Set $s = |\gamma|$, $b_1 = b$, $t = |a|$, and dead-zone biasing $b_m = -C$ ($m \geq 2$) with $C > \sup_{K_{\text{st}}} \|x\| + 1$. The block produces

$$F(x)_i = x_i + \gamma \text{ReLU}(ax_i + b), \quad F(x)_m = x_m \quad (m \neq i),$$

realizing signed P3 exactly with $c_{\text{prim}} = 1$.

Signed P3 with $a = 0$ (constant shift). $x_i \mapsto x_i + \gamma \text{ReLU}(b) = x_i + c$ with $c := \gamma \text{ReLU}(b)$. When $b \leq 0$ the shift is $c = 0$ and $s = 0$ realizes the identity exactly. When $b > 0$ and $\gamma \neq 0$, take $A = [e_i, v_2, \dots, v_k] \in V(k, d)$ with $\{v_2, \dots, v_k\}$ orthonormal in $e_i^\perp$, B any element of $V(k, d)$, and set $t = 0$ so that the ReLU argument is the constant bias vector:

$$\text{ReLU}(tB^\top x + b) = \text{ReLU}(b) = (\text{ReLU}(b_1), \text{ReLU}(b_2), \dots, \text{ReLU}(b_k)).$$

Choose $b_1 = b > 0$ and $b_m = -1$ for $m \geq 2$. Then $\text{ReLU}(b) = b e_1$, and $A \text{ReLU}(b) = b A e_1 = b e_i$. Setting $s = \gamma$ gives $F(x)_i = x_i + \gamma b = x_i + c$ on K_{st} , realizing signed P3 with $a = 0$ exactly in one block. Both signs of γ (hence c) are covered, including the negative-amplitude shift $\gamma < 0$ used in the cut-and-shift step of Theorem 18.

P4 register coupling. For $i \neq j$ with $x_j \geq 0$ on K_{st} : take $U = e_j^\top$, $V = e_i$, so $u_0 = e_j$, $v_0 = e_i$. Set $B = [e_j, q_2, \dots, q_k]$, $A = [e_i, v_2, \dots, v_k] \in V(k, d)$ with the usual orthonormal extensions, $s = 1$, $t = 1$, $b_1 = 0$, $b_m = -C$ for $m \geq 2$. The block produces $F(x)_i = x_i + \text{ReLU}(x_j) = x_i + x_j$ on $K_{\text{st}} \cap \{x_j \geq 0\}$, realizing P4 exactly.

All three primitives are realized exactly in one Stiefel block ($c_{\text{prim}} = 1$, $s, t \in \mathbb{R}$). Theorem 5 gives $L^p(K)$ -universality on every compact $K \subset \mathbb{R}^d$. $\square$

Corollary 36 (Noncompact Stiefel exact RIPR without scalar multipliers). *The $\text{St}(k, d)$ -constrained block family satisfies exact RIPR with primitive cost $c_{\text{prim}} \leq 2$ and no explicit scalar multipliers, for every $1 \leq k \leq d$.*

Proof. Fix register i and compact state set K_{st} . The nonconstant signed-P3 and P4 gadgets above become scalar-free over $\text{St}(k, d)$ by replacing the active read and write columns with the scaled columns themselves: for signed P3 with $a \neq 0$, use active read column ae_i and active write column γe_i ; for P4, use active read column e_j and active write column e_i . Complete these active columns to full column rank matrices and dead-zone all inactive preactivation coordinates. If the primitive has zero gain, it is the identity map and is omitted.

It remains to realize the signed-P3 constant shift $x_i \mapsto x_i + c$, where $c = \gamma \text{ReLU}(b) \neq 0$. First suppose $d \geq 2$. Choose a nonzero $r \in e_i^\perp$ and choose M so large that $r^\top x + M > 0$ on K_{st} . Use two scalar-free $\text{St}(k, d)$ blocks whose active read column is r , whose active write columns are ce_i and $-ce_i$, and whose active biases are $M + 1$ and M , respectively; complete all active columns to full column rank matrices and dead-zone inactive coordinates on the successive compact state sets. Because $r^\top e_i = 0$, the first block does not change the read value used by the second block, and the two active contributions are

$$ce_i(r^\top x + M + 1) - ce_i(r^\top x + M) = ce_i.$$

Thus the two-block gadget realizes the constant shift exactly.

For $d = 1$, choose M large enough on the compact interval K_{st} . The scalar-free full-rank blocks

$$x \mapsto x + \text{ReLU}(x + M) = 2x + M, \quad y \mapsto y - \text{ReLU}\left(\frac{1}{2}y + \frac{1}{2}M - c\right) = \frac{1}{2}y - \frac{1}{2}M + c$$

are both in their active ReLU regimes on the relevant compact sets and compose to $x \mapsto x + c$. Hence every RIPR primitive is realized with at most two scalar-free $\text{St}(k, d)$ blocks. $\square$

Theorem 37 (Orthonormal Stiefel universality). *$V(k, d)$ -constrained ResNets with both scalar multipliers $s, t \in \mathbb{R}$ and an unconstrained readout are $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$, with zero depth overhead.*

Proof. Corollary 35 (exact RIPR with $c_{\text{prim}} = 1$) + Theorem 5. $\square$

Proposition 38 (Orthonormal Stiefel satisfies the CAFC-fold condition). *Let $d \geq 2$ and $1 \leq k \leq d - 1$. Then $V(k, d)$ satisfies the CAFC-fold condition (Definition 26, rectangular variant). Consequently, $V(k, d)$ -constrained ResNets with outer scalar multiplier s only and an unconstrained terminal linear readout are $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$ (Theorem 28). The case $d = 1$ (or $k = d$) is covered by the exact RIH route of Corollary 35 instead.*

Proof. Fix a compact $K \subset \mathbb{R}^d$. Set $w_d = 1/\sqrt{d}$ and complete to a totally-nonzero ONB $\{w_1, \dots, w_d\}$ (as in Theorem 66; the set of orthonormal completions with some zero entry is a proper algebraic subvariety in the Haar measure on compatible frames, hence measure zero, so totally-nonzero completions exist). Take $\mathcal{D} = \{v \in w_d^\perp : \|v\| = 1\}$ as the read-direction family. Given write direction w_i and nonzero read direction $v \in w_i^\perp$:

Write matrix. Extend $\{w_i\}$ to an orthonormal set $\{w_i, q_2, \dots, q_k\} \subset \mathbb{R}^d$ and define $A = [w_i, q_2, \dots, q_k] \in V(k, d)$. Then $Ae_1 = w_i$ (so $\alpha_A = 1$ and $z = 0$ in Lemma 25).

Read matrix. Set $\hat{v} = v/\|v\|$. Extend $\{\hat{v}\}$ to an orthonormal set $\{\hat{v}, r_2, \dots, r_k\} \subset \mathbb{R}^d$ and define $B = [\hat{v}, r_2, \dots, r_k] \in V(k, d)$. The first preactivation component is $(B^\top x)_1 = \hat{v}^\top x = \|v\|^{-1}v^\top x$.

Dead-zone biasing. Apply Lemma 32 to the $m \geq 2$ preactivation coordinates, zeroing their ReLUs on K .

Assembly. $A \cdot \text{ReLU}(B^\top x + b) = Ae_1 \cdot \text{ReLU}(\|v\|^{-1}v^\top x + b_1) = w_i \cdot \text{ReLU}(\|v\|^{-1}v^\top x + b_1)$. The surviving coefficient is $\nu = \|v\|^{-1} > 0$. Setting $c = s\|v\|^{-1}$ and $\beta = \|v\|b_1$: $F(x) = x + c \cdot w_i \cdot \text{ReLU}(v^\top x + \beta)$. The map $(s, b_1) \mapsto (c, \beta)$ is a bijection $\mathbb{R}^2 \rightarrow \mathbb{R}^2$. This verifies Lemma 25(i)–(iii) with $z = 0$. $\square$

Remark 39 ($O(d)$ and the CAFC-fold condition). Since $O(d) = V(d, d)$, the same construction extends to $O(d)$ by using the full square frame: with $k = d$, take $A = [w_i, q_2, \dots, q_d]$ and $B = [\hat{v}, r_2, \dots, r_d]$ as full ONBs of $\mathbb{R}^d$; the dead-zone biasing argument applies unchanged to the remaining $d - 1$ rows, so $O(d)$ satisfies the CAFC-fold condition. For the square $O(d)$ case we use the transposed read matrix $B_{\text{sq}} = [\hat{v}, r_2, \dots, r_d]^\top$ to match the $B^\top x$ block-reduction convention of Lemma 25; otherwise the construction is identical to the rectangular case.

Corollary 40 (Noncompact Stiefel CAFC by absorption). *Let $1 \leq k \leq d$ and $\text{St}(k, d) = \{W \in \mathbb{R}^{d \times k} : \text{rank}(W) = k\}$. The $\text{St}(k, d)$ -constrained residual class is $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$ with no explicit scalar multipliers.*

Proof. Use the CAFC route above. Each nonzero ideal fold is realized over $V(k, d)$ by a rectangular block of the form

$$x + sA \text{ReLU}(B^\top x + b), \quad A, B \in V(k, d),$$

with no inner scalar multiplier. Replacing A by $\tilde{A} = sA$ absorbs the remaining scalar multiplier, and $\tilde{A} \in \text{St}(k, d)$ whenever $s \neq 0$. Zero-gain folds are omitted. The same argument applies to the square $O(d) = V(d, d)$ case, and the one-dimensional case is immediate because nonzero real weights can realize one-dimensional ReLU hinges directly. Hence the CAFC approximants are realized with $A, B \in \text{St}(k, d)$ and no explicit multipliers. $\square$

B.5.2 Grassmann $\text{Gr}(k, d)$ and PSD $\text{PSD}(k, d)$ ($c_{\text{prim}} = 1$, both s, t via RIH; s -only via CAFC; PSD inherited via Burer–Monteiro)

The Stiefel approach fails for Grassmann projectors due to a fundamental PSD constraint.

Proposition 41 (PSD diagonal constraint). *For any $\Pi \in \text{Gr}(k, d)$: $\Pi = QQ^\top$ for some $Q \in \mathbb{R}^{d \times k}$ with orthonormal columns, hence*

$$\Pi_{ii} = (QQ^\top)_{ii} = \|Q^\top e_i\|^2 \geq 0 \quad \text{for every } i.$$

In particular, $\text{row}_i(\Pi) \cdot e_i = \Pi_{ii} \geq 0$, so no Grassmann projector has $\text{row}_i(\Pi) = -e_i^\top$. Grassmann projectors do not satisfy directional richness.

Remark 42 (Stiefel reparametrization). Since $\text{Gr}(k, d) \cong V(k, d)/O(k)$, every projector $\Pi \in \text{Gr}(k, d)$ admits a factorization $\Pi = QQ^\top$ with $Q \in V(k, d)$. Lifting the Grassmann constraint to Stiefel does not circumvent the PSD obstruction: the weight matrix entering the residual block is still $\Pi = QQ^\top$, and $\Pi_{ii} = \|Q^\top e_i\|^2 \geq 0$ regardless of the parametrization. If instead one uses $Q \in V(k, d)$ directly as the weight matrix (rectangular $d \times k$ Stiefel blocks as in Remark 16), then directional richness holds and only the outer scalar multiplier s is needed (Appendix B.5.1). The two settings are genuinely distinct; a Stiefel ResNet is not a reparametrized Grassmann ResNet.

This motivates the following weaker condition.

Definition 43 (Projector directional richness). A matrix set $\mathcal{M} \subseteq \mathbb{R}^{d \times d}$ satisfies *projector directional richness* if for every nonzero $u \in \mathbb{R}^d$ and index i with $u_i \neq 0$, there exists $W \in \mathcal{M}$ and nonzero $c \in \mathbb{R}$ such that $\text{row}_i(W) = c \cdot u^\top$.

Lemma 44 (Grassmann satisfies projector directional richness). *For any $0 < k < d$, $\text{Gr}(k, d)$ satisfies projector directional richness: for every nonzero $u \in \mathbb{R}^d$ with $u_i \neq 0$, there is $\Pi \in \text{Gr}(k, d)$ with $\text{row}_i(\Pi) = c \cdot u^\top$ for some $c \neq 0$.*

Proof. Set $q_1 = u/\|u\|$. Since $\dim(e_i^\perp \cap q_1^\perp) \geq d - 2 \geq k - 1$ (for $d \geq k + 1$), choose $q_2, \dots, q_k \in e_i^\perp \cap q_1^\perp$ orthonormal, and set $\Pi = \sum_{m=1}^k q_m q_m^\top \in \text{Gr}(k, d)$. For $m \geq 2$, $(q_m)_i = 0$ since $q_m \perp e_i$, so $\text{row}_i(\Pi) = (q_1)_i q_1^\top = (u_i/\|u\|^2)u^\top = cu^\top$ with $c = u_i/\|u\|^2 \neq 0$. $\square$

Lemma 45 (Grassmann primitive gadget for signed P3). *Fix compact $K \subset \mathbb{R}^d$ with $\sup_K \|x\|_\infty \leq M$, register i , and $\gamma, a, b \in \mathbb{R}$. Choose $C > |a|M\sqrt{d} + |b|$, orthonormal $q_2, \dots, q_k \in e_i^\perp$, and define*

$$B = A = e_i e_i^\top + \sum_{m=2}^k q_m q_m^\top \in \text{Gr}(k, d),$$

with inner scalar multiplier $t = a$, outer scalar multiplier $s = \gamma$, active bias $b_i = b$, and dead-zone-biasing constants $b_j = -C$ for $j \neq i$. Then on K , the constrained block $F(x) = x + s \cdot A \cdot \text{ReLU}(t \cdot Bx + b)$ satisfies $F(x)_i = x_i + \gamma \text{ReLU}(ax_i + b)$ and $F(x)_j = x_j$ for all $j \neq i$. This realizes signed P3 on register i exactly with $c_{\text{prim}} = 1$.

Proof. Read. Since $q_m \perp e_i$ for $m \geq 2$, $(Bx)_i = e_i^\top Bx = x_i$, so the active preactivation is $(tBx + b)_i = ax_i + b$.

Dead-zone biasing. Apply Lemma 32 to the scaled matrix tB : since B is an orthogonal projector ($\|tB\|_{\text{op}} = |a|$), the choice $b_j = -C$ with $C > |a|M\sqrt{d}$ satisfies the lemma's hypothesis on the $j \neq i$ coordinates, so $\text{ReLU}(tBx + b)_j = 0$ on K . The ReLU output is $\text{ReLU}(ax_i + b) \cdot e_i$.

Write-back. $Ae_i = e_i e_i^\top e_i + \sum_{m \geq 2} q_m q_m^\top e_i = e_i$ (again $q_m \perp e_i$). Hence $F(x) = x + s \text{ReLU}(ax_i + b) \cdot e_i = x + \gamma \text{ReLU}(ax_i + b) \cdot e_i$. $\square$

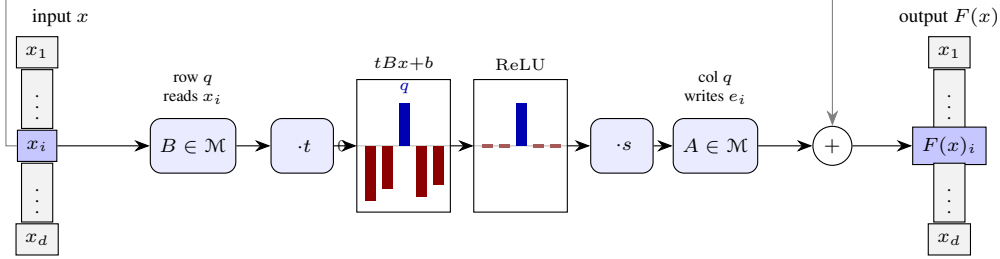


Figure 5: **RIH atom**: a single residual block $F(x) = x + s \cdot A \cdot \text{ReLU}(t \cdot Bx + b)$ with $A, B \in \mathcal{M}$. Witnesses are chosen so $\text{row}_q(B) = \nu e_i^\top$ (active row reads register i) and $Ae_q = e_i$ (write column deposits to e_i). The bias b has b_q free and $b_m = -C$ for $m \neq q$ (dead-zone biasing, Lemma 32); the bar charts before/after ReLU show only row q surviving. Net effect: $F(x)_i = x_i + s \text{ReLU}(\nu t x_i + b_q)$, $F(x)_m = x_m$ for $m \neq i$. The gray skip wire is the residual.

Lemma 46 (Grassmann coupling gadget for P4). *Let $d \geq 2$ and $1 \leq k \leq d - 1$. For every compact $K \subset \mathbb{R}^d$ and registers $i \neq j$, there exist $A, B \in \text{Gr}(k, d)$, biases $b \in \mathbb{R}^d$, and scalars $s, t \in \mathbb{R}$ such that the corresponding block satisfies the projected-primitive identities used by Corollary 22 on K : $F(x)_i = x_i + \text{ReLU}(x_j)$, $F(x)_m = x_m$ for $m \notin \{i, j\}$, and $F(x)_j = x_j + \text{ReLU}(x_j)$. In particular, on $K \cap \{x_j \geq 0\}$ it realizes the P4 update $F(x)_i = x_i + x_j$, with the contaminated j -component unrestricted per P4.*

Proof. Choice of projectors. Let $\sup_K \|x\|_\infty \leq M$ and choose $C > M\sqrt{d}$. Set $v = (e_i + e_j)/\sqrt{2}$. Choose orthonormal padding directions $q_2, \dots, q_k \in e_j^\perp$ and $p_2, \dots, p_k \in e_j^\perp \cap v^\perp$; these exist because $\dim e_j^\perp = d - 1 \geq k - 1$ and $\dim(e_j^\perp \cap v^\perp) = d - 2 \geq k - 1$. Define

$$B = e_j e_j^\top + \sum_{m=2}^k q_m q_m^\top \in \text{Gr}(k, d), \quad A = vv^\top + \sum_{m=2}^k p_m p_m^\top \in \text{Gr}(k, d),$$

and set $s = 2, t = 1, b_j = 0$, and $b_m = -C$ for $m \neq j$.

Read. $\text{row}_j(B) = e_j^\top$, so $(Bx)_j = x_j$. For $m \neq j$, $(Bx + b)_m \leq M\sqrt{d} - C < 0$ zeroes the ReLU. Dead-zone biasing of inactive coordinates gives $\text{ReLU}(Bx + b)|_{m \neq j} = 0$, and the active coordinate gives $\text{ReLU}(Bx + b)_j = \text{ReLU}(x_j)$.

Write-back. $Ae_j = vv^\top e_j = (1/\sqrt{2})v = (e_i + e_j)/2$. Multiplied by A and scaled by $s = 2$, this contributes $2 \cdot \frac{1}{2}(e_i + e_j) \text{ReLU}(x_j) = (e_i + e_j) \text{ReLU}(x_j)$ to $F(x) - x$. Therefore $F(x)_i =$

$x_i + \text{ReLU}(x_j)$ and $F(x)_j = x_j + \text{ReLU}(x_j)$, while $F(x)_m = x_m$ for $m \notin \{i, j\}$. On the slice $\{x_j \geq 0\}$ this specializes to $F(x)_i = x_i + x_j$ and $F(x)_j = 2x_j$. $\square$

Theorem 47 (Grassmann RIPR, $c_{\text{prim}} = 1$). *Let $d \geq 2$ and $1 \leq k \leq d - 1$. $\text{Gr}(k, d)$ -constrained ResNets with two scalar multipliers (s, t) and unconstrained readout are $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$, with zero depth overhead.*

Proof. P3: Lemma 45 ($c_{\text{prim}} = 1$). P4: Lemma 46 ($c_{\text{prim}} = 1$). Universal approximation: Theorem 5. $\square$

Why the inner scalar multiplier t is essential for RIH. The PSD constraint on Grassmann projectors forces every diagonal entry $B_{ii} \geq 0$, so the same-register P3 hinge with $\alpha < 1$ (negative slope) requires an inner scalar multiplier $t < 0$ to provide the sign flip. Proposition 65 (Appendix B.6) proves this rigorously: no single Grassmann block with $t = 1$ can realize an exact register-isolated negative-slope hinge. The CAFC construction (Theorem 66, Appendix B.6) bypasses the PSD obstruction by using a rotated frame and recovers scalar-output $L^p(K)$ -universality on every compact $K \subset \mathbb{R}^d$ with an unconstrained terminal readout using outer scalar multiplier s alone.

Verification. Lemma 45 ($c_{\text{prim}} = 1$ for P3 with both s, t) + Lemma 46 ($c_{\text{prim}} = 1$ for P4) + Theorem 5 give $\text{Gr}(k, d)$ -constrained ResNet $L^p(K)$ -universality on every compact $K \subset \mathbb{R}^d$ with both scalar multipliers and unconstrained readout, with zero depth overhead. The same ResNet class with *outer scalar multiplier s alone* also achieves scalar-output $L^p(K)$ -universality with an unconstrained terminal readout via Theorem 66 (CAFC). The Stiefel-factor implementation $\Pi = QQ^\top$ with $Q \in V(k, d)$ reduces per-layer parameter count from $2d^2 + d + 2$ to $2dk + d + 2$, a fractional savings of $(d - k)/d$ (e.g., 69% at $k = 160, d = 512$).

Corollary 48 (Fixed-rank PSD by Grassmann inclusion). *Let $d \geq 2$ and $1 \leq k \leq d - 1$. The $\text{PSD}(k, d)$ -constrained residual class is $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$ with both scalar multipliers s, t via the RIH route, and scalar-output $L^p(K)$ -universal with outer scalar multiplier s alone via the CAFC route.*

Proof. Every $\Pi \in \text{Gr}(k, d)$ is symmetric positive semidefinite and has rank k , hence $\text{Gr}(k, d) \subset \text{PSD}(k, d)$. Therefore every $\text{Gr}(k, d)$ -constrained block and network constructed above is also a valid $\text{PSD}(k, d)$ -constrained block and network. The RIH statement follows from Theorem 47, and the s -only scalar-output statement follows from Theorem 66. $\square$

Remark 49 (PSD inheritance is not absorption). The preceding corollary uses inclusion inheritance, not absorption: multiplying a projector by an arbitrary signed scalar can leave the PSD cone, and zero scaling drops rank, so the scalar multipliers remain external.

B.5.3 Fixed-spectrum symmetric orbits ($c_{\text{prim}} = 1$, both s, t via RIH)

Proposition 50 (Fixed-spectrum orbit, $c_{\text{prim}} = 1$). *Let $d \geq 2$ and let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ have at least two distinct eigenvalues, at least one nonzero. Then $\text{Sym}_\Lambda(d) = \{Q\Lambda Q^\top : Q \in O(d)\}$ satisfies RIPR with $c_{\text{prim}} = 1$, using scalar multipliers s, t , and is $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$ by Theorem 5.*

Proof. Fix a compact live state set K_{st} and let $M = \sup_{K_{\text{st}}} \|x\|_\infty$. For P3 on coordinate i , choose a nonzero eigenvalue λ and rotate its eigenvector to e_i . With $A = B = W = Q\Lambda Q^\top$, dead-zone biasing (Lemma 32) kills all inactive ReLU coordinates, while $We_i = \lambda e_i$ gives

$$F_i(x) = x_i + s\lambda \text{ReLU}(t\lambda x_i + b_i).$$

Taking $t = a/\lambda, s = \gamma/\lambda, b_i = b$ realizes the signed P3 update $x_i \mapsto x_i + \gamma \text{ReLU}(ax_i + b)$ exactly.

For P4, pick distinct eigenvalues $\lambda_1 \neq \lambda_2$ with $\lambda_1 \neq 0$. Rotate B so that $Be_j = \lambda_1 e_j$, bias all coordinates except j into the dead zone, and set $t = 1/\lambda_1$. Rotate A by $\pi/4$ in the (e_i, e_j) plane between the λ_1, λ_2 eigenspaces; then $A_{ij} = (\lambda_1 - \lambda_2)/2$ and $A_{mj} = 0$ for $m \notin \{i, j\}$. With $s = 1/A_{ij}$, the block gives $F_i(x) = x_i + \text{ReLU}(x_j)$ on K_{st} , hence the P4 update whenever $x_j \geq 0$, while all registers other than i, j are unchanged. This verifies RIPR with one block per primitive. $\square$

Remark 51 (Fixed-spectrum orbits and the CAFC-fold condition). For general $\text{Sym}_\Lambda(d)$ with distinct nonzero eigenvalues, the literal CAFC-fold condition remains open. The fixed-spectrum eigenvector gadget above still gives RIH universality using s, t .

B.5.4 Oblique projectors $E(d, k)$ ($c_{\text{prim}} = 1$, s -only RIH via ε -perturbation; s -only via CAFC)

Proposition 52 (Oblique projectors, $c_{\text{prim}} = 1$, outer scalar s). *For $0 < k < d$, the oblique projector family $E(d, k) = \{P : P^2 = P, \text{rank } P = k\}$ satisfies approximate RIPR with $c_{\text{prim}} = 1$ using the outer scalar multiplier s ; hence $E(d, k)$ -constrained ResNets are $L^p(K)$ -universal on every compact $K \subset \mathbb{R}^d$ by Corollary 22.*

Proof. For signed P3 on register i , choose $j \neq i$ and $S \subset \{1, \dots, d\} \setminus \{i, j\}$ with $|S| = k - 1$. Let $A = e_i e_i^\top + \sum_{m \in S} e_m e_m^\top$, so $Ae_i = e_i$ and $Ae_j = 0$. Put in the (i, j) block of B the rank-one idempotent

$$\begin{pmatrix} a & \varepsilon \\ a(1-a)/\varepsilon & 1-a \end{pmatrix},$$

and keep the same diagonal entries on S . After dead-zone biasing, the write map annihilates the auxiliary j component, giving

$$F_i(x) = x_i + \gamma \text{ReLU}(ax_i + \varepsilon x_j + b), \quad F_\ell(x) = x_\ell \ (\ell \neq i),$$

with $s = \gamma$. On the compact live set, choose $|\varepsilon|$ small enough that the live-coordinate error is below the budget in Corollary 22; the apparent $1/\varepsilon$ coefficient in B is killed by $Ae_j = 0$.

For P4, take $B = e_j e_j^\top + \sum_{m \in S} e_m e_m^\top$ and $A = e_i(e_i + e_j)^\top + \sum_{m \in S} e_m e_m^\top$. Both are rank- k idempotents, $Ae_j = e_i$, and dead-zone biasing leaves only $\text{ReLU}(x_j)e_j$, so the block realizes $F_i = x_i + \text{ReLU}(x_j)$ and fixes all registers outside $\{i, j\}$. $\square$

Proposition 53 ($E(d, k)$ satisfies the CAFC-fold condition). *Let $0 < k < d$. The oblique projector manifold $E(d, k)$ satisfies the CAFC-fold condition. Consequently, $E(d, k)$ -constrained ResNets with the outer scalar multiplier s satisfy Theorem 28.*

Proof. Fix a CAFC write direction w_i , a read direction $v \perp w_i$, and a compact set K ; choose an index j with $v_j \neq 0$ and $(w_i)_j \neq 0$. For the write side, choose biorthogonal columns/covectors $\widehat{U}, \widehat{V}$ with first pair $\widehat{u}_1 = w_i/(w_i)_j$, $\widehat{p}_1 = e_j$ and the remaining $k - 1$ pairs supported in $e_j^\perp \cap w_i^\perp$. Then $A = \widehat{U}\widehat{V}^\top$ is a rank- k idempotent and $Ae_j = w_i/(w_i)_j$.

For the read side, choose U, V with $V^\top U = I_k$ and $\text{row}_j(UV^\top) = \nu v^\top$ for some $\nu > 0$: if $v_j > 0$ take $u_1 = e_j$, and if $v_j < 0$ take $u_1 = -e_j - Mv$ with $|v_j|/\|v\|^2 < M < 1/|v_j|$; the remaining pairs live in $e_j^\perp \cap v^\perp$. Dead-zone biasing kills all ReLU components except j , leaving a positive read coefficient and a write vector parallel to w_i . Lemma 25 verifies the CAFC fold. $\square$

B.5.5 $\mathcal{J}_n$ ($c_{\text{prim}} = 1$, partner-pair RIH with both s, t ; macro-CAFC variant)

Proposition 54 ($\mathcal{J}_n$ exact RIH). *For even $d = 2n \geq 4$, $\mathcal{J}_n$ -constrained residual blocks satisfy exact RIPR with primitive cost $c_{\text{prim}} = 1$, using both scalar multipliers s, t . Theorem 5 gives scalar-output $L^p(K)$ -density on every compact $K \subset \mathbb{R}^{2n}$ for $1 \leq p < \infty$.*

Proof.

Setup. For even $d = 2n$ with $n \geq 2$, set $\mathcal{J}_n = \{J \in \text{SO}(2n) : J^2 = -I_{2n}\}$ (complex structures on $\mathbb{R}^{2n}$ compatible with the standard inner product). For odd d , $\det(J)^2 = \det(J^2) = \det(-I_d) = (-1)^d$ together with $|\det(J)| = 1$ forces d even, so $\mathcal{J}_n = \emptyset$. For $d = 2$ the manifold is the two-point set $\{J_0, -J_0\}$ which is too small to implement P4 (no cross-coordinate coupling); RIH-based universality therefore requires $n \geq 2$.

We exhibit the standard base point

$$J_0 = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix} \in \mathcal{J}_n.$$

Then $J_0^\top = -J_0$ (skew), $J_0^\top J_0 = -J_0^2 = I_{2n}$ (orthogonal), and $\det(J_0) = (i \cdot (-i))^n = 1$, so $J_0 \in \text{SO}(2n)$.

Partner pairing and signs. For $i \in \{1, \dots, 2n\}$, define the symplectic partner $\hat{i} = i + n$ (if $i \leq n$) or $\hat{i} = i - n$ (if $i > n$), and the sign sequences

$$\epsilon_i = \begin{cases} -1 & i \leq n, \\ +1 & i > n, \end{cases} \quad \delta_i = \begin{cases} +1 & i \leq n, \\ -1 & i > n. \end{cases}$$

Writing $x = (a, b) \in \mathbb{R}^n \oplus \mathbb{R}^n$ gives $J_0 x = (b, -a)$, from which the two key identities follow directly:

$$(J_0 x)_i = \epsilon_i x_i, \quad (J_0 e_i)_i = \delta_i. \quad (8)$$

Their product satisfies $\epsilon_i \delta_i = (J_0^2)_{ii} = -1$ for every i , so $J_0^2 = -I_{2n}$ holds verifiably on the explicit witness, and the two sign sequences *cancel* in the assembly step below. This anti-involution identity is what makes the partner-pair routing close on i regardless of α .

P3 block (single-coordinate ReLU; both scalar multipliers active). Fix signed-amplitude primitive P3, $x_i \mapsto x_i + \gamma \text{ReLU}(ax_i + b)$, for chosen $\gamma, a, b \in \mathbb{R}$, on a compact set K with $\sup_K \|x\|_\infty \leq M$. Choose

$$A = B = J_0, \quad t = a \epsilon_i, \quad s = \gamma \delta_i,$$

biases $b_{\hat{i}} = b$ and $b_j = -C$ for all $j \neq \hat{i}$, with $C > |t| M \sqrt{2n} + |b| + 1$. The resulting one-block residual $F(x) = x + s A \text{ReLU}(t B x + b)$ is computed coordinate-wise.

Kill step. For $j \neq \hat{i}$, every entry of $J_0 x$ is bounded by $\|J_0 x\|_\infty \leq \|J_0\|_{\text{op}} \|x\|_2 \leq \sqrt{2n} M$, so $(t J_0 x + b)_j \leq |t| M \sqrt{2n} - C < 0$ and $\text{ReLU}(t J_0 x + b)_j = 0$.

Active register. By (8) and $\epsilon_i^2 = 1$,

$$(t J_0 x + b)_i = t \epsilon_i x_i + b = a \epsilon_i^2 x_i + b = a x_i + b.$$

Hence $\text{ReLU}(t J_0 x + b) = \text{ReLU}(a x_i + b) \cdot e_{\hat{i}}$.

Write-back. Using (8) again,

$$J_0 [\text{ReLU}(a x_i + b) e_{\hat{i}}] = \text{ReLU}(a x_i + b) \delta_i e_i.$$

Assembly. Multiplying by the outer scalar multiplier $s = \gamma \delta_i$ gives $s \delta_i = \gamma \delta_i^2 = \gamma$, so

$$F(x)_i = x_i + \gamma \text{ReLU}(a x_i + b), \quad F(x)_j = x_j \quad (j \neq i).$$

This realises signed P3 with one block ($c_{\text{prim}} = 1$) and uses both scalar multipliers s and t .

P4 block (cross-coordinate addition; requires $n \geq 2$). Fix an arbitrary compact state set $K_{\text{st}} \subset \mathbb{R}^{2n}$ with $\sup_{K_{\text{st}}} \|x\|_\infty \leq M$. Fix $i \neq j$. We construct one block satisfying the $\hat{P}_r$ identities of Corollary 22: $F(x)_i = x_i + \text{ReLU}(x_j)$ on all of K_{st} , specializing to $x_i + x_j$ on $K_{\text{st}} \cap \{x_j \geq 0\}$, and $F(x)_m = x_m$ for $m \notin \{i, j\}$. Choose an auxiliary index $r \notin \{i, j\}$; this is possible because $d = 2n \geq 4$, leaving at least two coordinates outside $\{i, j\}$.

Choice of $A, B \in \mathcal{J}_n$. The homogeneous space $\mathcal{J}_n \cong \text{O}(2n)/\text{U}(n)$ acts transitively on ordered orthonormal pairs (u, v) with $u \perp v$ via $Ju = v$, $Jv = -u$. Concretely: extend $\{e_j, e_r\}$ to an orthonormal basis of $\mathbb{R}^{2n}$, declare the pair (e_j, e_r) to be a complex-structure pair, and assign arbitrary complex-structure pairings to the remaining $2n - 2$ basis vectors (possible because $2n - 2$ is even when $n \geq 2$). The resulting block-diagonal B satisfies $Be_j = e_r$, $Be_r = -e_j$, $B^\top = -B$, $B^2 = -I_{2n}$, $\det B = 1$, hence $B \in \mathcal{J}_n$. By the same construction with the pair (e_r, e_i) , choose $A \in \mathcal{J}_n$ with $Ae_r = e_i$ and $Ae_i = -e_r$.

Block parameters. Set $t = 1$, $s = 1$, $b_r = 0$, and $b_m = -C$ for $m \neq r$, with $C > M \sqrt{2n} + 1$.

Read. Since B is orthogonal and skew, $B^\top = B^{-1} = -B$, so $B^\top e_r = -Be_r = e_j$, i.e. $\text{row}_r(B) = e_j^\top$. Thus $(Bx)_r = e_j^\top x = x_j$.

Dead-zone biasing. Since B is orthogonal ($\|B\|_{\text{op}} = 1$), the block-parameters choice $C > M\sqrt{2n} + 1$ above satisfies the hypothesis of Lemma 32, giving $\text{ReLU}(Bx + b)_m = 0$ for all $m \neq r$.

Active register. Without any sign assumption on x_j , $\text{ReLU}(Bx + b)_r = \text{ReLU}(x_j)$, so $\text{ReLU}(Bx + b) = \text{ReLU}(x_j) \cdot e_r$.

Write. $A \cdot (\text{ReLU}(x_j)e_r) = \text{ReLU}(x_j) \cdot Ae_r = \text{ReLU}(x_j) \cdot e_i$. With $s = 1$, the residual update is $\text{ReLU}(x_j) \cdot e_i$, giving

$$F(x)_i = x_i + \text{ReLU}(x_j), \quad F(x)_m = x_m \quad (m \notin \{i, j\}),$$

which specializes to $F(x)_i = x_i + x_j$ on $K_{\text{st}} \cap \{x_j \geq 0\}$. This realises the $\hat{P}_r$ form of P4 with $c_{\text{prim}} = 1$.

Conclusion. Both primitives P3 and P4 are realised with a single residual block on $\mathcal{J}_n$, using both scalar multipliers s, t , and with the write-vector $J_0 e_i = \delta_i e_i$ (resp. $Ae_r = e_i$) genuinely supported on e_i . RIH at $c_{\text{prim}} = 1$ then gives scalar-output $L^p(K)$ -density on every compact $K \subset \mathbb{R}^{2n}$, $1 \leq p < \infty$, with zero depth overhead by the universal RIH $\Rightarrow$ universality theorem. $\square$

Why directional richness fails. The skew-symmetry constraint $J_{ii} = 0$ rules out the diagonal-row construction underlying both directional richness and its projector variant: no $J \in \mathcal{J}_n$ has $\text{row}_i(J) = c e_i^\top$. Universality is recovered through the partner register $\hat{i}$: the inner scalar multiplier t reads x_i via $(J_0 x)_i = \epsilon_i x_i$, and J_0 writes back to i via $J_0 e_i = \delta_i e_i$. The product $\epsilon_i \delta_i = -1$ (i.e. $J_0^2 = -I$) is precisely what cancels the signs uniformly in α .

Macro-CAFC variant. The literal one-block CAFC route fails for $\mathcal{J}_n$ because every J has zero diagonal (excluding j from $\text{row}_j(B)$). The self-contained macro-CAFC construction (Theorem 55) inflates each ideal fold to $O(d^2)$ $\mathcal{J}_n$ blocks via coordinate transvections generating $SL(d, \mathbb{R})$, and uses only the outer scalar multiplier s in the constituent blocks.

Theorem 55 ($\mathcal{J}_n$ exact macro-CAFC folds). *Let $d = 2n \geq 4$. For every rank-one ReLU fold*

$$\mathcal{F}(x) = x + c w \text{ReLU}(u^\top x + \beta)$$

and every nonempty compact state set $X \subset \mathbb{R}^d$, there is a finite $\mathcal{J}_n$ -constrained block program using only outer scalar multipliers such that

$$P(x) = \mathcal{F}(x) \quad \text{for every } x \in X.$$

For $c \neq 0$ and $u, w \neq 0$, the program uses $4N + 1$ blocks, where N is the transvection length of one chosen $SL(d, \mathbb{R})$ decomposition of the conjugating matrix. In particular the block count is $O(d^2)$ per ideal fold.

Proof. We use three elementary facts. First, every ordered orthonormal pairing $(u_h, v_h)_{h=1}^n$ defines

$$J_{\mathcal{P}} = \sum_h (v_h u_h^\top - u_h v_h^\top) \in \mathcal{J}_n, \quad J_{\mathcal{P}} u_h = v_h, \quad J_{\mathcal{P}} v_h = -u_h.$$

Second, for every transvection $T_{pq}(a)x = x + ax_q e_p$ and compact X , choose $r \notin \{p, q\}$, $A \in \mathcal{J}_n$ with $Ae_r = e_p$, and $B \in \mathcal{J}_n$ with $e_r^\top B = e_q^\top$. On $X_0 = X$ and $X_1 = \{x + s_1(x_q + \beta_1)e_p : x \in X\}$, choose inactive biases by dead-zone biasing and choose $\beta_1, \beta_2 > \sup_X |x_q|$ with

$$s_1 + s_2 = a, \quad s_1 \beta_1 + s_2 \beta_2 = 0.$$

The two blocks then give

$$x \mapsto x + [s_1(x_q + \beta_1) + s_2(x_q + \beta_2)]e_p = x + ax_q e_p$$

on X . The update writes coordinate $p \neq q$, so the second block reads the same x_q ; this is the compact-state bias enlargement used throughout the macro proof.

Third, elementary transvections generate $SL(d, \mathbb{R})$; choosing a finite transvection decomposition of length $N = O(d^2)$ and applying the two-block realization on successive compact images realizes any $L \in SL(d, \mathbb{R})$ on X with $2N$ $\mathcal{J}_n$ blocks. If $a, v \in e_1^\perp$, the pairing construction also gives one $\mathcal{J}_n$

block realizing $y \mapsto y + c a \text{ReLU}(v^\top y + \beta)$ on compact sets: take $Ae_1 = a/\|a\|$, $e_1^\top B = v^\top/\|v\|$, dead-zone the inactive rows, and set $s = c\|a\| \|v\|$.

For a nonzero fold choose $L \in SL(d, \mathbb{R})$ with $Lw, L^{-T}u \in e_1^\perp$; such an L is obtained by taking one row in $w^\perp$ and arranging $L^{-1}e_1 \in u^\perp$, then scaling a remaining row to determinant one. Realize L , the hyperplane fold, and L^{-1} on the successive compact state sets. The composition is exactly

$$L^{-1}(Lx + cLw \text{ReLU}((L^{-T}u)^\top Lx + \beta)) = x + cw \text{ReLU}(u^\top x + \beta).$$

If $u = 0$, the fold is a translation; realize each nonzero coordinate of $c \text{ReLU}(\beta)w$ by the same two-block transvection identity with the x_q -coefficient cancelled. Thus the nonzero fold costs $4N + 1 = O(d^2)$ blocks and uses only outer scalar multipliers. $\square$

B.5.6 Symplectic Grassmannian (*s*-only RIH + macro-CAFC)

For $\text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n})$ with even $d = 2n \geq 4$ and $1 \leq k \leq n - 1$, the elements are symplectic involutions of prescribed trace,

$$\text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n}) = \{X \in \mathbb{R}^{2n \times 2n} : X^\top J_{2n} X = J_{2n}, X^2 = I, \text{tr}(X) = 4k - 2n\},$$

where $J_{2n} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$.

Proposition 56 (Symplectic Grassmannian, $c_{\text{prim}} = 1$, outer scalar s). *Let $n \geq 2$ and $1 \leq k \leq n - 1$. Then $\text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n})$ -constrained ResNets with outer scalar multiplier $s \in \mathbb{R}$,*

$$F(x) = x + s \cdot A \cdot \text{ReLU}(Bx + b), \quad A, B \in \text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n}), b \in \mathbb{R}^{2n},$$

and unconstrained readout, satisfy approximate RIPR (Corollary 22) with primitive cost $c_{\text{prim}} = 1$, and are therefore scalar-output $L^p(K)$ -dense on every compact $K \subset \mathbb{R}^{2n}$, $1 \leq p < \infty$ (zero depth overhead).

Proof. Concretely, writing the active row coefficient as $a = \alpha - 1$, the live update on register i below is $x_i + s \cdot \text{ReLU}((\alpha - 1)x_i + \varepsilon x_j + \beta)$, an approximate signed-amplitude P3 with $|\varepsilon|$ -dependent error on T_{r-1} that is selected after T_{r-1} has been formed; signed-amplitude γ is realized by setting $s = \gamma$.

For P3, choose $A \in \text{Gr}_{\text{Sp}}$ with $Ae_i = e_i$. The companion 4×4 involution block

$$\begin{pmatrix} a & \varepsilon & 0 & 0 \\ (1 - a^2)/\varepsilon & -a & 0 & 0 \\ 0 & 0 & a & (1 - a^2)/\varepsilon \\ 0 & 0 & \varepsilon & -a \end{pmatrix}$$

has the form $\text{diag}(C, C^{-T})$, where

$$C = \begin{pmatrix} a & \varepsilon \\ (1 - a^2)/\varepsilon & -a \end{pmatrix}, \quad C^2 = I_2.$$

Thus the block is symplectic and involutive; its trace is zero, so replacing one positive and one negative symplectic pair preserves the prescribed trace. Embedded on a pair of symplectic coordinate pairs, this block gives $B \in \text{Gr}_{\text{Sp}}$ with active row $(Bx)_i = ax_i + \varepsilon x_j$. Dead-zone biasing on T_{r-1} kills all rows except i , and setting $s = \gamma$ gives

$$F_i(x) = x_i + \gamma \text{ReLU}(ax_i + \varepsilon x_j + b), \quad F_m(x) = x_m \ (m \neq i).$$

Choosing $|\gamma\varepsilon| \sup_{T_{r-1}} |x_j| \leq \eta_r$ gives the live-coordinate error required by Corollary 22. The large inactive entry $(1 - a^2)/\varepsilon$ is harmless because the tolerance is chosen after the compact state set is fixed.

For P4, use signed swaps between non-partner symplectic coordinate pairs: on $\{e_p, e_q, e_{\hat{p}}, e_{\hat{q}}\}$, swap e_p, e_q and swap their partners with the sign preserving the symplectic form, extending by $\pm I$ on the remaining pairs to keep trace $4k - 2n$. If $j \neq \hat{i}$, take B with row $i = e_j^\top$ and A with $Ae_i = e_i$; if $j = \hat{i}$, route through a disjoint pair r with B reading x_j into row r and $Ae_r = e_i$. Dead-zone biasing leaves only the active ReLU coordinate, so the block realizes $F_i = x_i + \text{ReLU}(x_j)$, which is P4 on the source-nonnegative states. The approximate-RIPR corollary completes the proof. $\square$

Remark 57 (GrSp and CAFC routes). Literal one-block CAFC fails for GrSp : $X^\top JX = J$ and $X^2 = I$ imply JX is skew, hence every column and row has a forced zero in its partner coordinate. The macro-CAFC route below removes this by realizing arbitrary folds through transvections and conjugation.

Theorem 58 (GrSp exact macro-CAFC folds). *Let $d = 2n \geq 4$ and $1 \leq k \leq n - 1$. For every rank-one ReLU fold*

$$\mathcal{F}(x) = x + cw \text{ReLU}(u^\top x + \beta)$$

and every nonempty compact state set $K \subset \mathbb{R}^d$, there is a finite $\text{GrSp}(2k, \mathbb{R}^{2n})$ -constrained block program using only outer scalar multipliers such that

$$P(x) = \mathcal{F}(x) \quad \text{for every } x \in K.$$

If $c \neq 0$ and $u, w \neq 0$, the program uses $4N + 1$ residual blocks, where $N = O(d^2)$ is the transvection length of one chosen $SL(d, \mathbb{R})$ normalizing matrix. If $u = 0$, the translation case uses at most $2d$ residual blocks.

Proof. Use two coordinate gadgets. Diagonal selectors $\text{diag}(\epsilon_1, \dots, \epsilon_n, \epsilon_1, \dots, \epsilon_n)$ with exactly k positive symplectic pairs lie in GrSp and provide rows $\pm e_q^\top$. Signed swaps of two non-partner symplectic pairs also lie in GrSp , preserve the trace $4k - 2n$, and can write a chosen coordinate into another. These gadgets give the same two-block transvection identity as in the $\mathcal{J}_n$ proof: choose A, B with $Ae_r = e_p$ and $e_r^\top B = e_q^\top$, choose two active biases above $\sup_K |x_q|$, choose s_1, s_2 with

$$s_1 + s_2 = a, \quad s_1\gamma_1 + s_2\gamma_2 = 0,$$

and dead-zone inactive rows on the compact sets before the first and second block. The composition is exactly $x \mapsto x + ax_q e_p$. Thus any $SL(d)$ map is realized on compact sets with $O(d^2)$ blocks, using only outer scalar multipliers.

Canonical folds are one block: diagonal selectors realize $y \mapsto y + Ce_1 \text{ReLU}(\pm y_1 + \theta)$, and a signed swap plus a selector realizes $y \mapsto y + Ce_1 \text{ReLU}(y_2 + \theta)$. If $u^\top w \neq 0$, choose $L \in SL(d)$ with $Lw = e_1$ and $L^{-T}u = (u^\top w)e_1$; if $u^\top w = 0$, choose $Lw = e_1$ and $L^{-T}u = e_2$. Realize L , the corresponding canonical fold, and L^{-1} on the successive compact sets. The result is the desired fold. The $u = 0$ case is a translation and is handled coordinatewise by the same two-block identity with the x_q -coefficient cancelled. The nonzero fold costs $4N + 1$ blocks with $N = O(d^2)$. $\square$

Corollary 59 (Macro-CAFC for $\mathcal{J}_n$ and GrSp). *The macro-CAFC hypothesis of Theorem 31 holds for $\mathcal{J}_n$ when $d = 2n \geq 4$, and it holds for $\text{GrSp}(2k, \mathbb{R}^{2n})$ when $n \geq 2$ and $1 \leq k \leq n - 1$. In both cases the construction uses only outer scalar multipliers. Consequently, the scalar-output CAFC proof gives $L^1(K)$ -universality with an unconstrained terminal linear readout and $O(d^2)$ constrained blocks per ideal fold for both families.*

Proof. Theorems 55 and 58 realize every rank-one fold required by Definition 30 on every compact state set, for $\mathcal{J}_n$ and GrSp respectively. Applying Theorem 31 to each gives the scalar-output $L^1(K)$ conclusion. Both constructions use blocks of the form $x \mapsto x + sA \text{ReLU}(Bx + b)$ with A, B in the constraint manifold, so only outer scalar multipliers appear and $t = 1$. $\square$

B.5.7 Lagrangian Grassmannian ($c_{\text{prim}} = 1$; s, t RIH via symmetric-unitary witnesses, or s -only CAFC)

We use the realified compact model

$$M(P, Q) = \begin{pmatrix} P & -Q \\ Q & P \end{pmatrix}, \quad P, Q = P^\top, Q^\top, \quad P^2 + Q^2 = I, \quad PQ = QP,$$

for $\text{LGr}(\mathbb{R}^{2n}) \cong \text{U}(n)/\text{O}(n)$. Its elements are orthogonal and $I_{2n} \in \text{LGr}$.

Proposition 60 (Lagrangian Grassmannian, $c_{\text{prim}} = 1$). *$\text{LGr}(\mathbb{R}^{2n})$ -constrained ResNets with scalar multipliers $s, t \in \mathbb{R}$ and unconstrained readout are $L^p(K)$ -dense on every compact $K \subset \mathbb{R}^{2n}$, $1 \leq p < \infty$, with $c_{\text{prim}} = 1$.*

Proof. For signed P3, take $A = B = I_{2n} \in \text{LGr}$. Dead-zone biasing all inactive coordinates gives $F_i(x) = x_i + s \text{ReLU}(tx_i + b_i)$ and $F_m(x) = x_m$ for $m \neq i$; setting $s = \gamma$, $t = a$, $b_i = b$ realizes $x_i \mapsto x_i + \gamma \text{ReLU}(ax_i + b)$ exactly.

For P4, take $B = I_{2n}$, bias all coordinates except j into the dead zone, and choose $A \in \text{LGr}$ with $Ae_j = e_i$. Such A is given by the following realification witnesses, with u_a the standard basis, $E_a = u_a u_a^\top$, $T_{ab} = u_a u_b^\top + u_b u_a^\top$, and P_{ab} the transposition matrix:

(i, j)	(P, Q)
$(a, b), (n+a, n+b), a \neq b$	$(P_{ab}, 0)$
$(a, n+b), a \neq b$	$(I - E_a - E_b, -T_{ab})$
$(n+a, b), a \neq b$	$(I - E_a - E_b, T_{ab})$
$(a, n+a)$	$(I - E_a, -E_a)$
$(n+a, a)$	$(I - E_a, E_a)$

Each row satisfies $P^2 + Q^2 = I$ and $PQ = QP = 0$, and direct substitution shows $M(P, Q)e_j = e_i$. Therefore $F_i = x_i + \text{ReLU}(x_j)$ and all registers outside $\{i, j\}$ are fixed, which verifies P4. $\square$

Lemma 61 (Symmetric-unitary column completion). *For every unit $z \in \mathbb{C}^n$ there exists a symmetric unitary $U_z \in \mathbb{C}^{n \times n}$ with $U_z e_1 = z$.*

Proof. For $n = 1$ take $U_z = [z]$. For $n \geq 2$, write $z = (z_1, z_{\text{tail}})$, $r = \|z_{\text{tail}}\|_2$. If $r = 0$, use $\text{diag}(z_1, 1, \dots, 1)$. If $r > 0$, choose $P \in \text{U}(n-1)$ with $Pe_1 = z_{\text{tail}}/r$, set $Q = \text{diag}(1, P)$, and define

$$S_0 = \begin{pmatrix} z_1 & r & 0 \\ r & -\bar{z}_1 & 0 \\ 0 & 0 & I_{n-2} \end{pmatrix}.$$

The leading block is symmetric unitary because $|z_1|^2 + r^2 = 1$ and the two columns are orthogonal. Then $U_z = QS_0Q^\top$ is symmetric unitary and maps e_1 to z . $\square$

Proposition 62 (LGr satisfies literal one-block CAFC-fold). *In the realified compact $\text{U}(n)/\text{O}(n)$ model, LGr realizes every rank-one fold $x \mapsto x + c w \text{ReLU}(u^\top x + \beta)$ on every compact $K \subset \mathbb{R}^{2n}$ using one constrained block and the outer scalar multiplier s only.*

Proof. Normalize $\hat{w} = w/\|w\|$, $\hat{u} = u/\|u\|$ and write $\hat{w} = (p, q)$, $\hat{u} = (r, s')$ in the $\mathbb{R}^n \times \mathbb{R}^n$ split. By Lemma 61, choose symmetric unitaries whose realifications $A, B \in \text{LGr}$ satisfy $Ae_1 = \hat{w}$ and $\text{row}_1(B) = \hat{u}^\top$. Bias every inactive preactivation below zero on K , set $b_1 = \beta/\|u\|$ and $s = c\|w\|\|u\|$, and use ReLU homogeneity. The surviving read coefficient is positive; replacing $\hat{u}$ by $-\hat{u}$ handles the opposite orientation. $\square$

Corollary 63 (LGr via CAFC-fold and the CAFC theorem). *Choosing any totally-nonzero ONB with $w_{2n} = 1/\sqrt{2n}$, Proposition 62 verifies both fold families in Definition 26. Hence Theorem 28 gives $L^1(K)$ -density for LGr-constrained ResNets with the outer scalar multiplier s only and terminal readout.*

Remark 64 (Convention scope). The CAFC proof uses the compact realified $\text{U}(n)/\text{O}(n)$ model of LGr; noncompact “real symmetric symplectic” variants would require a separate verification.

B.6 Single-Block Separation: t is necessary for Grassmann within RIH

RIH and CAFC can disagree on multiplier budget on the same manifold; the Grassmann entry of Table 2 exhibits a sharp such separation. The single-block RIH primitive on $\text{Gr}(k, d)$ requires the inner scalar multiplier t (Proposition 65), but the CAFC route makes it dispensable (Theorem 66).

Proposition 65 (Inner scalar multiplier t necessary for the single-block RIH primitive on $\text{Gr}(k, d)$). *Fix $0 < k < d$, a compact $K \subset \mathbb{R}^d$ with nonempty interior, and a register coordinate i . Let $a := \alpha - 1 < 0$ and assume the P3 hinge is nontrivial on K , i.e., $\Omega := \{x \in \text{int } K : ax_i + \beta > 0\} \neq \emptyset$. Then there are no $A, B \in \text{Gr}(k, d)$, $b \in \mathbb{R}^d$, $s \in \mathbb{R}$ such that the single block with inner scalar multiplier fixed to $t = 1$, $F(x) = x + s \cdot A \cdot \text{ReLU}(Bx + b)$, realizes the register-isolated negative-slope hinge $F(x)_i = x_i + \text{ReLU}(ax_i + \beta)$, $F(x)_m = x_m$ ($m \neq i$) on K . Hence a trainable inner scalar multiplier t is necessary for the single-block RIH primitive on $\text{Gr}(k, d)$.*

The PSD constraint on rank- k projectors forces the diagonal entry $(B)_{ii}$ to be nonnegative; with t held to 1, the active sign on register i is permanently fixed and the negative-slope hinge is unreachable in a single block. This is specific to the single-block construction; CAFC bypasses it.

Theorem 66 (Grassmann fold-and-cut without inner multiplier). *Let $d \geq 2$, $1 \leq k \leq d - 1$, and let $K \subset \mathbb{R}^d$ be compact. Consider residual blocks*

$$F_\ell(x) = x + s_\ell A_\ell \text{ReLU}(B_\ell x + b_\ell), \quad A_\ell, B_\ell \in \text{Gr}(k, d), \quad s_\ell \in \mathbb{R}, \quad b_\ell \in \mathbb{R}^d,$$

with the inner scalar multiplier fixed to $t_\ell = 1$, and allow an unconstrained terminal linear readout $L_{\text{out}} : \mathbb{R}^d \rightarrow \mathbb{R}$. Then the scalar-output class $\{L_{\text{out}} \circ F_N \circ \dots \circ F_1 : N \in \mathbb{N}\}$ is dense in $L^1(K)$, and in $L^p(K)$ for every $p \in [1, \infty)$ via the truncation argument of Remark 20.

Together, Proposition 65 and Theorem 66 show that the inner scalar multiplier is a *mechanism-specific* requirement of single-block RIH on Grassmann, not an architectural necessity: routing through a rotated frame removes it.

Proof of Proposition 65. Suppose for contradiction such a block exists, and write

$$h(x) := \text{ReLU}(ax_i + \beta), \quad y(x) := \text{ReLU}(Bx + b), \quad \Delta(x) := F(x) - x = sAy(x).$$

Step 1 (exact register isolation). The hypotheses give $\Delta(x) = h(x)e_i$ for all $x \in K$. Since $\Omega \neq \emptyset$, choose $x^0 \in \Omega$; then $h(x^0) > 0$, so $\Delta(x^0) = h(x^0)e_i$ is a nonzero vector in $\text{range}(A)$. Hence $e_i \in \text{range}(A)$.

Step 2 (write-side projector fixes e_i). Because $A \in \text{Gr}(k, d)$, $A = A^\top = A^2$ is the orthogonal projector onto $\text{range}(A)$. Since $e_i \in \text{range}(A)$, $Ae_i = e_i$, and by symmetry $e_i^\top A = e_i^\top$. Consequently for every $z \in \mathbb{R}^d$: $(Az)_i = e_i^\top Az = e_i^\top z = z_i$.

Step 3 (reduction to the i -th read coordinate). Taking the i -th component of $\Delta(x) = sAy(x)$ and using Step 2:

$$h(x) = \Delta_i(x) = s y_i(x) = s \text{ReLU}(e_i^\top Bx + b_i) \quad \text{for every } x \in K.$$

Since $h > 0$ on Ω and $\text{ReLU} \geq 0$, we have $s > 0$. On Ω the i -th preactivation is positive and the ReLU is linear, giving $s(e_i^\top Bx + b_i) = ax_i + \beta$ for all $x \in \Omega$. Equality of affine functions on a nonempty open subset of $\mathbb{R}^d$ implies equality of coefficients; in particular, $sB_{ii} = a$. Since $s > 0$ and $a < 0$, $B_{ii} < 0$.

Step 4 (Grassmann PSD obstruction). But $B \in \text{Gr}(k, d)$ is an orthogonal projector ($B = B^\top = B^2$), so

$$B_{ii} = e_i^\top B e_i = e_i^\top B^2 e_i = \|B e_i\|^2 \geq 0,$$

contradicting $B_{ii} < 0$. This contradiction proves the claim. $\square$

Proof of Theorem 66 (Grassmann fold-and-cut)

Proof of Theorem 66. We show that $\text{Gr}(k, d)$ satisfies the cross-axis fold-and-cut condition for $d \geq 2$, $k \in \{1, \dots, d - 1\}$, on every compact $K \subset \mathbb{R}^d$, then apply the CAFC machinery of §B.3.

Set $w_d = \mathbf{1}/\sqrt{d}$ (unit vector with all entries positive) and complete it to a totally-nonnzero orthonormal basis $\{w_1, \dots, w_{d-1}, w_d\}$. The set of orthonormal completions with some zero entry is a proper algebraic subvariety in the Haar measure on the compatible orthogonal frames, hence measure zero, so totally-nonnzero completions exist. It suffices to verify the two oriented fold families: writing w_i ($i < d$) reading $v = w_d$, and writing w_d reading any unit $v \in w_d^\perp$. Fix one requested (w_i, v) pair and a compact K .

Step 1 (surviving index with $\nu > 0$). By the totally-nonnzero property, $(w_i)_j \neq 0$ for all j . If $v = w_d$, every $v_j = (w_d)_j > 0$ so any j works. If $v \in w_d^\perp$ and $v \neq 0$, then $\sum_j v_j (w_d)_j = 0$ with all $(w_d)_j > 0$, so v has at least one positive standard-basis coordinate; pick such j , giving $\nu = v_j / \|v\|^2 > 0$.

Step 2 (write projector). Set $A = w_i w_i^\top \in \text{Gr}(1, d)$; then $Ae_j = (w_i)_j \cdot w_i$.

Step 3 (read projector). Set $B = vv^\top / \|v\|^2 \in \text{Gr}(1, d)$; then $\text{row}_j(B) = v_j \cdot v^\top / \|v\|^2$ is proportional to $v^\top$ with coefficient $\nu = v_j / \|v\|^2 \neq 0$.

Step 4 (dead-zone biasing). For $m \neq j$, the preactivation $(Bx)_m = \|v\|^{-2} v_m (v^\top x)$ satisfies $|(Bx)_m| \leq \|v\|^{-2} \|v\|^2 \sup_K \|x\| = \sup_K \|x\|$ by Cauchy–Schwarz. Choose $b_m < -(\sup_K \|x\| + 1)$ so that $(Bx + b)_m < 0$ on K , hence $\text{ReLU}((Bx + b)_m) = 0$. Only the j -th component survives.

Step 5 (assembly). Since $\text{sgn}(\nu) = +1$, the constrained block $F(x) = x + sA \text{ReLU}(Bx + b)$ realizes

$$F(x) = x + c \cdot w_i \cdot \text{ReLU}(v^\top x + \beta)$$

with $c = s \cdot (w_i)_j \cdot v_j / \|v\|^2$ and $\beta = b_j \|v\|^2 / v_j$. The map $(s, b_j) \mapsto (c, \beta)$ is bijective.

Step 6 (rank- k embedding). For $k \geq 2$ given w_i and v , build rank- k projectors by padding: $\tilde{A} = w_i w_i^\top + \sum_{m=2}^k q_m q_m^\top \in \text{Gr}(k, d)$ with $q_m \perp w_i$, $(q_m)_j = 0$ (possible since $w_i^\perp \cap e_j^\perp$ has dimension $d - 2 \geq k - 1$ when $k \leq d - 1$); the padding contributes nothing to column j , so $\tilde{A}e_j = (w_i)_j w_i$ as before. Similarly pad $\tilde{B} = vv^\top / \|v\|^2 + \sum_{m=2}^k r_m r_m^\top$ with $r_m \perp v$, $r_m \perp e_j$; orthogonality $r_m \perp v$ ensures $\tilde{B}$ is a valid orthogonal projector ($\tilde{B}^2 = \tilde{B}^\top = \tilde{B}$), and $r_m \perp e_j$ ensures $\text{row}_j(\tilde{B}) = v_j v^\top / \|v\|^2$ unchanged. Dead-zone biasing works identically on the extra coordinates.

The CAFC-fold condition (Definition 26) is therefore satisfied for $\text{Gr}(k, d)$, $d \geq 2$, $1 \leq k \leq d - 1$. Applying Theorem 28 yields scalar-output $L^1(K)$ -density of $\text{Gr}(k, d)$ -constrained ResNets with outer scalar multiplier s only (no inner scalar multiplier t), with an unconstrained terminal linear readout. $\square$

C Experimental Details

Scope of finite-scale studies. The CIFAR-10 and nanoGPT studies in this appendix examine regularization and optimization behavior in related constrained models, complementing the controlled synthetic diagnostic in Section 5. Tables report sample mean $\pm$ sample standard deviation over the seeds shown. The CIFAR-10 C-series has row-dependent coverage: primary completed rows use $n = 5$, C4 uses $n = 4$ from three complete pickles plus an old seed-42 partial trajectory, C13 has $n = 7$ compatible completed runs, and diverged C3/C5 are reported only as failure modes. The nanoGPT-scale tables use $n = 2$ for the 20,000-step convergence controls and $n = 4$ for the low-rank panels.

Compute resources. All training experiments used single-node SLURM jobs. The synthetic Lipschitz-separation arrays used up to four scheduler-assigned GPUs, 16 CPU cores, and 12–16 GB host memory; the main T1–T7 arrays used 2-hour allocations, while the s_0 sweep and intervention/factorial diagnostics used 20–30 minute allocations. The direct CAFC rate diagnostic is a CPU-only deterministic NumPy grid evaluation with no training. The S1q augmented-width unit diagnostic used CPU-local PyTorch runs, 5,000 iterations, batch size 256, and about 2–3 minutes per primary seed. The CIFAR-10 C-series used one GPU per condition/seed, 4 CPU cores, 16 GB host memory, and 6–12 hour allocations; primary arrays used H100/H200 GPUs, with A100 allowed in later completion runs. nanoGPT resources are reported in §C.5.

C.1 CIFAR-10 setup (C-series, 400 epochs)

8-layer constant-width ReLU MLP at width 512, dropout 0.05, biases enabled, and no residual skip connections. The C-series therefore tests finite-budget regularization effects of hidden square-weight constraints, not the residual architecture used in the universality theorems. Trained on the standard CIFAR-10 split [41] (50,000 training images, 10,000 test images, per-channel mean/std normalization, with RandomCrop(32, padding=4) and random horizontal flip augmentation). AdamW, batch size 256, base learning rate 10^{-3} , cosine schedule with 5-epoch linear warmup, 400 epochs, gradient clipping 1.0. Weight decay 0 on square block weights and $5 \cdot 10^{-4}$ on rectangular input/output projections. Rank target $r = 160$ (31% of 512). C0–C12 use seeds $\{11, 31, 42, 91, 137\}$ where the condition completed; C13 uses all compatible completed seeds $\{7, 11, 31, 42, 91, 123, 137\}$.

C.2 CIFAR-10 C-series main results

Table 4 reports the C-series ladder. All rows use the same non-residual MLP and differ only in the rule applied to each hidden 512×512 square weight after an optimizer step, or in the factorization used to generate that weight.

Table 4: CIFAR-10 C-series, 400 epochs. Entries are mean $\pm$ sample sd over the indicated seeds; for symmetrized $UV^\top$ rows, Rank gives factor rank. † C4 combines three complete saved-pickle runs with the seed-42 partial trajectory through epoch 372; final means use the last available value for that partial run. C3 and C5 diverged after early best snapshots; C5’s terminal failure was eigendecomposition nonconvergence after loss explosion.

Cond.	Form	Rank	n	Best test %	Final test %
C0	dense AdamW	full	5	58.60 $\pm$ 0.33	58.23 $\pm$ 0.45
C1	symmetrized dense	full	4	57.89 $\pm$ 0.12	57.50 $\pm$ 0.16
C2	polar factor $UV^\top$	full	4	61.58 $\pm$ 0.44	61.45 $\pm$ 0.37
C3	truncated SVD, free σ_i	160	diverged	–	–
C4	partial isometry $U_r V_r^\top$	160	4 †	66.15 $\pm$ 0.33	66.04 $\pm$ 0.28
C5	sym. eigentrunc., free λ_i	160	diverged	–	–
C6	projector $Q_r Q_r^\top$	160	5	66.95 $\pm$ 0.12	66.79 $\pm$ 0.14
C7	sym. $UV^\top$, lr 10^{-3}	factor 160	5	63.64 $\pm$ 0.24	63.41 $\pm$ 0.32
C8	sym. $UV^\top$, lr $2 \cdot 10^{-3}$	factor 160	5	63.39 $\pm$ 0.39	63.26 $\pm$ 0.39
C9	Stiefel $QQ^\top$, lr 10^{-3}	160	5	66.90 $\pm$ 0.06	66.68 $\pm$ 0.15
C10	Stiefel $QQ^\top$, lr $2 \cdot 10^{-3}$	160	5	66.67 $\pm$ 0.14	66.54 $\pm$ 0.16
C11	sym. $UV^\top$, lr 10^{-3}	factor 80	4	63.25 $\pm$ 0.20	63.06 $\pm$ 0.32
C12	sym. $UV^\top$, lr $2 \cdot 10^{-3}$	factor 80	4	63.59 $\pm$ 0.33	63.45 $\pm$ 0.31
C13	free $QQ^\top$ (no projection)	160	7	63.42 $\pm$ 0.38	63.28 $\pm$ 0.34

Observation. C4, C6, C9, and C10 form the top tier, showing that bounded rank-160 hidden-square weights are the stable high-performing variants. C7/C8 versus C11/C12 shows that the symmetric $UV^\top$ controls do not gain from the larger rank(W) ≤ 320 ceiling, and C13 drops by roughly 3.5 percentage points relative to C9, isolating the effect of enforcing $Q \in V(160, 512)$ rather than merely using a $QQ^\top$ factorization. The C-series therefore supports the qualitative claim that the projector constraint is the useful finite-budget inductive bias in this MLP; it is not evidence for the asymptotic universality theorem.

C.3 Synthetic Lipschitz separation experiments (full)

Architecture and training. 8-block residual stack at width 128, rank fraction $r/d = 0.5$ (so $r = 64$). Both input and output projections are constrained to $V(k, d)$ so the nonexpansive obstruction applies end-to-end pointwise; unlike the CIFAR-10 sweep, whose rectangular input/output layers are unconstrained and admit escape-from-1-Lipschitz at a global level. **Initial T1–T7 protocol:** Adam (lr 10^{-3} , cosine schedule, no warmup), batch 1024 fresh-sampled each epoch, 2,000 epochs, three seeds {42, 123, 7}. **Additional s_0 sweep + scalar multiplier-intervention re-runs:** AdamW (same lr / batch), cosine schedule with 5% linear warmup, plus added diagnostics (per-block scalar multiplier-gradient norms, e_1 -directional Lipschitz, constant-baseline and Lip-1-floor reference MSEs in JSON). The two protocols are otherwise identical.

Targets and Lipschitz floors. Lip-2 linear ($y = 2x_1$); sawtooth $k = 4$ on $[0, 1]$, Lipschitz constant $2k = 8$; Lip-0.94 sinusoid. The floor calculation is specific to these centered/scaled one-dimensional targets. The L^2 -optimal 1-Lipschitz fit reduces, after centering, to choosing a slope $|\alpha| \leq 1$ along the target’s support, giving MSE $(1 - 1/L)^2 \sigma_y^2$ where L is the target’s Lipschitz constant and $\sigma_y^2 = \text{Var}(y)$ on the support. This is not a floor formula for general Lip- L targets; it relies on the targets being one-dimensional and the fit class being 1-Lipschitz affine along the support. For our targets: *Lip-2 linear*: $\sigma_y^2 = 1/3$, floor $(1/2)^2 \cdot 1/3 = 1/12 \approx 0.083$; *sawtooth* $k = 4$ ($L = 8$): $\sigma_y^2 = 1/12$, floor $(7/8)^2 \cdot 1/12 = 49/768 \approx 0.064$. The constant-fit baselines are σ_y^2 themselves ($1/3$ and $1/12$).

Conditions and results.

Optimization gap and scalar multiplier-initialization sweep. T4’s failure is not an expressivity issue: an unconstrained- Q control with the same architecture (T7) fits the Lip-8 target with MSE $3 \cdot 10^{-4}$, and a tuned-init Grassmann run ($s_0 = 1.0$) recovers MSE 0.007 (next paragraph). The default $s_0 = 0.1$ leaves the scalar multipliers around $s \approx 0.13$ throughout training, and the model plateaus at the constant-fit basin (MSE $\approx 1/12$). The product-Lip needed for the Lip-8 target under uniform scalar multipliers is $(1 + s_0)^8 = 8 \Rightarrow s_0 \geq L^{1/D} - 1 \approx 0.297$ for $L = 8$, $D = 8$. The empirical robust-escape threshold is $\sim 3\times$ that:

Table 5: Synthetic Lipschitz-separation results (default scalar multiplier $s_0 = 0.1$). **prod. ub** is $\prod_l(1 + |s_l|) \cdot \prod_l \|A_l\|_{\text{op}} \|B_l\|_{\text{op}}$; **max layer** is the maximum per-layer operator norm and verifies constraint enforcement (Grassmann rows pinned at 1.0001, the Stiefel-projection floor). T1/T3 confirm the nonexpansive obstruction: prod. ub binds at 1.0008, MSE plateaus at the constant or Lip-1 basin. T4 is the only condition with structural capacity that fails to fit; the diagnostic sweep below characterizes the failure as an initialization-induced optimization gap.

ID	Middle blocks	Target	MSE (mean $\pm$ std, 3 seeds)	emp. Lip	prod. ub	max layer	Verdict
T1	Grass. comp. only	Lip-2 linear	0.135 ± 0.006	0.65	1.0008	1.0001	FAIL (predicted)
T2	Grass. resid.+s	Lip-2 linear	$3 \cdot 10^{-4} \pm 5 \cdot 10^{-4}$	1.79	4.0	1.0001	SUCCEED
T3	Grass. comp. only	sawtooth $k=4$	0.083 ± 0.002	0.017	1.0008	1.0001	FAIL (predicted)
T4	Grass. resid.+s	sawtooth $k=4$	0.081 ± 0.002	0.15	2.68	1.0001	FAIL (opt. gap)
T5	Uncon. comp. only	sawtooth $k=4$	$4 \cdot 10^{-4} \pm 1 \cdot 10^{-4}$	1.47	9596	3.25	SUCCEED (control)
T6	Grass. comp. only	Lip-0.94 sin	0.0037 ± 0.0006	0.14	1.0008	1.0001	SUCCEED (sanity)
T7	Uncon. resid.+s	sawtooth $k=4$	$3 \cdot 10^{-4} \pm 1 \cdot 10^{-4}$	1.41	42090	5.85	SUCCEED (control)

s_0	MSE (mean $\pm$ std, 3 seeds)	e_1 -Lip	s_{final}	prod. ub	Regime
0.10 (default)	0.0810 ± 0.002	1.29	0.13	2.7	stuck (constant basin)
0.50	0.0642 ± 0.021	5.49	0.71	87	stochastic transition (1/3 escape)
0.75	0.0280 ± 0.004	8.96	1.02	273	robust escape (3/3)
1.00	0.0071 ± 0.007	9.54	1.19	528	full success
2.00	0.0141 ± 0.023	9.77	1.90	5062	high-variance overshoot

Diagnostic columns (e_1 -Lip, s_{final} , prod. ub) report the seed-42 representative value at each s_0 ; for $s_0 \in \{0.50, 0.75, 1.00, 2.00\}$ the per-seed diagnostic values agree within $\pm 10\%$ except for $s_0 = 0.50$ (stochastic transition: 1/3 seeds escape) where the seed-42 “stuck” regime is reported. Full per-seed diagnostics will be released with the public code repository upon acceptance.

The transition between $s_0 = 0.5$ and $s_0 = 0.75$ is sharp: at $s_0 = 0.5$, only 1/3 seeds escape the constant basin; at $s_0 = 0.75$, 3/3 do. Diagnostic 1D slices along x_1 show that the failed seeds output an essentially flat function (amplitude ≈ 0.15 , single monotone bump), whereas the escaped seeds recover ~ 7 of 8 sawtooth peaks (amplitude ≈ 1.02 , ~ 7 sign changes).

Scalar multiplier-intervention experiment. A separate 9-seed experiment trains T4 from $s_0 = 0.1$ for 1,000 epochs (well past the MSE-0.08 plateau), then multiplies all scalar multiplier values by $k \in \{2, 5, 10\}$, resets Adam moments for the scalar multiplier parameters, and continues training for another 1,000 epochs. All 9/9 runs return to MSE ≈ 0.08 within a few hundred epochs. At factor $k = 10$, scalar multipliers stay at $s \approx 1.18$ (only slightly contracted from the boost), e_1 -Lip reaches ~ 10 (matching the Lip-8 target with overshoot), yet final MSE returns to the constant basin.

A factorial reset experiment (15 runs, varying which subset of {scalar multipliers, A , B , input projection, output projection, biases} is reset together) shows that *only* the full reset (everything except the trajectory) gives partial escape (MSE 0.069); no single-module reset suffices. Within the tested multiplier rescaling and per-module reset interventions, escape depends on the training path in the joint $(Q_A, Q_B, b, s, t, W_{\text{in}}, W_{\text{out}})$ parameter space rather than on any single component. We do not claim that this basin is a strict local minimum without Hessian analysis. The empirical claim is narrower: no post-hoc multiplier-only repair escapes within these tested interventions, so initializing scalar multipliers correctly remains necessary for trainability at this depth and target Lipschitz.

C.4 Direct CAFC construction at width $d = 2$: rate validation

To verify the CAFC construction numerically, we instantiate it explicitly for $\text{Gr}(1, 2)$ at width $d = 2$, with no training. Each residual block uses rank-1 orthogonal projectors $A = w_p w_p^\top$, $B = w_q w_q^\top \in \text{Gr}(1, 2)$ in the totally-nonnzero ONB $w_2 = (\cos \theta, \sin \theta)$ (all-positive), $w_1 = (-\sin \theta, \cos \theta)$ (totally nonzero) at $\theta = \pi/3$; the inner scalar multiplier is fixed to $t = 1$ (the CAFC route uses only the outer scalar multiplier s); the bias is set by Lemma 32 (dead-zone biasing) so a single coordinate of $Bx + b$ survives ReLU on a compact ball. We approximate the read-direction integral $g(y)$ on $y \in [0, 1]$ by feeding $x_{\text{init}} = (0, y)$ through N such blocks and reading $L_{\text{out}}(\text{state}) = g(0) + w_1^\top (\text{state} - x_{\text{init}})$. The construction realizes the canonical N -piece CPWL interpolant of g exactly, as a sum of N ReLU ridges with breakpoints at $y_k = k/N$. We measure the L^1 error against the ground truth g on an 8192-point grid for two targets:

- **Smooth** (C^∞): two-scale $g(y) = 0.5 + 0.15 \sin(2\pi y) + 0.10 \sin(10\pi y)$ on $[0, 1]$.
- **BV staircase**: a 7-jump piecewise-constant function with non-dyadic jump locations $y_j \in \{0.11, 0.23, 0.34, 0.47, 0.58, 0.71, 0.83\}$ and mixed-sign amplitudes $\{+.20, -.15, +.30, -.25, +.20, -.20, +.10\}$ ($\text{TV}(g) = 1.40$). The non-dyadic locations deliberately avoid alignment with the uniform $1/N$ grid for the N used below.

Figure 6 plots the empirical L^1 error vs depth $N \in \{8, 16, 32, 64, 128, 256\}$. Linear log-log fits give slopes -1.915 for the smooth target and -0.956 for the BV staircase (neither exactly matches the asymptotic $-2 / -1$: the smooth target’s $5\times$ harmonic introduces finite- N Nyquist effects below $N \sim 16$, and non-dyadic jump locations in the BV target introduce finite- N misalignment with the grid).

The approximant panels show $N \in \{4, 16, 64, 256\}$ (log-spaced across this range; $N = 4$ exposes the smooth-target Nyquist undersampling); the rate fit excludes $N = 4$. The BV slope exhibits the expected N^{-1} decay consistent with Theorem 7’s $O(N^{-(d-1)}) = O(N^{-1})$ bound at $d = 2$; the smooth slope reflects the C^2 -rate $O(N^{-2})$ of CPWL interpolation, faster than the BV worst case the proof bounds and consistent with the proof being an upper bound that is loose for regularity above BV.

This measures a univariate read-direction slice on $[0, 1]$, not full $L^1(K)$ approximation of a 2D target on a compact $K \subset \mathbb{R}^2$, so it is a constructive diagnostic on the CAFC atom / ridge-sum implementation rather than evidence for the full theorem. The construction verifies that every A, B matrix is a rank-1 orthogonal projector ($P = P^\top$, $P^2 = P$, $\text{tr } P = 1$), confirming that the network executes through $\text{Gr}(1, 2)$ -constrained blocks throughout.

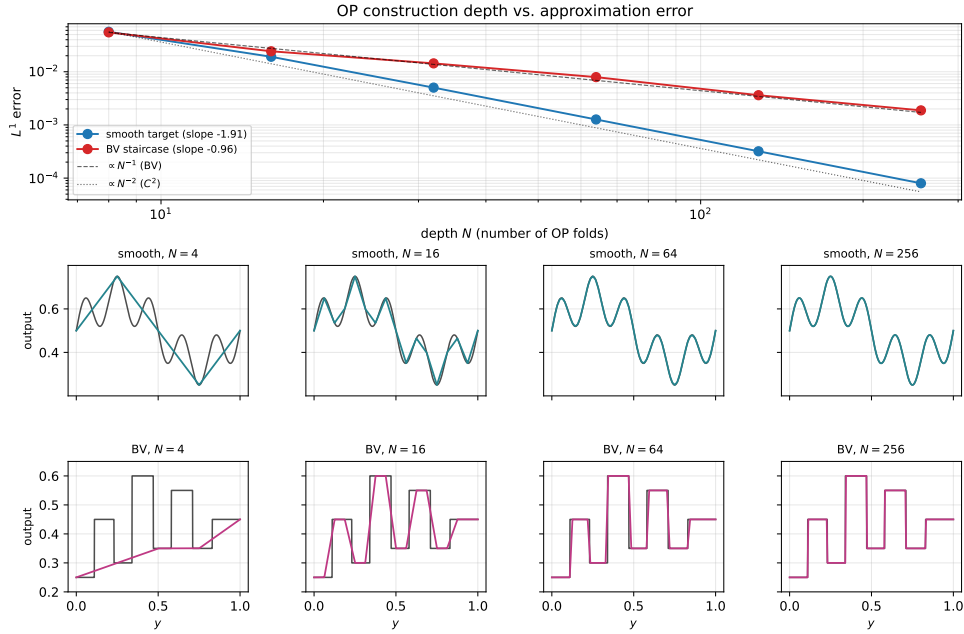


Figure 6: Direct CAFC construction for $\text{Gr}(1, 2)$ at width $d = 2$ (no training, univariate read-direction slice). **Top row:** L^1 error vs depth N on a log-log scale, with reference slopes -1 (BV) and -2 (C^2). The BV target exhibits the expected N^{-1} decay consistent with Theorem 7’s $O(N^{-(d-1)})$ rate at $d = 2$; the smooth two-scale sin target exhibits the faster C^2 -rate N^{-2} of CPWL interpolation (rate fit on $N \geq 8$ to exclude Nyquist undersampling of the $5\times$ harmonic). **Middle row:** smooth-target approximant (viridis) at $N \in \{4, 16, 64, 256\}$ overlaid on the target (black); **bottom row:** BV staircase approximant (plasma) at the same N values. Per- N panels visualize the convergence: $N = 4$ undersamples both targets (smooth: $5\times$ harmonic aliased; BV: misses most jumps), $N = 16$ resolves both, and $N \geq 64$ overlays the target essentially exactly.

C.5 nanoGPT-scale setup

Scope. The nanoGPT-scale tables and figures below are small finite-scale studies of constraint behavior at GPT-2 scale. Per-layer constraint costs are modest, and we report observations as regularization observations specific to each constraint site.

Model. $d_{\text{model}} = 384$, $n_{\text{layer}} = 6$, $n_{\text{head}} = 2$, sequence length 512, vocabulary 50,304 (BPE/GPT-2 tokenizer). Standard nanoGPT initialization ($\mathcal{N}(0, 0.02)$ on Linear/Embedding weights, zero biases, residual scaling $1/\sqrt{2n_{\text{layer}}}$ on c_{proj} and W_O). Weight tying: `lm_head.weight = tok_emb.weight`.

Training. AdamW, learning rate 6×10^{-4} peak with cosine decay to 6×10^{-5} over 20,000 iterations. Warmup 2,000 iters, weight decay 0.1, gradient clipping 1.0. Batch size 32, gradient accumulation 8, effective batch 256. The convergence and $UV^\top$ control panel uses seeds {42, 123}; the low-rank panels use seeds {42, 123, 7, 1234}. bfloat16 forward/backward, fp32 master.

Polar projection. Constrained Stiefel factors Q_Q^h, Q_K^h (and FFN Q_n^A, Q_n^B) are polar-projected onto $V(D, k)$ after each AdamW step. The projection uses fp32 SVD with TF32 disabled. Observed $\|Q^\top Q - I\|_\infty$ stays at $6\text{--}10 \cdot 10^{-5}$ for the nanoGPT-scale $D = 384, 386$ matrices at $k = 128, 192$ (the fp32 polar floor for these shapes). The low-rank panels use the same projection pipeline. The synthetic augmented-width test (§C.6) uses fp64 polar with floor $\sim 10^{-7}$.

Scalar multiplier initialization. Following the synthetic-test calibration sweep (§C.6), all constrained-FFN residual atoms initialize the outer scalar multiplier $s_0 = 0.1$ and inner scalar multiplier $t_0 = 1.0$. Lowering s_0 from the default 1.0 used for balanced per-block writes to 0.1 reduces the per-atom residual write magnitude $10\times$ and prevents the init blow-up observed at high rank in the synthetic test (initial-loss reduction by a factor of $\sim 5 \cdot 10^5$ at $k = 4$ after this change). All reported 20,000-iteration nanoGPT panels use this tuned default.

Data. OpenWebText [43] preprocessed via the standard nanoGPT data pipeline [44]. Validation cross-entropy is reported every 500 iterations on 100 held-out batches; Tables 6 and 7 use the final 20,000-iteration evaluation.

Hardware and wallclock. Runs on H100 / H200 / A100 (whichever scheduler-available). 20,000-iteration arrays use 12-hour SLURM allocations; the 10-run $d_{\text{model}} = 384$ convergence panel completed in about 9.5 hours wallclock. Constrained-FFN conditions add $\sim 5\%$ wallclock overhead for the per-step polar projection (fp32 SVD on $D \times k$ matrices, no TF32).

nanoGPT-scale provenance and compact results

Table 6 lists the nanoGPT-scale conditions cited from the main text. All rows use the setup above, and tables report sample mean $\pm$ sample standard deviation across seeds. ID suffixes follow the architecture legend: q is projected Stiefel $QQ^\top$, u is free $QQ^\top$, and lr is free $UV^\top$.

Table 6: Compact 20,000-iteration nanoGPT convergence/control panel. Validation cross-entropy is mean $\pm$ sample sd across seeds {42, 123}; lower is better. Projected Stiefel rows sit at the fp32 polar floor, while free rows are intentionally unconstrained. IDs match experiment logs and result directories.

ID	Configuration	k	n	Val CE @20K	Stiefel viol
B0	dense baseline	—	2	3.6831 ± 0.0079	—
G1q	Q/K Stiefel $QQ^\top$	128	2	3.7098 ± 0.0024	$7.7 \cdot 10^{-5}$
G1u	Q/K free $QQ^\top$	128	2	3.6985 ± 0.0034	—
G1lr	Q/K free $UV^\top$	128	2	3.6962 ± 0.0129	—
G6A	block-Grassmann GELU MLP	128	2	3.8377 ± 0.0249	$6.2 \cdot 10^{-5}$
G6Au	free $QQ^\top$ block-FFN	128	2	3.8162 ± 0.0155	0.95 (free)
G6Alr	free $UV^\top$ block-FFN	128	2	3.7488 ± 0.0007	1.14 (free)

Table 7: Low-rank nanoGPT panels. All rows use seeds $\{42, 123, 7, 1234\}$ and report final validation cross-entropy as mean $\pm$ sample sd.

Site	Family	k	n	Val CE @20K
Q/K	free $QQ^\top$	64	4	3.6992 ± 0.0143
Q/K	Stiefel $QQ^\top$	64	4	3.7074 ± 0.0108
Q/K	free $UV^\top$	64	4	3.7053 ± 0.0142
Q/K	free $QQ^\top$	32	4	3.7143 ± 0.0052
Q/K	Stiefel $QQ^\top$	32	4	3.7276 ± 0.0032
Q/K	free $UV^\top$	32	4	3.7107 ± 0.0202
Q/K	free $UV^\top$	16	4	3.7251 ± 0.0081
FFN	Stiefel $QQ^\top$	64	4	3.8396 ± 0.0072
FFN	free $QQ^\top$	64	4	3.8312 ± 0.0065
FFN	free $UV^\top$	64	4	3.7992 ± 0.0126
FFN	Stiefel $QQ^\top$	32	4	3.8626 ± 0.0182
FFN	free $QQ^\top$	32	4	3.8605 ± 0.0106
FFN	free $UV^\top$	32	4	3.8325 ± 0.0115
FFN	free $UV^\top$	16	4	3.8775 ± 0.0090

Auxiliary controls. The $n = 2$ convergence controls show nonzero but low-seed constraint costs. The $n = 4$ low-rank panels refine the comparison: at Q/K, Stiefel $QQ^\top$ at $k = 64$ is close to the same-rank $UV^\top$ control, while the FFN site benefits more from the extra freedom of $UV^\top$ at matched rank. These studies are empirical diagnostics for constraint behavior, not evidence for the universality theorems.

C.6 Synthetic augmented-width test (ID: S1q)

This unit-level diagnostic calibrates the square-Grassmann residual atom used by the atom-stack and block-Grassmann FFN diagnostics. Setup: $d_{\text{in}} = 3$, augmented width $D = d_{\text{in}} + 2 = 5$, 16 atoms, target ϕ_{smooth} on $[-1, 1]^3$, AdamW lr 10^{-3} , weight decay 0, fp64 polar projection, 5,000 iterations, and $A = Q_A Q_A^\top$, $B = Q_B Q_B^\top$ with trainable scalar multipliers s, t . At $k = 2$, the constrained atom stack drops MSE by 99.25% over three seeds (mean final MSE 0.0998) and pins $\|Q^\top Q - I\|_\infty$ at $1.19 \cdot 10^{-7}$. A scalar multiplier initialization sweep identifies $(s_0, t_0) = (0.1, 1.0)$ as the stable nanoGPT default: it gives mean final MSE 0.0183 over three seeds and reduces the high-rank initial-loss blow-up by a factor of $\sim 5 \cdot 10^5$.

D Additional Limitations

Activation regime for the block-Grassmann FFN

The single-atom containment construction in §D.3 is stated for ReLU, while the empirical block-Grassmann GELU implementation (ID: G6A) uses GELU activation. The GELU setting is not covered by the theorem, so those empirical results are reported as comparison studies rather than constructions covered by the theorem. A scaling observation suggests a controlled approximation when the multiplier budget is large enough: with $g_\alpha(z) = \text{GELU}(\alpha z)/\alpha$, standard asymptotics give $g_\alpha(z) \rightarrow \text{ReLU}(z)$ as $\alpha \rightarrow \infty$ uniformly on compact intervals. In a block-Grassmann branch, simulating a ReLU atom $x \mapsto x + s \cdot P_A \cdot \text{ReLU}(t \cdot P_B x + b)$ by setting $t' = \alpha t$, $b' = \alpha b$, $s' = s/\alpha$ yields

$$s' \cdot P_A \cdot \text{GELU}(t' \cdot P_B x + b') \rightarrow s \cdot P_A \cdot \text{ReLU}(t \cdot P_B x + b)$$

uniformly on compact state tubes as $\alpha \rightarrow \infty$. The state-restoring boundary becomes approximate rather than exact under this substitution.

D.1 Augmented-width sketch at width $d + 2$

The augmented-width route, not used by the main theorems, would lift $x \in K \subset \mathbb{R}^d$ to a residual state in $\mathbb{R}^{d+2}$, use one scratch register for an active scalar code and one frozen scratch register for storage between phases, and read the first d coordinates without a terminal unconstrained readout. The intended program is: build a finite-grid scalar code, reset the active scratch coordinate between phases via same-register two-hinge identities, and decode functions of the frozen code into visible

output coordinates. A theorem for this route would require per-manifold reset identities and an L^p bookkeeping argument, which are not part of the present results.

D.2 Constrained- Q/K attention note

This paper contains no constrained-attention universality theorem. The finite-softmax route of Kajitsuka–Sato [40] should extend to manifolds in Table 2 that provide a scalar-register witness $W_* \in \mathcal{M}$ with $W_*^\top e \neq 0$, but the composition with a constrained FFN theorem and the permutation-equivariant target assumptions are not part of the present results.

D.3 Block-Grassmann FFN containment

The block-Grassmann GELU MLP condition (ID: G6A) used in the nanoGPT comparison contains a single square-Grassmann residual atom: set all but one branch multiplier to zero, choose the active read/write projectors $P_B = Q_B Q_B^\top$, $P_A = Q_A Q_A^\top$, and keep the active branch’s s, t, b . The layer then equals $x \mapsto x + s P_A \text{ReLU}(t P_B x + b)$ for that branch. This containment does not by itself prove universality of the full GELU block-Grassmann stack; the experiments are reported as comparison studies, while the theorem would require the augmented-width route above plus an activation approximation argument.

D.4 Generalization-gap heuristic via manifold covering numbers

As a heuristic capacity calculation, compare operator-norm *covering numbers* of the constrained matrix class. For compact manifold rows, and for bounded pieces of noncompact rows after imposing an operator-norm radius, a smooth dimension-counting cover gives $\log \mathcal{N}(\mathcal{M}, \delta) = O(\dim(\mathcal{M}) \log(1/\delta))$, where $\dim(\mathcal{M})$ is the dimension of $\mathcal{M}$ as a smooth submanifold of $\mathbb{R}^{d \times d}$ (or $\mathbb{R}^{d \times k}$ for rectangular Stiefel). If a generalization bound depends on such layerwise covers, this suggests a capacity reduction of order $\sqrt{\dim(\mathcal{M})/d^2}$ relative to the unconstrained spectral-norm ball at matched operator norm. For $\text{Gr}(160, 512)$ the reduction is $\sqrt{56,320/262,144} \approx 0.46$. We present this as a heuristic interpretation, not as a theorem for the full pipeline at matched norms.

Further limitations

Several refinements remain outside the present theorem statements: exact-width vector-output universality using only constrained blocks; literal CAFC and scalar multiplier minimality for $\mathcal{J}_n$; quantitative theory for the optimization gap in Section 5.1; tight head-count bounds for constrained attention; a manifold-compatible lift from L^p to $C(K)$ density in the style of Liu et al. [39]; a matched-norm version of the covering-number heuristic in §D.4; and a finite-depth GELU analogue of the ReLU containment in §D.3.

E Unit-Bounded Multiplier Residual Universality

This appendix records what remains of the residual universality constructions after imposing $|s_\ell|, |t_\ell| \leq 1$ on every residual block. The subdivision argument applies to cross-axis CAFC folds: literal CAFC manifolds are handled fold by fold, and macro-CAFC manifolds by subdividing the transvections in the macro program. It does not give a unit-bounded version of every RIPR construction, since current same-coordinate RIPR gadgets may require coefficients larger than one.

E.1 Preliminaries

For square matrix families $\mathcal{M} \subset \mathbb{R}^{d \times d}$, the residual blocks considered here are

$$\Phi_{A,B,b,s,t}(x) = x + s A \text{ReLU}(t B x + b), \quad A, B \in \mathcal{M}, \quad b \in \mathbb{R}^d, \quad (9)$$

with coordinatewise ReLU. For rectangular Stiefel $V(k, d)$, the read side is $B^\top x$ and $b \in \mathbb{R}^k$, as in the main paper. Rows that use only the outer scalar set $t = 1$.

Definition 67 (unit-bounded multiplier block class). The unit-bounded multiplier block family associated with $\mathcal{M}$ is

$$\mathcal{B}_{\mathcal{M}}^{\leq 1} := \{ \Phi_{A,B,b,s,t} : A, B \in \mathcal{M}, b \in \mathbb{R}^d, |s| \leq 1, |t| \leq 1 \}.$$

For rectangular Stiefel this notation means the analogous set of $\Phi_{A,B,b,s,t}^{\text{rect}}$ blocks with $b \in \mathbb{R}^k$. A depth- D unit-bounded multiplier ResNet is a composition

$$x \mapsto L_{\text{out}} \Phi_D \circ \dots \circ \Phi_1(\iota x), \quad \Phi_\ell \in \mathcal{B}_{\mathcal{M}}^{\leq 1},$$

with the same input embedding and final affine readout conventions as in the paper’s RIPR and CAFC universality theorems. Biases and the final readout are not bounded.

E.2 Unit-scalar multiplier subdivision for cross-axis folds

The proof mechanism under this constraint is additivity along an unchanged read coordinate.

Definition 68 (cross-axis residual fold). Let $u, v \in \mathbb{R}^d$ satisfy $v^\top u = 0$. For $c, \beta \in \mathbb{R}$, define

$$F_{u,v,c,\beta}(x) := x + c u \text{ReLU}(v^\top x + \beta).$$

The direction v is the read axis and u is the write axis.

Lemma 69 (exact subdivision of a cross-axis fold). Let $v^\top u = 0$, $c \in \mathbb{R}$, and $m \in \mathbb{N}$. Then

$$F_{u,v,c/m,\beta}^m = F_{u,v,c,\beta}.$$

Consequently every cross-axis fold can be realized as a composition of $\lceil |c| \rceil$ cross-axis folds whose scalar amplitudes have absolute value at most one.

Proof. For any x ,

$$v^\top F_{u,v,c/m,\beta}(x) = v^\top x + \frac{c}{m} v^\top u \text{ReLU}(v^\top x + \beta) = v^\top x.$$

The scalar $\text{ReLU}(v^\top x + \beta)$ is therefore identical at every copy of the subdivided fold. After r copies the state is

$$x + r \frac{c}{m} u \text{ReLU}(v^\top x + \beta),$$

and the case $r = m$ is the claimed fold. $\square$

The paper’s CAFC block does not always expose the coefficient c directly. For a fixed manifold witness it has the form

$$x \mapsto x + s \gamma u \text{ReLU}(\eta v^\top x + b)$$

after dead-zone biasing, where $\gamma \neq 0$ is the nonzero coefficient provided by the witness and $\eta \in \{+1, -1\}$ is absorbed into the choice of read direction and bias. A desired fold coefficient c is implemented by $s = c/\gamma$. With unit-bounded scalar multipliers, choose

$$m = \max\{1, \lceil |c/\gamma| \rceil\}, \quad s' = c/(m\gamma).$$

Then $|s'| \leq 1$, and Lemma 69 shows that the m -block composition is the original fold. If a row has several witness gains, the depth bound uses the smallest nonzero gain selected by the proof.

E.3 Unit-bounded multiplier adaptation of CAFC fold programs

The paper’s CAFC construction (§4.2) produces finite cross-axis fold programs over manifolds satisfying the CAFC-fold condition. Imposing the unit bound changes depth: replace each fold with its unit-scalar multiplier subdivision.

Definition 70 (fold program size). Consider a finite fold program

$$P = F_N \circ \dots \circ F_1, \quad F_r(x) = x + c_r u_r \text{ReLU}(v_r^\top x + \beta_r),$$

where $v_r^\top u_r = 0$ for every r . Given nonzero manifold gains γ_r , define the unit-bounded multiplier subdivision depth

$$D_{\text{bg}}(P; \gamma) := \sum_{r=1}^N \max\{1, \lceil |c_r/\gamma_r| \rceil\}.$$

Lemma 71 (unit-bounded multiplier adaptation of CAFC fold programs). Let P be a finite cross-axis fold program on a compact state set, and assume that every fold of P is realizable by a CAFC block over $\mathcal{M}$ with nonzero gain γ_r . Then P is realized exactly by a unit-bounded multiplier $\mathcal{M}$ -constrained ResNet of depth $D_{\text{bg}}(P; \gamma)$.

Proof. Apply the replacement in (E.2) to each fold F_r . Each replacement realizes F_r exactly by Lemma 69. Composing the replacements in the same order gives the same program P . Every outer scalar in the replacement has absolute value at most one, and the CAFC constructions use $t = 1$ after the read-sign convention has been fixed. Hence all scalar multipliers are bounded by one. $\square$

E.4 Macro-CAFC subdivision

The $\mathcal{J}_n$ and Gr_{Sp} rows use macro-CAFC: each ideal fold is replaced by a finite program of coordinate transvections

$$T_{pq}(a)x = x + a x_q e_p, \quad p \neq q,$$

and canonical cross-axis folds. The transvection template has two internal scalars

$$s_1 = \frac{a\beta_2}{\beta_2 - \beta_1}, \quad s_2 = -\frac{a\beta_1}{\beta_2 - \beta_1}, \quad \beta_1, \beta_2 > \sup |x_q|.$$

Thus the unit-bounded replacement must control both the transvection coefficient and these internal scalars.

Lemma 72 (bounded transvection subdivision). *Let $p \neq q$. For every $a \in \mathbb{R}$ and $m \in \mathbb{N}$,*

$$T_{pq}(a/m)^m = T_{pq}(a).$$

Thus a nonzero transvection of coefficient a is a composition of

$$m_{\text{tr}}(a) := \lfloor |a| \rfloor + 1$$

transvections of coefficient $a/m_{\text{tr}}(a)$, whose absolute value is strictly smaller than one.

Proof. The source coordinate is unchanged: $(T_{pq}(a/m)x)_q = x_q$. After r copies the p -coordinate is $x_p + r(a/m)x_q$, while every coordinate used as a source remains the same for this transvection. Taking $r = m$ gives $T_{pq}(a)$. $\square$

Lemma 73 (bounded two-block transvection). *Let $p \neq q$, $a \in \mathbb{R}$, and let $X \subset \mathbb{R}^d$ be compact. Assume that the macro row supplies a two-block transvection template: for each coefficient a' , after dead-zone biasing, two residual blocks with $t = 1$ and*

$$s_1 = \frac{a'\beta_2}{\beta_2 - \beta_1}, \quad s_2 = -\frac{a'\beta_1}{\beta_2 - \beta_1}$$

realize $T_{pq}(a')$ on the chosen compact input set whenever $\beta_1, \beta_2 > \sup |x_q|$. Then $T_{pq}(a)$ is realized on X by a unit-bounded multiplier program. If $a \neq 0$, the program uses

$$2m_{\text{tr}}(a) = 2(\lfloor |a| \rfloor + 1)$$

residual blocks; if $a = 0$, no block is needed.

Proof. The case $a = 0$ is the identity. Assume $a \neq 0$, put

$$m = m_{\text{tr}}(a), \quad a' = a/m,$$

so $|a'| < 1$. Let $M = \sup_{x \in X} |x_q|$ and choose

$$\beta_1 = M + 1, \quad \beta_2 = \beta_1 + \delta, \quad \delta > \frac{|a'|\beta_1}{1 - |a'|}.$$

Then $\beta_1, \beta_2 > M$, and

$$|s_2| = \frac{|a'|\beta_1}{\delta} < 1, \quad |s_1| = \frac{|a'|(\beta_1 + \delta)}{\delta} = |a'| + \frac{|a'|\beta_1}{\delta} < 1.$$

It remains to choose the inactive-row biases on the correct compact set. Define the partial transvection states and the first-block intermediate states by

$$X_r := T_{pq}(ra')X, \quad Y_r := \{x + s_1(x_q + \beta_1)e_p : x \in X_r\}, \quad r = 0, \dots, m-1.$$

The enlarged set

$$X^+ := \bigcup_{r=0}^{m-1} (X_r \cup Y_r)$$

is a finite union of compact sets, hence compact. Every state in X^+ has the same q -coordinate as its ancestor in X , because all intermediate updates write only coordinate $p \neq q$. Thus $|z_q| \leq M$ on X^+ , so the active preactivations $z_q + \beta_h$ are strictly positive for $h = 1, 2$. Choose all inactive dead-zone biases using X^+ . The two-block template then realizes $T_{pq}(a')$ on each X_r , maps X_r to X_{r+1} , and has unit-bounded scalar multipliers. Composing over $r = 0, \dots, m-1$ gives

$$T_{pq}(a')^m = T_{pq}(a)$$

by Lemma 72. $\square$

Lemma 74 (unit-bounded multiplier macro-CAFC replacement). *Suppose a macro-CAFC construction over $\mathcal{M}$ replaces every ideal cross-axis fold used by the CAFC proof by a finite program consisting of:*

- (i) *coordinate transvections $T_{p_\rho q_\rho}(a_\rho)$, each realized by the two-block template of Lemma 73; and*
- (ii) *canonical one-block cross-axis folds with actual outer scalars θ_κ , read/write directions still orthogonal, and $t = 1$.*

Then the same ideal CAFC fold is realized on the same compact state set by a unit-bounded multiplier $\mathcal{M}$ -constrained program. One valid depth bound for this fixed macro realization is

$$D_{\text{macro}}^{\leq 1} \leq \sum_{\rho: a_\rho \neq 0} 2(\lceil |a_\rho| \rceil + 1) + \sum_{\kappa} \max\{1, \lceil |\theta_\kappa| \rceil\}.$$

Proof. Proceed through the finite macro program in order; each current input set is compact because it is the continuous image of the original compact state set. Transvections are handled by Lemma 73 on that current compact set. For a canonical cross-axis fold with outer scalar θ_κ , split

$$m_\kappa = \max\{1, \lceil |\theta_\kappa| \rceil\}, \quad \theta'_\kappa = \theta_\kappa / m_\kappa.$$

Since the read coordinate is unchanged by the write update, Lemma 69 realizes the original canonical fold exactly after choosing inactive biases on the finite union of partial-fold trajectories. Each replacement is exact on the current compact set, so induction over the macro program gives the same ideal fold. Summing the transvection and canonical-fold costs gives (74). $\square$

Limits of same-coordinate RIH subdivision

The subdivision identities above do not apply to same-coordinate RIPR gadgets, including fixed-spectrum symmetric orbits and the approximate RIPR witnesses for $E(d, k)$ and Gr_{Sp} . Current coefficients such as $t = a/\lambda$, $s = \gamma/\lambda$, and $s = 2/(\lambda_1 - \lambda_2)$ can exceed one; more importantly, $R \mapsto R + \gamma \text{ReLU}(aR + b)$ changes the coordinate later copies would read. A unit-bounded treatment of these gadgets would require a different approximation argument. The manifold conclusions used in Theorems 75–76 are unaffected: $E(d, k)$ is covered by literal CAFC and Gr_{Sp} by macro-CAFC.

E.5 Main unit-bounded multiplier theorems

The depth accounting under this constraint is route-specific. A single formula cannot cover both literal CAFC folds and macro-CAFC transvection programs.

Theorem A: literal CAFC manifolds

For a compact set K and scalar target $f \in L^1(K)$, let

$$P_{\text{CAFC}}(K, f, \varepsilon) = \{(c_h, u_h, v_h, \beta_h, \gamma_h)\}_{h=1}^{N_{\text{CAFC}}}$$

be the finite ideal CAFC fold schedule supplied by the paper's CAFC proof after the analytic approximation tolerances have been chosen. Here

$$x \mapsto x + c_h u_h \text{ReLU}(v_h^\top x + \beta_h), \quad v_h^\top u_h = 0,$$

is the h -th ideal fold, and $\gamma_h \neq 0$ is the corresponding nonzero manifold witness gain. Define

$$D_{\text{CAFC}}^{\leq 1}(K, f, \varepsilon) := \sum_{h=1}^{N_{\text{CAFC}}} \max\{1, \lceil |c_h / \gamma_h| \rceil\}.$$

The literal CAFC manifolds covered here are $V(k, d)$ and $\text{St}(k, d)$ with $1 \leq k \leq d$, $\text{Gr}(k, d)$, $\text{PSD}(k, d)$ with $0 < k < d$, $\text{LGr}(\mathbb{R}^{2n})$, and $E(d, k)$ with $0 < k < d$, all with the paper’s dimensional conventions and ambient residual dimension denoted by d .

Theorem 75 (literal CAFC unit-bounded multiplier universality). *For every manifold family $\mathcal{M}$ listed above, every compact $K \subset \mathbb{R}^d$, every $f \in L^1(K)$, and every $\varepsilon > 0$, there is a scalar-output unit-bounded multiplier $\mathcal{M}$ -constrained ResNet with unconstrained terminal linear readout and depth at most $D_{\text{CAFC}}^{\leq 1}(K, f, \varepsilon)$ such that*

$$\|F - f\|_{L^1(K)} \leq \varepsilon.$$

Proof. Use the finite CAFC fold schedule selected by the paper’s CAFC proof on the compact set K . Each listed manifold admits a literal CAFC fold realization with nonzero gain $\gamma_h \neq 0$ and $t = 1$ on every compact state set: $V(k, d)$, $\text{Gr}(k, d)$, LGr , and $E(d, k)$ are verified in the per-manifold sections of Appendix B.5; $\text{St}(k, d)$ inherits by absorption, and $\text{PSD}(k, d)$ via the inclusion $\text{Gr}(k, d) \subset \text{PSD}(k, d)$. Replace the h -th fold by the exact unit-bounded multiplier subdivision from Lemma 71: the replacement introduces no additional analytic error and has depth as in (E.5). The final readout is the one used by the CAFC proof. $\square$

Theorem B: macro-CAFC manifolds

For the macro-CAFC manifolds, first fix the finite CAFC fold schedule selected by the CAFC proof on K . Then fix, for each ideal fold in that schedule, the paper’s macro realization: a finite list of transvections with coefficients $\{a_\rho\}$, plus the canonical cross-axis fold blocks with actual outer scalars $\{\theta_\kappa\}$. Define

$$D_{\text{macro}}^{\leq 1}(K, f, \varepsilon) := \sum_{\rho: a_\rho \neq 0} 2(\lfloor |a_\rho| \rfloor + 1) + \sum_{\kappa} \max\{1, \lceil |\theta_\kappa| \rceil\}.$$

The coefficients $\{a_\rho\}$ and $\{\theta_\kappa\}$ depend on the chosen macro decompositions. The dead-zone biases and the displayed internal two-block transvection biases are chosen after the enlarged compact subdivision state sets have been formed; the biases may be large.

Theorem 76 (macro-CAFC unit-bounded multiplier universality). *For $\mathcal{J}_n$ with $d = 2n \geq 4$, and for $\text{Gr}_{\text{Sp}}(2k, \mathbb{R}^{2n})$ with $n \geq 2$ and $1 \leq k \leq n - 1$, the scalar-output unit-bounded multiplier constrained ResNet class is dense in $L^1(K)$ for every compact K . More precisely, for every $f \in L^1(K)$ and $\varepsilon > 0$, the macro-CAFC route gives a unit-bounded multiplier ResNet of depth at most $D_{\text{macro}}^{\leq 1}(K, f, \varepsilon)$ with*

$$\|F - f\|_{L^1(K)} \leq \varepsilon.$$

Proof. Use the paper’s CAFC approximant and replace each ideal CAFC fold by the macro realization from Appendix B.5: a finite transvection program plus canonical cross-axis fold blocks (using only outer scalar multipliers and $t = 1$). Lemma 74 subdivides every transvection and canonical fold into an exact unit-bounded multiplier program on the enlarged compact subdivision state sets, with inactive biases chosen on those enlarged sets so each subdivided operation is exact on the states where it is applied. The analytic approximation and final readout are unchanged, so the final $L^1(K)$ error is the one already budgeted in the CAFC proof. Summing the route-specific costs gives (E.5). $\square$

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